

The Metaphysics

Aristotle

Being as being, substance, and the unmoved mover. The Oxford translation of W. D. Ross, 1908.

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Book I

Layout 2
BOOK I (A)

CHAPTER I

980a All men by nature desire to know. An indication of this is the delight we take in our senses; for even apart from their usefulness they are loved for themselves; and above all others the sense of sight. For not only with a view to action, but even when we are not going to do anything, we prefer sight to almost everything else. The reason is that this, most of all the senses, makes us know and brings to light many differences between things.

By nature animals are born with the faculty of sensation, and from sensation memory is produced in some of them, though not in others. 980b And therefore the former are more intelligent and apt at learning than those which cannot remember; those which are incapable of hearing sounds are intelligent though they cannot be taught, e.g. the bee, and any other race of animals that may be like it; and those which besides memory have this sense of hearing, can be taught.

The animals other than man live by appearances and memories, and have but little of connected experience; but the human race lives also by art and reasonings. And from memory experience is produced in men; for many memories of the same thing produce finally the capacity for a single experience. 981a Experience is almost identified with science and art, but really science and art come to men *through* experience; for 'experience made art', as Polus says, and rightly, 'but inexperience luck.' And art arises, when from many notions gained by experience one universal judgement about a class of objects is produced. For to have a judgement that when Callias was ill of this disease this did him good, and similarly in the case of Socrates and in many individual cases, is a matter of experience; but to judge that it has done good to all persons of a certain constitution, marked off in one class, when they were ill of this disease, e.g. to phlegmatic or bilious people when burning with fever,—this is a matter of art.

With a view to action experience seems in no respect inferior to art, and we even see men of experience succeeding more than those who have theory without experience. The reason is that experience is knowledge of individuals, art of universals, and actions and productions are all concerned with the individual; for the physician does not cure *man*, except in an incidental way, but Callias or Socrates or some other called by some such individual name, who happens to be a man. If, then, a man has the theory without the experience, and knows the universal but does not know the individual included in this, he will often fail to cure; for it is the individual that is to be cured. But yet we think that *knowledge* and *understanding* belong to art rather than to experience, and we suppose artists to be wiser than men of experience (which implies that Wisdom depends in all cases rather on knowledge); and this because the former know the cause, but the latter do not. For men of experience know that the thing is so, but do not know why, while the others know the 'why' and the cause. Hence we think that the master-workers in each craft are more honourable and know in a truer sense and are wiser than the manual workers, 981b because they know the causes of the things that are done [we think the manual workers are like certain lifeless things which act indeed, but act without knowing what they do, as fire burns,—but while the lifeless things perform each of their functions by a natural tendency, the labourers perform them through habit]; thus we view them as being wiser not in virtue of being able to act, but of having the theory for themselves and knowing the causes. And in general it is a sign of the man who knows, that he can teach, and therefore we think art more truly knowledge than experience is; for artists can teach, and men of

mere experience cannot.

Again, we do not regard any of the senses as Wisdom; 10 yet surely these give the most authoritative knowledge of particulars. But they do not tell us the 'why' of anything—e. g. why fire is hot; they only say that it is hot.

At first he who invented any art that went beyond the common perceptions of man was naturally admired by men, not only because there was something useful in the inventions, 15 but because he was thought wise and superior to the rest. But as more arts were invented, and some were directed to the necessities of life, others to its recreation, the inventors of the latter were naturally always regarded as wiser than the inventors of the former, because their branches of knowledge did not aim at utility. Hence when all such inventions were 20 already established, the sciences which do not aim at giving pleasure or at the necessities of life were discovered, and first in the places where men first began to have leisure. This is why the mathematical arts were founded in Egypt; for there the priestly caste was allowed to be at leisure.

We have said in the *Ethics* what the difference is between 25 art and science and the other kindred faculties; but the point of our present discussion is this, that all men suppose what is called Wisdom to deal with the first causes and the principles of things. This is why, as has been said before, the man of 30 experience is thought to be wiser than the possessors of any perception whatever, the artist wiser than the men of experience, the master-worker than the mechanic, and the theoretical kinds of knowledge to be more of the nature of Wisdom than the productive. 982a Clearly then Wisdom is knowledge about certain causes and principles.

CHAPTER II

Since we are seeking this knowledge, we must inquire of 5 what kind are the causes and the principles, the knowledge of which is Wisdom. If we were to take the notions we have about the wise man, this might perhaps make the answer more evident. We suppose first, then, that the wise man knows all things, as far as possible, although he has not 10 knowledge of each of them in detail; secondly, that he who can learn things that are difficult, and not easy for man to know, is wise (sense-perception is common to all, and therefore easy and no mark of Wisdom); again, he who is more exact and more capable of teaching the causes is wiser, in every branch of knowledge; and of the sciences, also, that 15 which is desirable on its own account and for the sake of knowing it is more of the nature of Wisdom than that which is desirable on account of its results, and the superior science is more of the nature of Wisdom than the ancillary; for the wise man must not be ordered but must order, and he must not obey another, but the less wise must obey *him*.

20 Such and so many are the notions, then, which we have about Wisdom and the wise. Now of these characteristics that of knowing all things must belong to him who has in the highest degree universal knowledge; for he knows in a sense all the subordinate objects. And these things, the most universal, are on the whole the hardest for men to know; 25 for they are furthest from the senses. And the most exact of the sciences are those which deal most with first principles; for those which involve fewer principles are more exact than those which involve additional principles, e.g. arithmetic than geometry. But the science which investigates causes is also the more communicable, for the people who teach are 30 those who tell the causes of each thing. And understanding and knowledge pursued for their own sake are found most in the knowledge of that which is most knowable; for he who chooses to know for the sake of knowing will choose most readily that which is most truly knowledge, 982b and such is the knowledge of that which is most knowable; and the first principles and the causes are most knowable; for by reason of these, and from these, all

other things are known, but these are not known by means of the things subordinate to them. And the science which knows to what end each thing must be done is the most authoritative of the sciences, and more authoritative than any ancillary science; and this end is the good in each class, and in general the supreme good in the whole of nature. Judged by all the tests we have mentioned, then, the name in question 'Wisdom' falls to the same science; this must be a science that investigates the first principles and causes; for the good, i. e. the end and aim, is one of the causes.

That it is not a science of production is clear even from the history of the earliest philosophers. For it is owing to their wonder that men both now begin and at first began to philosophize; they wondered originally at the obvious difficulties, then advanced little by little and stated difficulties about the greater matters, e.g. about the phenomena of the moon and those of the sun, and about the stars and about the genesis of the universe. And a man who is puzzled and wonders thinks himself ignorant (whence even the lover of myth is in a sense a lover of Wisdom, for the myth is composed of wonders); therefore since they philosophized in order to escape from ignorance, evidently they were pursuing science in order to know, and not for any utilitarian end. And this is confirmed by the facts; for it was when almost all the necessities of life and the things that make for comfort and recreation were present, that such knowledge began to be sought. Evidently then we do not seek it for the sake of any other advantage; but as the man is free, we say, who exists for himself and not for another, so we pursue this as the only free science, for it alone exists for itself.

Hence the possession of it might be justly regarded as beyond human power; for in many ways human nature is in bondage, so that according to Simonides 'God alone can have this privilege', and it is unfitting that man should not be content to seek the knowledge that is suited to him. If, then, there is something in what the poets say, and jealousy is natural to the divine power, it would probably occur in this case above all, and all who excelled in this knowledge would be unfortunate. But the divine power cannot be jealous (nay, according to the proverb, 'bards tell many a lie'), nor should any science be thought more honourable than one of this sort. For the most divine science is also most honourable; and this science alone is, in two ways, most divine. For the science which it would be most meet for God to have is a divine science, and so is any science that deals with divine objects; and this science alone has both these qualities; for (1) God is thought to be among the causes of all things and to be a first principle, and (2) such a science either God alone can have, or God above all others. All the sciences, indeed, are more necessary than this, but none is better.

Yet the acquisition of it must in a sense end in something which is the opposite of our original inquiries. For all men begin, as we said, by wondering that the matter is so (as those who have not yet perceived the explanation marvel at automatic marionettes)—whether the object of their wonder be the solstices or the incommensurability of the diagonal of a square with the side; for it seems wonderful to all men that there is a thing which cannot be measured even by the smallest unit. But we must end in the contrary and, according to the proverb, the better state, as is the case in these instances when men learn the cause; for there is nothing which would surprise a geometer so much as if the diagonal turned out to be commensurable.

We have stated, then, what is the nature of the science we are searching for, and what is the mark which our search and our whole investigation must reach.

Chapter 3

Evidently we have to acquire knowledge of the original causes (for we say we know each thing only when we think we recognize its first cause), and causes are spoken of in four senses. In one of these we mean the substance, i.e. the essence (for the 'why' is reducible finally to the formula, and the ultimate 'why' is a cause and principle); in another the matter or substratum, in a third the source of the change, and in a fourth the cause opposed to this, the purpose and the good (for this is the end of all generation and change). We have studied these causes sufficiently in our work on nature, but yet let us call to our aid those who have attacked the investigation of being and philosophized about reality before us. For obviously they too speak of certain principles and causes; to go over their views, then, will be of profit to the present inquiry, for we shall either find another kind of cause, or be more convinced of the correctness of those which we now maintain.

Of the first philosophers, most thought the principles which were of the nature of matter were the only principles of all things; that of which all things that are consist, and from which they first come to be, and into which they are finally resolved (the substance remaining, but changing in its modifications), this they say is the element and the principle of things, and therefore they think nothing is either generated or destroyed, since this sort of entity is always conserved, as we say Socrates neither comes to be absolutely when he comes to be beautiful or musical, nor ceases to be when he loses these characteristics, because the substratum, Socrates himself, remains. So they say nothing else comes to be or ceases to be; for there must be some entity—either one or more than one—from which all other things come to be, it being conserved.

Yet they do not all agree as to the number and the nature of these principles. Thales, the founder of this school of philosophy, says the principle is water (for which reason he declared that the earth rests on water), getting the notion perhaps from seeing that the nutriment of all things is moist, and that heat itself is generated from the moist and kept alive by it (and that from which they come to be is a principle of all things). He got his notion from this fact, and from the fact that the seeds of all things have a moist nature, and that water is the origin of the nature of moist things.

Some think that the ancients who lived long before the present generation, and first framed accounts of the gods, had a similar view of nature; for they made Ocean and Tethys the parents of creation, and described the oath of the gods as being by water, which the poets themselves call Styx; for what is oldest is most honourable, and the most honourable thing is that by which one swears. It may perhaps be uncertain whether this opinion about nature is primitive and ancient, but Thales at any rate is said to have declared himself thus about the first cause. Hippo no one would think fit to include among these thinkers, because of the paltriness of his thought.

Anaximenes and Diogenes make air prior to water, and the most primary of the simple bodies, while Hippasus of Metapontium and Heraclitus of Ephesus say this of fire, and Empedocles says it of the four elements, adding a fourth—earth—to those which have been named; for these, he says, always remain and do not come to be, except that they come to be more or fewer, being aggregated into one and segregated out of one.

Anaxagoras of Clazomenae, who, though older than Empedocles, was later in his philosophical activity, says the principles are infinite in number; for he says almost all the things that are made of parts like themselves are generated and destroyed (as water or fire is) only by aggregation and segregation, and are not in any other sense generated or destroyed, but remain eternally.

From these facts one might think that the only cause is the so-called material cause; but as men thus advanced, the very facts showed them the way and joined in forcing them to investigate the subject. However true it may be that all generation and destruction proceed from some one or

more elements, why does this happen and what is the cause? For at least the substratum itself does not make itself change; e.g. neither the wood nor the bronze causes the change of either of them, nor does the wood manufacture a bed and the bronze a statue, but something else is the cause of the change. And to seek this is to seek the second cause, as *we* should say,—that from which comes the beginning of the movement. Now those who at the very beginning set themselves to this kind of inquiry, and said the substratum was one, were not at all dissatisfied with themselves; but some at least of those who maintain it to be one—as though defeated by this search for the second cause—say the one and nature as a whole is unchangeable not only in respect of generation and destruction (for this is a primitive belief, and all agreed in it), but also of all other change; and this view is peculiar to them. Of those who said the universe was one, none succeeded in discovering a cause of this sort, except perhaps Parmenides, and he only inasmuch that he supposes that there is not only one but in some sense two causes. But for those who make more elements it is more possible to state the second cause, e.g. for those who make hot and cold, or fire and earth, the elements; for they treat fire as having a nature which fits it to move things, and water and earth and such things they treat in the contrary way.

When these men and the principles of this kind had had their day, as the latter were found inadequate to generate the nature of things, men were again forced by the truth itself, as we said, to inquire into the next kind of cause. For surely it is not likely either that fire or earth or any such element should be the reason why things manifest goodness and beauty both in their being and in their coming to be, or that those thinkers should have supposed it was; nor again could it be right to ascribe so great a matter to spontaneity and luck. When one man said, then, that reason was present—as in animals, so throughout nature—as the cause of the world and of all its order, he seemed like a sober man in contrast with the random talk of his predecessors. We know that Anaxagoras certainly adopted these views, but Hermotimus of Clazomenae is credited with expressing them earlier. Those who thought thus stated that there is a principle of things which is at the same time the cause of beauty, and that sort of cause from which things acquire movement.

Chapter 4

One might suspect that Hesiod was the first to look for such a thing—or some one else who put love or desire among existing things as a principle, as Parmenides does; for he, in constructing the genesis of the universe, says:—

Love first of all the Gods she planned.

And Hesiod says:—

First of all things was chaos made, and then

Broad-breasted earth, and love that foremost is

Among all the immortals,

which implies that among existing things there must be from the first a cause which will move things and bring them together. How these thinkers should be arranged with regard to priority of discovery let us be allowed to decide later; but since the contraries of the various forms of good were also perceived to be present in nature—not only order and the beautiful, but also disorder and the ugly, and bad things in greater number than good, and ignoble things than beautiful, therefore another thinker introduced friendship and strife, each of the two the cause of one of these two sets of qualities. For if we were to follow out the view of Empedocles, and interpret it according to its meaning and not to its lisping expression, we should find that friendship is the cause of good things, and strife of bad. Therefore, if we said that Empedocles in a sense both mentions, and is the

first to mention, the bad and the good as principles, we should perhaps be right, since the cause of all goods is the good itself.

These thinkers, as we say, evidently grasped, and to this extent, two of the causes which we distinguished in our work on nature—the matter and the source of the movement,—vaguely, however, and with no clearness, but as untrained men behave in fights; for they go round their opponents and often strike fine blows, but they do not fight on scientific principles, and so these thinkers do not seem to know what they say; for it is evident that, as a rule, they make no use of their causes except to a small extent. For Anaxagoras uses reason as a *deus ex machina* for the making of the world, and when he is at a loss to tell for what cause something necessarily is, then he drags reason in, but in all other cases ascribes events to anything rather than to reason. And Empedocles, though he uses the causes to a greater extent than this, neither does so sufficiently nor attains consistency in their use. At least, in many cases he makes friendship segregate things, and strife aggregate them. For when the universe is dissolved into its elements by strife, fire is aggregated into one, and so is each of the other elements; but when again under the influence of friendship they come together into one, the parts must again be segregated out of each element.

Empedocles, then, in contrast with his predecessors, was the first to introduce this cause in a divided form, not positing one source of movement, but different and contrary sources. Again, he was the first to speak of four material elements; yet he does not *use* four, but treats them as two only; he treats fire by itself, and its opposites—earth, air, and water—as one kind of thing. We may learn this by study of his verses.

This philosopher then, as we say, spoke of the principles in this way, and made them of this number. Leucippus and his associate Democritus say that the full and the empty are the elements, calling the one being and the other non-being—the full and solid being being, the empty and the rare non-being (whence they say being is no more being than non-being is, because the solid is no more being than the empty); and they make these the material causes of things. And as those who make the underlying substance one generate all other things by its modifications, supposing the rare and the dense to be the sources of the modifications, in the same way these philosophers say the differences in the elements are the causes of all other qualities. These differences, they say, are three—shape and order and position. For they say the real is differentiated only by ‘rhythm’ and ‘inter-contact’ and ‘turning’; and of these rhythm is shape, inter-contact is order, and turning is position; for A differs from N in shape, AN from NA in order, Γ from H in position. The question of movement—whence or how it belongs to things—these thinkers, like the others, lazily neglected.

Regarding the two causes, then, as we say, the inquiry seems to have been pushed thus far by the early philosophers.

Chapter 5

Contemporaneously with these philosophers and before them, the Pythagoreans, as they are called, devoted themselves to mathematics; they were the first to advance this study, and having been brought up in it they thought its principles were the principles of all things. Since of these principles numbers are by nature the first, and in numbers they seemed to see many resemblances to the things that exist and come into being—more than in fire and earth and water (such and such a modification of numbers being justice, another being soul and reason, another being opportunity—and similarly almost all other things being numerically expressible); since, again, they saw that the attributes and the ratios of the musical scales were expressible in numbers; since, then, all other things seemed in their whole nature to be modelled after numbers, and numbers

seemed to be the first things in the whole of nature, they supposed the elements of numbers to be the elements of all things, and the whole heaven to be a musical scale and a number. And all the properties of numbers and scales which they could show to agree with the attributes and parts and the whole arrangement of the heavens, they collected and fitted into their scheme; and if there was a gap anywhere, they readily made additions so as to make their whole theory coherent. E.g. as the number 10 is thought to be perfect and to comprise the whole nature of numbers, they say that the bodies which move through the heavens are ten, but as the visible bodies are only nine, to meet this they invent a tenth—the 'counter-earth'. We have discussed these matters more exactly elsewhere.

But the object of our discussion is that we may learn from these philosophers also what they suppose to be the principles and how these fall under the causes we have named. Evidently, then, these thinkers also consider that number is the principle both as matter for things and as forming their modifications and their permanent states, and hold that the elements of number are the even and the odd, and of these the former is unlimited, and the latter limited; and the 1 proceeds from both of these (for it is both even and odd), and number from the 1; and the whole heaven, as has been said, is numbers.

Other members of this same school say there are ten principles, which they arrange in two columns of cognates—limit and unlimited, odd and even, one and plurality, right and left, male and female, resting and moving, straight and curved, light and darkness, good and bad, square and oblong. In this way Alcmaeon of Croton seems also to have conceived the matter, and either he got this view from them or they got it from him; for he expressed himself similarly to them. For he says most human affairs go in pairs, meaning not definite contraries such as the Pythagoreans speak of, but any chance contraries, e. g. white and black, sweet and bitter, good and bad, great and small. He threw out indefinite suggestions about the other contraries, but the Pythagoreans declared both how many and which their contraries are.

From both these schools, then, we can learn this much, that the contraries are the principles of things; and how many these principles are and which they are, we can learn from one of the two schools. But how these principles can be brought together under the causes we have named has not been clearly and articulately stated by them; they seem, however, to range the elements under the head of matter; for out of these as immanent parts they say substance is composed and moulded.

From these facts we may sufficiently perceive the meaning of the ancients who said the elements of nature were more than one; but there are some who spoke of the universe as if it were one entity, though they were not all alike either in the excellence of their statement or in regard to the nature of the entity. The discussion of them is in no way appropriate to our present investigation of causes, for they do not, like some of the natural philosophers, assume being to be one and yet generate it out of the one as out of matter, but they speak in another way; those others add change, since they generate the universe, but these thinkers say the universe is unchangeable. Yet this much is appropriate to the present inquiry: Parmenides seems to fasten on that which is one in formula, Melissus on that which is one in matter, for which reason the former says that it is limited, the latter that it is unlimited; while Xenophanes, the first of this school of monists (for Parmenides is said to have been his pupil), gave no clear statement, nor does he seem to have grasped either of these two kinds of unity, but he contemplates the whole heaven and says the One is, namely God. Now these thinkers, as we said, must be neglected for the purposes of the present inquiry—two of them entirely, as being a little too naïve, viz. Xenophanes and Melissus; but Parmenides seems to speak with somewhat more insight. For, claiming that, besides the existent, nothing non-existent exists, he thinks that the existent is of necessity one and that nothing else exists (on this we have spoken

more clearly in our work on nature), but being forced to follow the observed facts, and supposing the existence of that which is one in formula, but more than one according to our sensations, he now posits two causes and two principles, calling them hot and cold, i. e. fire and earth; and of these he ranges the hot with the existent, and the other with the non-existent.

From what has been said, then, and from the wise men who have now sat in council with us, we have got thus much—both from the earliest philosophers, who regard the first principle as corporeal (for water and fire and such things are bodies), and of whom some suppose that there is one corporeal principle, others that there are more than one, but both put these under the head of matter; and from some others who posit both this cause and besides this the source of movement, which is stated by some as one and by others as two.

Down to the Italian school, then, and apart from it, philosophers have treated these subjects rather obscurely, except that, as we said, they have used two kinds of cause, and one of these—the source of movement—some treat as one and others as two. But the Pythagoreans have said in the same way that there are two principles, but added this much, which is peculiar to them, that they thought finitude and infinity and unity were not attributes of certain other things, e. g. of fire or earth or anything else of this kind, but that infinity itself and unity itself were the substance of the things of which they are predicated. This is why number was the substance of all things. On this subject, then, they expressed themselves thus; and regarding the question of essence they began to make statements and definitions, but treated the matter too simply. For they both defined superficially and thought that the first subject of which a given term would be predicable, was the substance of the thing, as if one supposed that 'double' and '2' were the same, because 2 is the first thing of which 'double' is predicable. But surely to be double and to be 2 are not the same; if they are, one thing will be many—a consequence which they actually drew. From the earlier philosophers, then, and from their successors we can learn thus much.

Chapter 6

After the systems we have named came the philosophy of Plato, which in most respects followed these thinkers, but had peculiarities that distinguished it from the philosophy of the Italians. For, having in his youth first become familiar with Cratylus and with the Heraclitean doctrines (that all sensible things are ever in a state of flux and there is no knowledge about them), these views he held even in later years. Socrates, however, was busying himself about ethical matters and neglecting the world of nature as a whole but seeking the universal in these ethical matters, and fixed thought for the first time on definitions; Plato accepted his teaching, but held that the problem applied not to any sensible thing but to entities of another kind—for this reason, that the common definition could not be a definition of any sensible thing, as they were always changing. Things of this other sort, then, he called Ideas, and sensible things, he said, were apart from these, and were all called after these; for the multitude of things which have the same name as the Form exist by participation in it. Only the name 'participation' was new; for the Pythagoreans say that things exist by 'imitation' of numbers, and Plato says they exist by participation, changing the name. But what the participation or the imitation of the Forms could be they left an open question.

Further, besides sensible things and Forms he says there are the objects of mathematics, which occupy an intermediate position, differing from sensible things in being eternal and unchangeable, from Forms in that there are many alike, while the Form itself is in each case unique.

Since the Forms are the causes of all other things, he thought their elements were the elements of all things. As matter, the great and the small were principles; as essential reality, the One; for

from the great and the small, by participation in the One, come the Forms—i. e. the Numbers.

But he agreed with the Pythagoreans in saying that the One is substance and not a predicate of something else; and in saying that the Numbers are the causes of the reality of other things, he also agreed with them; but positing a dyad and constructing the infinite out of great and small, instead of treating the infinite as one, is peculiar to him; and so is his view that the Numbers exist apart from sensible things, while *they* say that the things themselves are Numbers, and do not place the objects of mathematics between Ideas and sensible things. His divergence from the Pythagoreans in making the One and the Numbers separate from things, and his introduction of the Forms, were due to his inquiries in the region of definitory formulae (for the earlier thinkers had no tincture of dialectic), and his making the other entity besides the One a dyad was due to the belief that the numbers, except those which were prime, could be neatly produced out of the dyad as out of a plastic material.

Yet what happens is the contrary; the theory is not a reasonable one. For they make many things out of the matter, and the form generates only once, but what we observe is that one table is made from one matter, while the man who applies the form, though he is one, makes many tables. And the relation of the male to the female is similar; for the latter is impregnated by one copulation, but the male impregnates many females; yet these are analogues of those first principles.

Plato, then, declared himself thus on the points in question; it is evident from what has been said that he has used only two causes, that of the essence and the material cause (for the Forms are the cause of the essence of all other things, and the One is the cause of the essence of the Forms); and it is evident what the underlying matter is, of which the Forms are predicated in the case of sensible things, and the One in the case of Forms, viz. that this is a dyad, the great and the small. Further, he has assigned the cause of good and that of evil to the elements, one to each of the two, as we say some of his predecessors sought to do, e. g. Empedocles and Anaxagoras.

Chapter 7

Our account of those who have spoken about first principles and reality and of the way in which they have spoken, has been concise and summary; but yet we have learnt this much from them, that of those who speak about 'principle' and 'cause' no one has mentioned any principle except those which have been distinguished in our work on nature, but all evidently have some inkling of *them*, though only vaguely. For some speak of the first principle as matter, whether they suppose one or more first principles, and whether they suppose this to be a body or to be incorporeal; e. g. Plato spoke of the great and the small, the Italians of the infinite, Empedocles of fire, earth, water, and air, Anaxagoras of the infinity of things composed of similar parts. These, then, have all had a notion of this kind of cause, and so have all who speak of air or fire or water, or something denser than fire and rarer than air; for some have said the prime element is of this kind. These thinkers grasped this cause only; but certain others have mentioned the source of movement, e.g. those who make friendship and strife, or reason, or love, a principle.

The essence, i. e. the substantial reality, no one has expressed distinctly. It is mentioned chiefly by those who believe in the Forms; for they do not suppose either that the Forms are the matter of sensible things, and the One the matter of the Forms, or that they are the source of movement (for they say these are causes rather of immobility and of beings at rest), but they furnish the Forms as the essence of every other thing, and the One as the essence of the Forms.

That which is the end for which actions and changes and movements take place, they assert to be a cause in a way, but not in this way, i. e. not in the way in which it is its *nature* to be a cause. For

those who speak of reason or friendship class these causes as goods; they do not speak, however, as if anything that exists either existed or came into being for the sake of these, but as if movements started from these. In the same way those who say the One or the existent is the good, say that it is the cause of substance, but not that substance either is or comes to be for the sake of this. Therefore it turns out that in a sense they both say and do not say the good is a cause; for they do not call it a cause *qua* good but only incidentally.

All these thinkers, then, as they cannot pitch on another cause, seem to testify that we have determined rightly both how many and of what sort the causes are. Besides this it is plain that when the causes are being looked for, either all four must be sought thus or they must be sought in one of these four ways. Let us next discuss the possible difficulties with regard to the way in which each of these thinkers has spoken, and with regard to his views about the first principles.

Chapter 8

Those, then, who say the universe is one and posit one kind of thing as matter, and as corporeal matter which has spatial magnitude, evidently go astray in many ways. For they posit the elements of bodies only, not of incorporeal things, though there are incorporeal things. And in trying to state the causes of generation and destruction, and in giving an account of the nature of all things, they do away with the cause of movement. Further, they err in not positing the substance, i. e. the essence, as the cause of anything, and besides this in lightly calling any of the simple bodies except earth the first principle, without inquiring how they are produced out of one another,—I mean fire, water, earth, and air. For some things are produced out of others by combination, others by separation, and this makes the greatest difference to their priority and posteriority. For (1) in a way the property of being most elementary of all would seem to belong to the first thing from which they are produced by combination, and *this* property would belong to the most fine-grained and subtle of bodies. Therefore those who make fire the principle would be most in agreement with this argument. But each of the other thinkers agrees that the element of corporeal things is of this sort. At least none of the later philosophers who said the world was one claimed that earth was the element, evidently because of the coarseness of its grain. (Of the other three elements each has found some judge on its side; for some maintain that fire, others that water, others that air is the element. Yet why, after all, do they not name earth also, as most men do—for people say all things are earth. And Hesiod says earth was produced first of corporeal things; so ancient and popular has the opinion been.) According to this argument, then, no one would be right who either says the first principle is any of the elements other than fire, or supposes it to be denser than air but rarer than water. But (2) if that which is later in generation is prior in nature, and that which is concocted and compounded is later in generation, the contrary of what we have been saying must be true,—water must be prior to air, and earth to water.

Let this suffice, then, as our statement about those who posit one cause such as we mentioned; but the same is true if we suppose more of these, as Empedocles says the matter of things is four bodies. For he too is confronted by consequences some of which are the same as have been mentioned, while others are peculiar to him. For we see these bodies produced from one another, which implies that the same body does not always remain fire or earth (we have spoken about this in our works on nature); and regarding the moving cause and the question whether we must suppose one or two, he must be thought to have spoken neither correctly nor altogether plausibly. [And in general those who speak in this way must do away with change of quality, for on their view cold will not come from hot nor hot from cold. For if it did there would be something that

accepted those very contraries, and there would be some one entity that became fire and water, which Empedocles denies.]

As regards Anaxagoras, if one were to suppose that he said there were two elements, the supposition would accord thoroughly with a view which Anaxagoras himself did not state articulately, but which he must have accepted if any one had developed his view. True, to say that in the beginning all things were mixed is absurd both on other grounds and because it follows that they must have existed before in an unmixed form, and because nature does not allow any chance thing to be mixed with any chance thing, and also because on this view modifications and accidents could be separated from substances (for the same things which are mixed can be separated); yet if one were to follow him up, piecing together what he means, he would perhaps be seen to be somewhat modern in his views. For when nothing was separated out, evidently nothing could be truly asserted of the substance that then existed. I mean, e. g. that it was neither white nor black, nor grey nor any other colour, but of necessity colourless; for if it had been coloured, it would have had one of these colours. And similarly, by this same argument, it was flavourless, nor had it any similar attribute; for it could not be either of any quality or of any size, nor could it be any definite kind of thing. For if it were, one of the particular forms would have belonged to it, and this is impossible, since all were mixed together; for the particular form would necessarily have been already separated out, but he says all were mixed except mind, and this alone was unmixed and pure. From this it follows, then, that he must say the principles are the One (for this is simple and unmixed) and the Other, which is of such a nature as we suppose the indefinite to be before it is defined and partakes of some form. Therefore, while expressing himself neither rightly nor clearly, he means something like what the later thinkers say and what is now more clearly seen to be the case.

But these thinkers are, after all, at home only in arguments about generation and destruction and movement; for it is practically only of this sort of substance that they seek the principles and the causes. But those who extend their vision to all things that exist, and of existing things suppose some to be perceptible and others not perceptible, evidently study both classes, which is all the more reason why one should devote some time to seeing what is good in their views and what bad from the stand-point of the inquiry we have now before us.

The 'Pythagoreans' treat the principles and the elements more strangely than the physical philosophers (the reason is that they got the principles from non-sensible things, for the objects of mathematics, except those of astronomy, are of the class of things without movement); yet their discussions and investigations are all about nature; for they generate the heavens, and with regard to their parts and attributes and functions they observe the phenomena, and use up the principles and the causes in explaining these, which implies that they agree with the others, the physical philosophers, that the real is just all that which is perceptible and contained by the so-called 'heavens'. But the causes and the principles which they mention are, as we said, sufficient to act as steps even up to the higher realms of reality, and are more suited to these than to theories about nature. They do not tell us at all, however, how there can be movement if limit and unlimited and odd and even are the only things assumed, or how without process and change there can be generation and destruction, or how the bodies that move through the heavens can do what they do. Further, if we either granted them that spatial magnitude consists of these elements, or this were proved, still how would some bodies be light and others have weight? To judge from what they assume and maintain, they speak no more of mathematical bodies than of perceptible; hence they have said nothing whatever about fire or earth or the other bodies of this sort, I suppose because

they have nothing to say which applies *peculiarly* to perceptible things.

Further, how are we to combine the beliefs that the modifications of number, and number itself, are causes of what exists and happens in the heavens both from the beginning and now, and that there is no other number than this number out of which the world is composed? When in one particular region they place opinion and opportunity, and, a little above or below, injustice and sifting or mixture, and allege as proof of this that each one of these is a number, but when there happens to be already in each place a plurality of the extended bodies composed of numbers, because these modifications of number attach to the various groups of places,—this being so, is this number, which we must suppose each of these abstractions to be, the same number which is exhibited in the material universe, or is it another than this? Plato says it is different; yet even he thinks that both these bodies and their causes are numbers, but that the *intelligible* numbers are causes, while the others are *sensible*.

Chapter 9

Let us leave the Pythagoreans for the present; for it is enough to have touched on them as much as we have done. But as for those who posit the Ideas as causes, firstly, in seeking to grasp the causes of the things around us, they introduced others equal in number to these, as if a man who wanted to count things thought he could not do it while they were few, but tried to count them when he had added to their number. For the Forms are practically equal to or not fewer than the things, in trying to explain which these thinkers proceeded from them to the Forms. For to each set of substances there answers a Form which has the same name and exists apart from the substances, and so also in the case of all other groups in which there is one character common to many things, whether the things are in this changeable world or are eternal.

Further, of the ways in which we prove that the Forms exist, none is convincing; for from some no inference necessarily follows, and from some it follows that there are Forms of things of which we think there are no Forms.

For according to the arguments from the existence of the sciences there will be Forms of all things of which there are sciences, and according to the argument that there is one attribute common to many things there will be Forms even of negations, and according to the argument that there is an object for thought even when the thing has perished, there will be Forms of perishable things; for we can have an image of these.

Further, of the more accurate arguments, some lead to Ideas of relations, of which we say there is no independent class, and others involve the difficulty of the 'third man'.

And in general the arguments for the Forms destroy the things for whose existence those who maintain the Forms are more anxious than for the existence of the Ideas; for it follows that not the dyad but number is first, i. e. that the relative is prior to the absolute,—besides all the other points on which certain people by following out the opinions held about the Ideas have come into conflict with the principles of the theory.

Further, according to the assumption on which our belief in the Ideas rests, there will be Forms not only of substances but also of many other things (for the concept is single not only in the case of substances but also in the other cases, and there are sciences not only of substance but also of other things, and a thousand other such conclusions also follow). But according to the necessities of the case and the opinions held about the Forms, if they can be shared there must be Ideas of substances only. For they are not shared incidentally, but a thing must share in its Form as in something not predicated of a subject (e. g. if a thing shares in 'double itself', it shares also in 'eternal', but

incidentally; for 'eternal' happens to be predicable of the 'double'). Therefore the Forms will be substance; and the same terms indicate substance in this and in the ideal world (or what will be the meaning of saying that there is something apart from the particulars—the one over many?). And if the Ideas and the particulars that share them have the same Form, there will be something common to these; for why should '2' be one and the same in the perishable 2's or in those which are many but eternal, and not the same in the '2 itself' as in the particular 2? But if they have not the same Form, they must have only the name in common, and it is as if one were to call both Callias and a wooden image a 'man', without observing any community between them.

Above all one might discuss the question what on earth the Forms contribute to sensible things, either to those that are eternal or to those that come into being and cease to be. For they cause neither movement nor any change in them. But again they help in no wise towards the *knowledge* of the other things (for they are not even the substance of these, else they would have been in them), nor towards their being, if they are not *in* the particulars which share in them; though if they were, they might be thought to be causes, as white causes whiteness in that with which it is mixed. But this argument, which first Anaxagoras and later Eudoxus and certain others used, is too easily upset; for it is not difficult to collect many insuperable objections to such a view.

But further all other things cannot come from the Forms in any of the usual senses of 'from'. And to say that they are patterns and the other things share them is to use empty words and poetical metaphors. For what is it that works, looking to the Ideas? Anything can either be, or become, like another without being copied from it, so that whether Socrates exists or not a man might come to be like Socrates; and evidently this might be so even if Socrates were eternal. And according to the Platonist's principles there will be several patterns of the same thing, and therefore several Forms, e.g. 'animal' and 'two-footed' and also 'man himself' will be Forms of man. Again, the Forms are patterns not only of sensible things, but of things-in-themselves also, i. e. the genus as genus is the pattern of the species; therefore the same thing will be pattern and copy.

Again it must be held to be impossible that the substance and that of which it is the substance should exist apart; how, therefore, can the Ideas, being the substances of things, exist apart?

In the *Phaedo* the case is stated in this way—that the Forms are causes both of being and of becoming; yet when the Forms exist, still the things that share in them do not come into being, unless there is some efficient cause; and many other things come into being (e. g. a house or a ring), of which we say there are no Forms. Clearly, therefore, even the other things can both be and come into being owing to such causes as produce the things just mentioned.

Again, if the Forms are numbers, how can they be causes? Is it because existing things are other numbers, e.g. one number is man, another is Socrates, another Callias? Why then are the one set of numbers causes of the other set? It will not make any difference even if the former are eternal and the latter are not. But if it is because things in this sensible world (e. g. harmony) are ratios of numbers, evidently there is some one class of things of which they are ratios. If, then, this—the matter—is some definite thing, evidently the numbers themselves too will be ratios of something to something else. E. g. if Callias is a numerical ratio between fire and earth and water and air, his Idea also will be a number of certain other underlying things; and the Idea of man, whether it is a number in a sense or not, will still be a numerical ratio of certain things and not a number proper, nor will it be a number merely because it is a numerical ratio.

Again, from many numbers one number is produced, but how can one Form come from many Forms? And if the number comes not from the many numbers themselves but from the units in them, e.g. in 10,000, how is it with the units? If they are specifically alike, numerous absurdities will

follow, and also if they are not alike (neither the units in the same number being like one another nor those in different numbers being all like to all); for in what will they differ, as they are without quality? This is not a plausible view, nor can it be consistently thought out. Further, the Platonists must set up a second kind of number (with which arithmetic deals), and all the objects which are called 'intermediate' by some thinkers; and how do these exist or from what principles do they proceed? Or why must they be intermediate between the things in this sensible world and the things-in-themselves? Further, the units in 2 must each come from a prior 2; but this is impossible. Further, why is a number, when taken all together, one? Again, besides what has been said, if the units are *diverse* the Platonists should have spoken like those who say there are four, or two, elements; for each of these thinkers gives the name of element not to that which is common, e. g. to body, but to fire and earth, whether there is something common to them, viz. body, or not. But in fact the Platonists speak as if the One were *homogeneous* like fire or water; and if this is so, the numbers will not be substances. Evidently, if there is a One-in-itself and this is a first principle, 'one' is being used in more than one sense; for otherwise the theory is impossible.

When we Platonists wish to reduce substances to their principles, we state that lines come from the short and long (i. e. from a kind of small and great), and the plane from the broad and narrow, and the solid from the deep and shallow. Yet how then can the plane contain a line, or the solid a line or a plane? For the broad and narrow is a different class of things from the deep and shallow. Therefore, just as number is not present in these, because the many and few are different from these, evidently no other of the higher <more abstract> classes will be present in the lower <more concrete>. But again the broad is not a genus which includes the deep, for then the solid would have been a species of plane. Further, from what principle will the presence of the points in the line be derived? Plato even used to object to this class of things <sc. points> as being a geometrical fiction. What we call the point he called the principle of the line, and this is what he meant by the indivisible lines which he often posited. Yet these must have a limit; therefore the argument from which the existence of the line follows proves also the existence of the point.

In general, though philosophy seeks the cause of perceptible things, we Platonists have given this up (for we say nothing of the cause from which change takes its start), but while we fancy we are stating the substance of perceptible things, we assert the existence of a second class of substances, while our account of the way in which they are the substances of perceptible things is empty talk; for 'sharing', as we said before, means nothing. Nor have the Forms any connexion with that which we see to be the cause in the case of the arts, and for whose sake mind and nature produce all that they *do* produce,—with this cause which we assert to be one of the first principles; but mathematics has come to be the whole of philosophy for modern thinkers, though they say that it should be studied for the sake of other things. Further, one might suppose that the substance which according to them underlies as matter is too mathematical, and is a predicate and differentia of the substance, i. e. of the matter, rather than the matter itself; i. e. the great and the small are like the rare and the dense which the physical philosophers speak of, calling these the primary differentiae of the substratum; for these are a kind of excess and defect. And regarding movement, if the great and the small are to *be* movement, evidently the Forms will be moved; but if they are not, whence did movement come? If we cannot answer this the whole study of nature has been annihilated.

And what is thought to be easy—to show that all things are one—is not done; for by the separating of the universal from particulars all things do not come to be one but there comes to be a One-in-itself, if we grant all the assumptions. And not even this follows, if we do not grant that

the universal is a class; and this in some cases it cannot be.

Nor can it be explained either how the lines and planes and solids that come after the numbers exist or can exist, or what meaning they have; for these can neither be Forms (for they are not numbers), nor the intermediates (for those are the objects of mathematics), nor the perishable things. This is evidently a distinct fourth class.

In general, if we search for the elements of existing things without distinguishing the many senses in which things are said to exist, we cannot succeed, especially if the search for the elements of which things are made is conducted in the Platonic manner. For it is surely impossible to discover what 'acting' or 'being acted on', or 'the straight', is made of, but if elements can be discovered at all, it is only the elements of substances; therefore to seek the elements of all existing things or to think one has them is incorrect. And how could we *learn* the elements of all things? Evidently we cannot start by knowing something before. For as he who is learning geometry, though he may know other things before, knows none of the things with which the science deals and about which he is to learn, so is it in all other cases. Therefore if there is a science of all things, as some maintain, he who is learning this will know nothing before. Yet all learning is by means of premises which are (either all or some of them) known before,—whether the learning be by demonstration or by definitions; for the elements of the definition must be known before and be familiar; and learning by induction proceeds similarly. But again, if the science is innate, it is wonderful that we are unaware of our possession of the greatest of sciences. Again, how is one to *know* what all things are made of, and how is this to be made *evident*? This also affords a difficulty; for there might be a conflict of opinion, as there is about certain syllables; some say *za* is made out of *s* and *d* and *a*, while others say it is a distinct sound and none of those that are familiar. Further, how could we know the objects of sense without having the sense in question? Yet we should, if the elements of which all things consist, as complex sounds consist of their proper elements, are the same.

Chapter 10

It is evident, then, even from what we have said before, that all men seem to seek the causes named in the Physics, and that we cannot name any beyond these; but they seek these vaguely; and though in a sense they have all been described before, in a sense they have not been described at all. For the earliest philosophy is, on all subjects, like one who lisps, since in its beginnings it is but a child. For even Empedocles says bone exists by virtue of the ratio in it. Now this is the essence and the substance of the thing. But it is similarly necessary that the ratio should be the substance of flesh and of everything else, or of none; therefore it is on account of this that flesh and bone and everything else will exist, and not on account of the matter, which *he* names,—fire and earth and water and air. But while he would necessarily have agreed if another had said this, he has not said it clearly.

On such questions our views have been expressed before; but let us return to enumerate the difficulties that might be raised on these same points; for perhaps we may get some help towards our later difficulties.

Book II

Layout 2

Chapter 1

The investigation of the truth is in one way hard, in another easy. An indication of this is found in the fact that no one is able to attain the truth adequately, while, on the other hand, no one fails entirely, but every one says something true about the nature of things, and while individually they contribute little or nothing to the truth, by the union of all a considerable amount is amassed. Therefore, since the truth seems to be like the proverbial door, which no one can fail to hit, in this way it is easy, but the fact that we can have a whole truth and not the particular part we aim at shows the difficulty of it.

Perhaps, as difficulties are of two kinds, the cause of the present difficulty is not in the facts but in us. For as the eyes of bats are to the blaze of day, so is the reason in our soul to the things which are by nature most evident of all.

It is just that we should be grateful, not only to those whose opinions we may share, but also to those who have expressed more superficial views; for these also contributed something, by developing before us the powers of thought. It is true that if there had been no Timotheus we should have been without much of our lyric poetry; but if there had been no Phrynis there would have been no Timotheus. The same holds good of those who have expressed views about the truth; for from the better thinkers we have inherited certain opinions, while the others have been responsible for the appearance of the better thinkers.

It is right also that philosophy should be called knowledge of the truth. For the end of theoretical knowledge is truth, while that of practical knowledge is action (for even if they consider how things are, practical men do not study the cause in itself, but in some relation and at some time). Now we do not know a truth without its cause; and a thing has a quality in a higher degree than other things if in virtue of it the similar quality belongs to the other things (e. g. fire is the hottest of things; for it is the cause of the heat of all other things); so that that which causes derivative truths to be true is most true. Therefore the principles of eternal things must be always most true; for they are not merely sometimes true, nor is there any cause of their being, but they themselves are the cause of the being of other things, so that as each thing is in respect of being, so is it in respect of truth.

Chapter 2

Evidently there is a first principle, and the causes of things are neither an infinite series nor infinitely various in kind. For (1), on the one hand, one thing cannot proceed from another, as from matter, *ad infinitum*, e. g. flesh from earth, earth from air, air from fire, and so on without stopping; nor on the other hand can the efficient causes form an endless series, man for instance being acted on by air, air by the sun, the sun by Strife, and so on without limit. Similarly the final causes cannot go on *ad infinitum*,—walking for the sake of health, this for the sake of happiness, happiness for the sake of something else, and so one thing always for the sake of another. And the case of the formal cause is similar. For in the case of an intermediate, which has a last term and a prior term outside it, the prior must be the cause of the later terms. For if we had to say which of the three is the cause, we should say the first; surely not the last, for the final term is the cause of none; nor even the intermediate, for it is the cause only of one. It makes no difference whether there is one intermediate or more, nor whether they are infinite or finite in number. But of series which are

infinite in this way, and of the infinite in general, all the parts down to that now present are alike intermediates; so that if there is no first there is no cause at all.

Nor can there be an infinite process downwards, with a beginning in the upper direction, so that water should proceed from fire, earth from water, and so always some other kind should be produced. For one thing comes *from* another in two ways (if we exclude the sense in which 'from' means 'after' as we say 'from the Isthmian games come the Olympian'), (a) as the man comes from the boy, by the boy's changing, or (b) as air comes from water. By 'as the man comes from the boy' we mean 'as that which has come to be from that which is coming to be, or as that which is finished from that which is being achieved' (for as becoming is between being and not being, so that which is becoming is always between that which is and that which is not; and the learner is a man of science in the making, and this is what is meant when we say that *from* a learner a man of science is being made); on the other hand, coming from another thing as water comes from air implies the destruction of the other thing. This is why changes of the former kind are not reversible,—the boy does not come from the man (for that which comes to be does not come to be *from* the process of coming to be, but exists *after* the process of coming to be; for it is thus that the day comes from the morning—in the sense that it comes after the morning; and therefore the morning cannot come from the day); but changes of the other kind are reversible. But in both cases it is impossible that the number of terms should be infinite. For terms of the former kind being intermediates must have an end, and terms of the latter kind change into *one another*; for the destruction of either is the generation of the other.

At the same time it is impossible that the first cause, being eternal, should be destroyed; for while the process of becoming is not infinite in the upward direction, a first cause by whose destruction something came to be could not be eternal.

Further, the *final cause* is an end, and that sort of end which is not for the sake of something else, but for whose sake everything else is; so that if there is to be a last term of this sort, the process will not be infinite; but if there is no such term there will be no final cause. But those who maintain the infinite series destroy the Good without knowing it. Yet no one would try to do anything if he were not going to come to a limit. Nor would there be reason in the world; the reasonable man, at least, always acts for a purpose; and this is a limit, for the end is a limit.

But the *formal cause*, also, cannot be reduced always to another definition which is fuller in expression. For the original definition is always more of a definition, and not the later one; and in a series in which the first term is not correct, the next is not so either.—Further, those who speak thus destroy science; for it is not possible to have this till one comes to the indivisible concepts. And knowledge becomes impossible; for how can one think things that are infinite in this way? For this is not like the case of the line, to whose divisibility there is no stop, but which we cannot think if we do not make a stop; so that one who is tracing the infinitely divisible line cannot be counting the possibilities of section.

But further, the *matter* in a changeable thing must be cognized.

Again, nothing infinite can exist; and if it could, at least being infinite is not infinite.

But (2) if the *kinds* of causes had been infinite in number, then also knowledge would have been impossible; for we think we know, only when we have ascertained all the causes, but that which is infinite by addition cannot be gone through in a finite time.

Chapter 3

The effect which lectures produce on a hearer depends on his habits; for we demand the language we are accustomed to, and that which is different from this seems not in keeping but somewhat unintelligible and foreign because of its unwontedness. For the customary is more intelligible. The force of habit is shown by the laws, in whose case, with regard to the legendary and childish elements in them, habit has more influence than our knowledge about them. Some people do not listen to a speaker unless he speaks mathematically, others unless he gives instances, while others expect him to cite a poet as witness. And some want to have everything done accurately, while others are annoyed by accuracy, either because they cannot follow the connexion of thought or because they regard it as pettifoggery. For accuracy has something of this character, so that as in trade so in argument some people think it mean. Therefore one must be already trained to know how to take each sort of argument, since it is absurd to seek at the same time knowledge and the way of attaining knowledge; and neither is easy to get.

The minute accuracy of mathematics is not to be demanded in all cases, but only in the case of things which have no matter. Therefore its method is not that of natural science; for presumably all nature has matter. Hence we must inquire first what nature is: for thus we shall also see what natural science treats of [and whether it belongs to one science or to more to investigate the causes and the principles of things].

Book III

Layout 2

Chapter 1

We must, with a view to the science which we are seeking, first recount the subjects that should be first discussed. These include both the other opinions that some have held on certain points, and any points besides these that happen to have been overlooked. For those who wish to get clear of difficulties it is advantageous to state the difficulties well; for the subsequent free play of thought implies the solution of the previous difficulties, and it is not possible to untie a knot which one does not know. But the difficulty of our thinking points to a 'knot' in the object; for in so far as our thought is in difficulties, it is in like case with those who are bound; for in either case it is impossible to go forward. Therefore one should have surveyed all the difficulties beforehand, both for the reasons we have stated and because people who inquire without first stating the difficulties are like those who do not know where they have to go; besides, a man does not otherwise know even whether he has found what he is looking for or not; for the end is not clear to such a man, while to him who has first discussed the difficulties it is clear. Further, he who has heard all the contending arguments, as if they were the parties to a case, must be in a better position for judging.

The first problem concerns the subject which we discussed in our prefatory remarks. It is this—whether the investigation of the causes belongs to one or to more sciences, and, if to one, whether this should survey only the first principles of substance, or also the principles on which all men base their proofs, e. g. whether it is possible at the same time to assert and deny one and the same thing or not, and all other such questions. And if the science in question deals with substance, whether does *one* science deal with all substances, or more than one, and if more, whether are all akin, or must some of them be called forms of Wisdom and the others something else? And this itself is also one of the things that must be discussed—whether sensible substances alone should be said to exist or others also besides them, and whether these others are of one kind or there are several classes of substances, as is supposed by those who believe both in Forms and in mathematical objects intermediate between these and sensible things. We must inquire, then, as we say, into these questions, and also whether our investigation is concerned only with substances or also with the essential attributes of substances. Further, with regard to the same and other and like and unlike and contrariety, and with regard to prior and posterior and all other such terms, about which the dialecticians try to inquire, starting their investigation from probable premises only,—whose business is it to inquire into all these? Further, we must discuss the essential attributes of these themselves; and we must ask not only what each of these is, but also whether one thing always has one contrary. Again, whether are the principles and elements of things the *classes*, or the *parts* present in each thing, into which it is divided; and if they are the classes, whether are they the classes that are predicated proximately of the individuals, or the highest classes, e. g. whether is animal or man the first principle and the more independent of the individual instance? And we must inquire and discuss especially whether there is, besides the matter, any thing that is a cause in itself or not, and whether this can exist apart or not, and whether it is one or more in number. Once more, is there something apart from the concrete thing (by the concrete thing I mean the matter with something predicated of it), or is there nothing apart, or is there something in some cases though not in others, and what sort of cases are these? Again we ask whether the principles are limited in number or in kind, both those in the formulae and those in the substratum; and whether the principles of perishable and of imperishable things are the same or different; and whether they

are all imperishable or those of perishable things are perishable. Further there is the question which is hardest of all and most perplexing, whether unity and being, as the Pythagoreans and Plato said, are not attributes of something else but are the substance of existing things, or this is not the case, but the substratum is something else,—as Empedocles says, love; as someone else says, fire; while one says water and one, air. Again we ask whether the principles are universal or like individual things, and whether they exist potentially or actually; further, whether they are potential or actual in any other sense than in reference to movement; for these questions also would present much difficulty. Further, whether are numbers and lines and figures and points a kind of substance or not, and if they are substances whether are they separate from sensible things or present in them? With regard to all these matters not only is it hard to get possession of the truth, but it is not easy even to think out the difficulties well.

Chapter 2

First then with regard to what we mentioned first, whether does it belong to one or to more sciences to investigate all the kinds of causes? How could it belong to one science to know the principles if these are not contrary?

Further, there are many things to which not all the principles pertain. For how can a principle of change or the nature of the good be present in unchangeable things, since everything that in itself and by its own nature is good is an end, and a cause in the sense that for its sake the other things both come to be and are, and since an end or purpose is the end of some action, and all actions imply change; so that in unchangeable things this principle could not exist nor could there be a good-in-itself. This is why in mathematics nothing is proved by means of this kind of cause, nor is there any demonstration of this kind—'because it is better, or worse'; indeed no one even mentions anything of the kind. And so for this reason some of the Sophists, e.g. Aristippus, ridiculed mathematics; for in the arts, even in the industrial arts, e. g. in carpentry and cobbling, the reason always given is 'because it is better, or worse', but the mathematical sciences take no account of goods and evils.

But if there are several sciences of the causes, and a different science for each different principle, which of these sciences should be said to be that which we seek, or which of the people who possess them has the most scientific knowledge of the object in question? The same thing may have all the kinds of causes, e. g. the moving cause of a house is the art or the builder, the final cause is the function it fulfils, the matter is earth and stones, and the form is the definitory formula. To judge from our previous discussion of the question which of the sciences should be called Wisdom, there is reason for applying the name to each of them. For inasmuch as it is most architectonic and authoritative and the other sciences, like slave-women, may not even contradict it, the science of the *end* and of the *good* is of the nature of Wisdom (for the other things are for the sake of the end). But inasmuch as it was described as dealing with the first causes and that which is in the highest sense object of knowledge, the science of *substance* must be of the nature of Wisdom. For as men may know the same thing in many ways, we say that he who knows what a thing is by the characteristics it has knows more fully than he who knows it by the characteristics it has not, and in the former class itself one knows more fully than another, and he knows most fully who knows what a thing is, not he who knows its quantity or quality or what it can by nature do or have done to it; and further in all other cases also (i. e. where demonstration is possible) we think that the knowledge of each thing is present when we know what it is, e. g. what squaring a rectangle is, viz. that it is the finding of a mean; and similarly in all other cases. And we know about becomings

and actions and about every change when we know the *source of the movement*; and this is other than and opposed to the end. Therefore it would seem to belong to different sciences to investigate these causes severally.

But, regarding the starting-points of demonstration also, it is a disputable question whether they are the object of one science or of more. By the starting-points of demonstration I mean the common beliefs, on which all men base their proofs, e. g. that everything must be either affirmed or denied, and that a thing cannot at the same time be and not be, and all other such premises; the question is whether the same science deals with them as with substance, or a different science, and if it is not one science, which of the two must be identified with that which we now seek.—It is not reasonable that these topics should be the object of one science; for why should it be peculiarly appropriate to geometry or to any other science to understand these matters? If then it belongs to every science alike, and cannot belong to all, it is not peculiar to the science which investigates substances, any more than to any other science, to know about these topics.—And, at the same time, in what way can there be a *science* of the first principles? For we are aware even now what each of them is; at least even other sciences use them as familiar. And if there is a demonstrative science which deals with them, there will have to be an underlying kind, and some of them must be demonstrable attributes and others must be axioms (for it is impossible that there should be demonstration about all things); for the demonstration must start from certain premises and be about a certain subject and prove certain attributes. Therefore it follows that all attributes that are proved must belong to one class; for all demonstrative sciences use the axioms.—But if the science of substance and the science which deals with the axioms are different, which of them is more authoritative and prior? The *axioms* are most universal and are principles of all things. And if it is not the business of the philosopher, to whom else will it belong to inquire what is true and what is untrue about them?

In general, do all substances fall under one science or under more than one? If the latter, to what sort of substance is the present science to be assigned? On the other hand, it is not reasonable that one science should deal with all. For then there would be one demonstrative science dealing with all attributes. For every demonstrative science investigates with regard to some subject its essential attributes, starting from the common beliefs. Therefore to investigate the essential attributes of one subject, starting from one set of beliefs, is the business of one science. For the subject belongs to one science, and the premises belong to one, whether to the same or to another; so that the attributes also are investigated either by these sciences or by one derived from them.

Further, does our investigation deal with substances alone or also with their attributes? I mean for instance, if the solid is a substance and so are lines and planes, is it the business of the same science to know these and to know the attributes of each of these classes (the attributes which the mathematical sciences prove), or of a different science? If of the *same*, the science of substance also must be a demonstrative science; but it is thought that there is *no* demonstration of the essence of things. And if of *another*, what will be the science that investigates the attributes of substance? This is a very difficult question.

Further, must we say that sensible substances alone exist, or that there are others besides these? And are substances of one kind or are there several kinds of substances, as those say who assert the existence both of the Forms and of the intermediates with which they say the mathematical sciences deal?—In what sense we say the Forms are causes and substances in themselves has been explained in our first remarks about them; while the ideal theory presents difficulties in many ways, the most paradoxical thing of all is the statement that there are certain things besides those in the material

universe, and that these are the same as sensible things except that they are eternal while the latter are perishable. For they say there is a man-in-himself and a horse-in-itself and health-in-itself, with no further qualification,—a procedure like that of the people who said there are gods, but in human form. For they were positing nothing but eternal men, nor are the Platonists making the Forms anything other than eternal sensible things.—Further, if we are to posit besides the Forms and the sensibles the intermediates between them, we shall have many difficulties. For clearly on the same principle there will be lines besides the lines-in-themselves and the sensible lines, and so with each of the other classes of things; so that since astronomy is one of these mathematical sciences there will also be a heaven besides the sensible heaven, and a sun and a moon (and so with the other heavenly bodies) besides the sensible. Yet how are we to believe these things? It is not reasonable even to suppose these bodies immovable, but to suppose their *moving* is quite impossible. And similarly with the things of which optics and mathematical harmonics treat. For these also cannot exist apart from the sensible things, for the same reasons. For if there are sensible things and sensations intermediate between Form and individual, evidently there will also be animals intermediate between animals-in-themselves and the perishable animals.—We might as also raise the question, with reference to *which kind* of existing things we must look for these additional sciences. If geometry is to differ from mensuration only in this, that the latter of these deals with things that we perceive, and the former with things that are not perceptible, evidently there will be a science other than medicine, intermediate between medical-science-in-itself and this individual medical science, and so with each of the other sciences. Yet how is this possible? There would have to be also healthy things besides the perceptible healthy things and the healthy-in-itself. And at the same time not even this is true, that mensuration deals with perceptible and perishable magnitudes; for then it would have perished, when they perished. And astronomy also cannot be dealing with perceptible magnitudes nor with this heaven above us. For neither are perceptible lines such lines as the geometer speaks of (for no perceptible thing is straight or curved in the way in which he defines 'straight' and 'curved'; for a hoop touches a straight edge not at a point, but as Protagoras said it did, in his refutation of the geometers), nor are the movements and complex orbits in the heavens like those of which astronomy treats, nor have geometrical points the same nature as the actual stars.—Now there are some who say that these so-called intermediates between the Forms and the perceptible things exist, not apart from the perceptible things, however, but in these; the impossible results of this view would take too long to enumerate, but it is enough to consider such points as the following:—It is not reasonable that this should be so only in the case of these *intermediates* but clearly the *Forms* also might be in the perceptible things; for the same account applies to both. Further, it follows from this theory that there are two solids in the same place, and that the intermediates are not immovable, since they are in the moving perceptible things. And in general to what purpose would one suppose them to *exist* indeed, but to exist *in* perceptible things? For the same paradoxical results will follow which we have already mentioned; there will be a heaven besides *the* heaven, only it will be not apart but in the same place; which is still more impossible.

Chapter 3

Apart from the difficulty of stating the case truly with regard to these matters, it is hard to say, with regard to the first principles, whether it is the genera that should be taken as elements and principles, or rather the primary constituents of a thing; e.g. it is the primary parts of which all articulate sounds consist that are thought to be elements and principles of articulate sound, not

the common genus—articulate sound; and we give the name of 'elements' to those geometrical propositions, the proofs of which are implied in the proofs of the others, either of all or of most. Further, both those who say there are several elements of corporeal things and those who say there is one, say the parts of which bodies consist and are compounded are principles, e.g. Empedocles says fire and water and the intermediates between these are the constituent elements of things, but does not describe these as genera of existing things. Besides this, if we want to examine the nature of anything else, we examine the parts of which, e.g., a bed consists and how they are put together, and then we know its nature. To judge from these arguments, then, the principles of things would not be the genera; but in so far as we know each thing by its definition, and the genera are the principles or starting-points of definitions, the genera must also be the principles of definable things. And if to get the knowledge of things is to get the knowledge of the species according to which they are named, the genera are at least starting-points of the *species*. And some also of those who say unity and being, or the great and the small, are elements of things, seem to treat them as genera.—But, again, it is not possible to describe the principles in *both* ways. For the formula of the essence is one; but definition by genera will be different from that which states the constituent parts of a thing.

Besides this, even if the genera are in the highest degree principles, whether should one regard the first of the genera as principles, or those which are predicated directly of the individuals? This also admits of dispute. For if the universal is always more of a principle, evidently the uppermost of the genera are the principles; for these are predicated of all things. There will, then, be as many principles of things as there are primary genera, so that both being and unity will be principles and substances; for these are most of all predicated of all things. But it is not possible that either unity or being should be a genus of things; for the differentiae of any genus must each of them both have being and be one, but it is not possible for the genus to be predicated of the differentiae taken apart from the species (any more than for the species of the genus to be predicated of the proper differentiae of the genus); so that if unity or being is a genus, no differentia will either be one or have being. But if unity and being are not genera, neither will they be principles, if the genera are the principles.—Again, the intermediate classes, whose concepts include the differentiae, will on this theory be genera, down to the individuals; but as it is, some are thought to be genera and others are not thought to be so. Besides this, the differentiae are principles even more than the genera; and if these also are principles, there comes to be practically an infinite number of principles, especially if we suppose the highest genus to be a principle.—But again, if unity *is* more of the nature of a principle, and the indivisible is one, and everything indivisible is so either in quantity or in species, and that which is so in species is prior to the divisible, and genera are divisible into species (for man is not the *genus* of individual men), that which is predicated directly of the individuals will have more unity.—Further, in the case of things in which the distinction of prior and posterior is present, that which is predicable of these things cannot be something apart from them; e.g. if two is the first of numbers, there will not be a Number apart from the kinds of numbers; and similarly there will not be a Figure apart from the kinds of figures; and if the genera of these things do not exist apart from the species, the genera of other things will scarcely do so; for genera of these things are thought to exist if any do. But in the indivisible species one member is not prior and another posterior. Further, where one is better and another worse, the better is always prior; so that of these also no genus can exist. From these considerations, then, the species predicated of individuals seem to be principles rather than the genera.—But again, it is not easy to say in what sense these are to be taken as principles. For the principle or cause must exist alongside of the things of which it is the

principle, and must be capable of existing in separation from them; and for what reason should we suppose any such thing to exist alongside of the individual, except that it is predicated universally and of all? But if this is the reason, the more universal must be supposed to be more of a principle; so that the highest genera would be the principles.

Chapter 4

There is a difficulty connected with these, the hardest of all and the most necessary to examine, and to this our argument has now brought us. If, on the one hand, there is nothing apart from individual things, and the individuals are infinite in number, how is it possible to get knowledge of the *infinite* individuals? For all things that we know, we know in so far as they have some unity and identity, and in so far as some attribute belongs to them universally.—But if this is necessary, and there must be something apart from the individuals, it will be necessary that the classes exist apart from the individuals,—either the lowest or the highest classes; but we found by discussion just now that this is impossible.—Further, if we admit in the fullest sense that something exists apart from the concrete thing, whenever something is predicated of the matter, must there, if there is something apart, be something corresponding to each set of individuals, or to some and not to others, or to none? (1) If there is nothing apart from individuals, there will be no object of thought, but all things will be objects of sense, and there will not be knowledge of anything, unless we say that sensation is knowledge. Further, nothing will be eternal or unmovable; for all perceptible things perish and are in movement. But if there is nothing eternal, neither can there be a process of coming to be; for that which comes to be, and that from which it comes to be, must be something, and the ultimate term in this series cannot have come to be, since the series has a limit and nothing can come to be out of that which is not.—Further, if generation and movement exist there must also be a limit; for no movement is infinite, but every movement has an end, and that which is incapable of completing its coming to be cannot be in process of coming to be; and that which has completed its coming to be must *be* as soon as it has come to be.—Further, since the matter exists, because it is ungenerated, it is *a fortiori* reasonable that the substance or essence, that which the matter is at any time coming to be, should exist; for if neither essence nor matter is, nothing will be at all. And since this is impossible there must be something besides the concrete thing, viz. the shape or form.—But again (2) if we are to suppose this, it is hard to say in which cases we are to suppose it and in which not. For evidently it is not possible to suppose it in all cases; we could not suppose that there is a house besides the particular houses.—Besides this, will the substance of all the individuals, e.g. of all men, be one? This is paradoxical, for all the things whose substance is on this view one are not one. But are they many and different? This also is unreasonable.—At the same time, how does the matter become each of the individuals, and how *is* the concrete thing these two elements?

Again, one might ask the following question also about the first principles. If they are one *in kind* only, nothing will be numerically one, not even unity-itself and being-itself. And how will it be possible to know, if there is not to be something common to a whole set of individuals? But if there is a common element which is *numerically* one, and each of the principles is one, and the principles are not as in the case of perceptible things different for different things (e.g. since this particular syllable is the same in kind whenever it occurs, the elements of it are also the same in kind; only in kind, for these also, like the syllable, are numerically different in different contexts),—if the principles of things are not one in this sense, but are numerically one, there will be nothing else besides the elements; for there is no difference of meaning between ‘numerically one’ and ‘individual’. For this is just what we mean by the individual—the numerically one, and by the

universal we mean that which is predicable of the individuals. Therefore just as, if the elements of articulate sound were limited in number, all the literature in the world would be confined to the ABC, since there could not be two or more letters of the same kind, so is it with all existing things and their first principles.

One difficulty which is as great as any has been omitted both by modern philosophers and by their predecessors—whether the principles of perishable and those of imperishable things are the same or different. If they are the same, how are some things imperishable and others perishable, and for what reason? The school of Hesiod and all the mythologists thought only of what was plausible to themselves, and had no regard to us. For asserting the first principles to be gods and born of gods, they say that the beings which did not taste of nectar and ambrosia became mortal; and clearly they are using words which are familiar to themselves, yet what they have said even about the very application of these causes is above our comprehension. For if the gods taste nectar and ambrosia for their pleasure, these are in no wise the causes of their existence; and if they taste them to maintain their existence, how can gods who need food be eternal?—But into the subtleties of the mythologists it is not worth our while to inquire seriously; those, however, who use the language of proof we must cross-examine and ask why, after all, things which consist of the same elements are, some of them, eternal in nature, while others perish. Since these philosophers mention no cause, and it is unreasonable that things should be as they say, evidently the principles or causes of things cannot be the same. Even the man whom one might suppose to speak most consistently—Empedocles,—even he has made the same mistake; for he maintains that strife is a principle that causes *destruction*, but strife would seem none the less to *produce* everything, except the One; for all things excepting God proceed from strife. At least he says:—

From which grew all that was and is and will be hereafter—

Trees, and men and women, and beasts and birds

And water-nourished fish, and long-aged gods.

The implication is evident even apart from these words; for if strife had not been present in things, all things would have been one, as he says—‘when they have come together, strife stands outermost’ Hence it also follows on his theory that God most blessed is less wise than all others; for he does not know all the elements; for he has in him no strife, and knowledge is of the like by the like. ‘For by earth,’ he says,

we see earth, by water water,

By ether godlike ether, by fire wasting fire,

Love by love, and strife by gloomy strife.

But—and this is the point we started from—this at least is evident, that on his theory it follows that strife is as much the cause of existence as of destruction. And similarly friendship is not specially the cause of existence; for in collecting things into the One it destroys all other things.—And at the same time Empedocles mentions no cause of the change itself, except that things are so by nature.

But when strife waxed great in the limbs,

And sprang to honour as the time was fulfilled

Which is fixed for them in turn by a mighty oath.

This implies that change was necessary; but he shows no cause of the necessity. But yet so far at least he alone speaks consistently; for he does not make some things perishable and others imperishable, but makes all perishable except the elements. The difficulty we are speaking of now is, why some things are perishable and others are not, if they consist of the same principles.

Let this suffice as proof of the fact that the principles cannot be the same. But if there are different principles, one difficulty is whether these themselves will be imperishable or perishable. For if they are *perishable* evidently these also must consist of certain elements; for all things that perish, perish by being resolved into the elements of which they consist; so that it follows that prior to the principles there are other principles. And this is impossible, whether the process has a limit or proceeds to infinity. Further, how will perishable things exist, if their principles are to be destroyed? But if the principles are *imperishable*, why will things composed of *some* imperishable principles be perishable, while those composed of the others are imperishable? This is not probable, but is either impossible or needs much proof. Further, no one has even tried to maintain different principles; they maintain the same principles for all things. But they swallow the difficulty we stated first as if they took it to be something trifling.

The hardest inquiry of all, and the one most necessary for knowledge of the truth, is whether being and unity are the substances of things, and whether each of them, without being anything else, is being or unity respectively, *or* we must inquire what being and unity are, with the implication that they have some other underlying nature. For some people think they are of the former, others think they are of the latter character. Plato and the Pythagoreans thought being and unity were nothing else, but this was their nature, their essence being just unity and being. But the natural philosophers take a different line; e. g. Empedocles—as though reducing it to something more intelligible—says what unity is; for he would seem to say it is love: at least, this is for all things the cause of their being one. Others say this unity and being, of which things consist and have been made, is fire, and others say it is air. A similar view is expressed by those who make the elements more than one; for these also must say that being and unity are precisely all the things which they say are principles, (1) If we do not suppose unity and being to be substances, it follows that none of the other universals is a substance; for these are most universal of all. For if there is no unity-itself or being-itself, there will scarcely be in any other case anything apart from what are called the individuals. Further, if unity is not a substance, evidently number also will not exist as an entity separate from the individual things; for number is units, and the unit is something whose essence it is to be one.—But (2) if there is a unity-itself and a being-itself, their substance must be unity and being; for it is not something else that is predicated universally of them, but just unity and being. But if there is to be a being-itself and a unity-itself, there is much difficulty in seeing how there will be anything else besides these,—I mean, how things will be more than one in number. For what is different from being does not exist, so that it necessarily follows, according to the argument of Parmenides, that all things that are are one and this is being.—There are objections to both views. For whether unity is not a substance or there *is* a unity-itself, number cannot be a substance. We have already said why this result follows if unity is not a substance: and if it is, the same difficulty arises as arose with regard to being. For whence is there to be another one besides the unity-itself? It must be not-one; but all things are either one or many, and of the many each is one.—Further, if the unity-itself is indivisible, according to Zeno's doctrine it will be nothing. For that which neither when added makes a thing greater nor when subtracted makes it less, he asserts to have no being, evidently assuming that whatever has being is a spatial magnitude. And if it is a magnitude, it is corporeal; for the corporeal has being in every dimension, while the other objects of mathematics, e.g. a plane or a line, added in one way will increase what they are added to, but in another way will not do so, and a point or a unit does so in no way. But if he argues thus, his argument is of a low order; and an indivisible thing *can* exist, so that in this way too the position may be defended even against him; for the indivisible when added will make the number,

though not the size, greater. But how can a *magnitude* proceed from one such indivisible or from many? It is like saying that the line is made out of points. But even if one supposes the case to be such that, as some say, number proceeds from the unity-itself and something else which is not one, none the less we must inquire why and how the product will be sometimes a number and sometimes a magnitude, if the not-one was inequality and was the same principle in either case. For it is not evident how magnitudes could proceed either from the one and this principle, or from some number and this principle.

Chapter 5

A question connected with these is whether numbers and bodies and planes and points are substances or not. If they are not, it baffles us to say what being is and what the substances of things are. For modifications and movements and relations and dispositions and ratios do not seem to indicate the substance of anything; for all are predicated of a subject, and none is a 'this'. And as to the things which might seem most of all to indicate substance, water and earth and fire and air, of which composite bodies consist, heat and cold and the like are modifications of these, not substances, and the body which is thus modified alone persists as something real and as a substance. But, on the other hand, the body is surely less of a substance than the surface, and the surface less than the line, and the line less than the unit and the point. For the body is bounded by these; and they are thought to be capable of existing without body, but the body cannot exist without these. This is why, while most of the philosophers and the earlier among them thought that substance and being were identical with *corporeal matter*, and that all other things were attributes of this, so that the first principles of bodies were the first principles of being, the more recent and those who were held to be wiser thought *numbers* were the first principles. As we said, then, if these are not substance, there is no substance and no being at all; for surely it is not proper to call the accidents of these beings. But if this is admitted, that lines and points are substance more than bodies, but we do not see to what sort of bodies these could belong (for they cannot be in perceptible bodies), there can be no substance.—Further, these are all evidently divisions of the body,—one a division in breadth, another in depth, another in length.—Besides this, no sort of shape is present in the solid more than any other; so that if the Hermes is not in the stone, neither is the half of the cube in the cube as something determinate; therefore the surface is not in it either; for if any sort of surface were in it, the surface which marks off the half of the cube would be in it too. And the same account applies to the line and to the point and the unit. Therefore, if on the one hand the body is in the highest degree substance, and on the other hand these things are so more than the body, but these are not even instances of substance, it baffles us to say what being is and what the substance of things is.—For besides what has been said, the questions of generation and destruction confront us with further paradoxes. For if substance, not having existed before, now exists, or having existed before, afterwards does not exist, this change is thought to be accompanied by a process of becoming or perishing; but points and lines and surfaces cannot be in process of becoming nor of perishing, though they at one time exist and at another do not. For when bodies come into contact or are separated, their boundaries instantaneously become one at one time—when they touch, and two at another time—when they are separated; so that when they have been put together one boundary does not exist but has perished, and when they have been separated the boundaries exist which before did not exist. For it cannot be said that the point (which is indivisible) was divided into two. And if the boundaries come into being and cease to be, from what do they come into being? A similar account may also be given of the 'now' in time; for

this also cannot be in process of coming into being or of ceasing to be, but yet seems to be always different, which shows that it is not a substance. And evidently the same is true of points and lines and planes; for the same argument applies, as they are all alike either limits or divisions.

Chapter 6

In general one might raise the question why, besides perceptible things and the intermediates, we have to look for another class of things, such as the Forms which we posit. If it is for this reason, because the objects of mathematics, while they differ from the things in this world in some other respect, differ not at all in that there are many of the same kind, so that their first principles cannot be limited in number (just as the elements of all the language in this sensible world are not limited in number, but in kind, unless one takes the elements of this individual syllable or of this individual articulate sound—whose elements will be limited even in number—, so is it also in the case of the intermediates; for there also the members of the same kind are infinite in number), so that if there are not—besides perceptible and mathematical objects—others such as some maintain the Forms to be, there will be no substance which is one in number as well as in kind, nor will the first principles of things be determinate in number, but only in kind,—if then this must be so, the Forms also must therefore be held to exist. Even if those who support this view do not express it distinctly, still this is what they mean, and they must be maintaining the Forms just because each of the Forms is a substance and none is by accident. But if we are to suppose that the Forms exist and the principles are one in number, not in kind, the impossible results that we have mentioned necessarily follow.

Closely connected with this is the question whether the elements exist potentially or in some other way. If in some other way, there will be something else prior to the first principles; for the potency is prior to the actual cause, and it is not necessary for everything potential to be actual.—But if the elements exist potentially, it is possible that everything that is should not be. For even that which is not yet is capable of being; for that which is not comes to be, but nothing that is incapable of being comes to be.

We must not only raise these questions about the first principles, but also ask whether they are universal or what we call individuals. If they are universal, they will not be substances; for everything that is common indicates not a 'this' but a 'such', but substance is a 'this'.—And if we can actually hypostatize the common predicate as an individual, Socrates will be several animals—himself and 'man' and 'animal,' if each of these indicates a 'this' and a single thing.—If, then, the principles are universals, these results follow; if they are not universals but of the nature of individuals, they will not be knowable; for the knowledge of anything is universal. Therefore if there is to be knowledge of the principles there must be other principles prior to them, which are universally predicated of them.

Book IV

Layout 2

Chapter 1

There is a science which investigates being as being and the attributes which belong to this in virtue of its own nature. Now this is not the same as any of the so-called special sciences; for none of these others deals generally with being as being. They cut off a part of being and investigate the attributes of this part—this is what the mathematical sciences for instance do. Now since we are seeking the first principles and the highest causes, clearly there must be some thing to which these belong in virtue of its own nature. If then our predecessors who sought the elements of existing things were seeking these same principles, it is necessary that the elements must be elements of being not by accident but just because it is being. Therefore it is of being as being that we also must grasp the first causes.

Chapter 2

There are many senses in which a thing may be said to 'be', but they are related to one central point, one definite kind of thing, and have not merely the *epithet* 'being' in common. Everything which is healthy is related to health, one thing in the sense that it preserves health, another in the sense that it produces it, another in the sense that it is a symptom of health, another because it is capable of it. And that which is medical is relative to the medical art, one thing in the sense that it possesses it, another in the sense that it is naturally adapted to it, another in the sense that it is a function of the medical art. And we shall find other words used similarly to these. So, too, there are many senses in which a thing is said to be, but all refer to one starting-point; some things are said to be because they are substances, others because they are affections of substance, others because they are a process towards substance, or destructions or privations or qualities of substance, or productive or generative of substance, or of things which are relative to substance, or negations of some of these things or of substance itself. It is for this reason that we say even of non-being that it *is* non-being. As, then, there is one science which deals with all healthy things, the same applies in the other cases also. For not only in the case of things which have one common notion does the investigation belong to one science, but also in the case of things which are related to one common nature; for even these in a sense have one common notion. It is clear then that it is the work of one science also to study all things that are, *qua* being.—But everywhere science deals chiefly with that which is primary, and on which the other things depend, and in virtue of which they get their names. If, then, this is substance, it is of substances that the philosopher must grasp the principles and the causes.

Now for every single class of things, as there is one perception, so there is one science, as for instance grammar, being one science, investigates all articulate sounds. Therefore to investigate all the species of being *qua* being, is the work of a science which is generically one, and to investigate the several species is the work of the specific parts of the science.

If, now, being and unity are the same and are one thing in the sense that they are implied in one another as principle and cause are, not in the sense that they are explained by the same definition (though it makes no difference even if we interpret them similarly—in fact this would strengthen our case); for 'one man' and 'existent man' and 'man' are the same thing, and the doubling of the words in 'the man is *one* and is a *man* does not give any new meaning (it is clear that his unity is not severed from his humanity either in coming to be or in ceasing to be; and similarly his being

is not severed), so that it is obvious that the addition in these cases means the same thing, and unity is nothing apart from being; and if, further, the essence of each thing is one in no merely accidental way, and similarly is from its very nature something that *is*:—all this being so, there must be exactly as many species of being as of unity. And to investigate the essence of these is the work of a science which is generically one—I mean, for instance, the discussion of the same and the similar and the other concepts of this sort; and nearly all contraries are reducible to this source; but let us take them as having been investigated in the ‘Selection of Contraries’.—And there are as many parts of philosophy as there are kinds of substance, so that there must necessarily be among them a first philosophy and one which follows this. For being and unity fall immediately into genera; and therefore the sciences too will correspond to these genera. For ‘philosopher’ is like ‘mathematician’ with its variety of meanings; for mathematics also has parts, and there is a first and a second science and other successive ones within the sphere of mathematics.

Now since it is the work of one science to investigate opposites, and plurality is opposite to unity, and it belongs to one science to investigate the negation and the privation because in both cases we are really investigating unity, to which the negation or the privation refers (for we either say simply that unity is not present, or that it is not present in some particular class; in the latter case the characteristic difference of the class modifies the meaning of ‘unity’, as compared with the meaning conveyed in the bare negation; for the negation means just the absence of unity, while in privation there is also implied an underlying nature of which the privation is predicated),—since, then, plurality is opposite to unity, in view of all these facts, the contraries of the concepts we named above, the other and the dissimilar and the unequal, and everything else which is derived either from these or from plurality and unity, must fall within the province of the science above-named.—And contrariety is one of these concepts, for contrariety is a kind of difference, and difference is a kind of otherness. Therefore, since there are many senses in which a thing is said to be one, these terms also will have many senses, but yet it belongs to one science to consider them all; for a term belongs to different sciences not if it has different senses, but if its definitions neither are identical nor can be referred to one central meaning. And since all things are referred to that which is primary, as for instance all things which are one are referred to the primary one, we must say that this holds good also of the same and the other and of contraries in general; so that after distinguishing the various senses of each, we must then explain by reference to what is primary in each term, saying how they are related to it; some in the sense that they possess it, others in the sense that they produce it, and others in other such ways.

It is evident then that it belongs to one science to be able to give an account of these concepts as well as of substance. This was one of the questions in our book of problems.

And it is the function of the philosopher to be able to investigate all things. For if it is not the function of the philosopher, who is it who will inquire whether Socrates and Socrates seated are the same thing, or whether one thing has one contrary, or what contrariety is, or how many meanings it has? And similarly with all other such questions. Since, then, these are essential modifications of unity *qua* unity and of being *qua* being, not *qua* numbers or lines or fire, it is clear that it belongs to this science to investigate both the essence of these concepts and their properties. And those who study these properties err not by leaving the sphere of philosophy, but by forgetting that substance, of which they have no correct idea, is prior to these other things. For number *qua* number has peculiar attributes, such as oddness and evenness, commensurability and equality, excess and defect, and these belong to numbers either in themselves or in relation to one another. And similarly the solid and the motionless and that which is in motion and the weightless and that which has

weight have other peculiar properties. So too certain properties are peculiar to being as such, and it is about these that the philosopher has to investigate the truth.—An indication of this may be mentioned:—dialecticians and sophists assume the same guise as the philosopher, for sophistic is philosophy which exists only in semblance, and dialecticians embrace all things in their dialectic, and being is common to all things; but evidently their dialectic embraces these subjects because these are proper to philosophy.—For sophistic and dialectic turn on the same class of things as philosophy, but this differs from dialectic in the nature of the faculty required and from sophistic in respect of the purpose of the philosophic life. Dialectic is merely critical where philosophy claims to know, and sophistic is what appears to be philosophy but is not.

Again, in the list of contraries one of the two columns is privative, and all contraries are reducible to being and non-being, and to unity and plurality, as for instance rest belongs to unity and movement to plurality. And nearly all thinkers agree that being and substance are composed of contraries; at least all name contraries as their first principles—some name odd and even, some hot and cold, some limit and the unlimited, some love and strife. And everything else is evidently reducible to unity and plurality (this reduction we must take for granted), and the principles stated by other thinkers fall entirely under these as their genera. It is obvious then from these considerations too that it belongs to one science to examine being *qua* being. For all things are either contraries or composed of contraries, and unity and plurality are the starting-points of all contraries. And these belong to one science, whether they have or have not one common notion. Probably they have not; yet even if 'one' has several meanings, the other meanings will be related to the primary meaning—and similarly in the case of the contraries.—And if being or unity is not a universal and the same in every instance, or is not separable from the particular instances (as in fact it probably is not; the unity is in some cases that of common reference, in some cases that of serial succession),—just for this reason it does not belong to the geometer to inquire what is contrariety or completeness or being or unity or the same or the other, but only to presuppose these concepts and reason from this starting-point.—Obviously then it is the work of one science to examine being *qua* being, and the attributes which belong to it *qua* being, and the same science will examine not only substances but also their attributes, both those above named and the concepts 'prior' and 'posterior', genus and species, whole and part, and the others of this sort.

Chapter 3

We must state whether it belongs to one or to different sciences to inquire into the truths which are in mathematics called axioms, and into substance. Evidently the inquiry into these also belongs to one science, and that the science of the philosopher; for these truths hold good for everything that is, and not for some special genus apart from others. And all men use them, for they are true of being *qua* being, and each genus has being. But men use them just so far as to satisfy their purposes; that is, as far as the genus, whose attributes they are proving, extends. Therefore since these truths clearly hold good for all things *qua* being (for this is what is common to them), he who studies being *qua* being will inquire into them too.—And for this reason no one who is conducting a special inquiry tries to say anything about their truth or falsehood,—neither the geometer nor the arithmetician. Some natural philosophers indeed have done so, and their procedure was intelligible enough; for they thought that they alone were inquiring about the whole of nature and of being. But since there is one kind of thinker who is even above the natural philosopher (for nature is only one particular genus of being), the discussion of these truths also will belong to him whose inquiry is universal and deals with primary substance. Physics also is a kind of Wisdom, but it is not the

first kind.—And the attempts of some who discuss the terms on which truth should be accepted, are due to a want of training in logic; for they should know these things already when they come to a special study, and not be inquiring into them while they are pursuing it.—Evidently then the philosopher, who is studying the nature of all substance, must inquire also into the principles of syllogism.

But he who knows best about each genus must be able to state the most certain principles of his subject, so that he whose subject is being *qua* being must be able to state the most certain principles of all things. This is the philosopher, and the most certain principle of all is that regarding which it is impossible to be mistaken; for such a principle must be both the best known (for all men may be mistaken about things which they do not know), and non-hypothetical. For a principle which every one must have who knows anything about being, is not a hypothesis; and that which every one must know who knows anything, he must already have when he comes to a special study. Evidently then such a principle is the most certain of all; which principle this is, we proceed to say. It is, that the same attribute cannot at the same time belong and not belong to the same subject in the same respect; we must presuppose, in face of dialectical objections, any further qualifications which might be added. This, then, is the most certain of all principles, since it answers to the definition given above. For it is impossible for any one to believe the same thing to be and not to be, as some think Heraclitus says; for what a man says he does not necessarily believe. If it is impossible that contrary attributes should belong at the same time to the same subject (the usual qualifications must be presupposed in this premise too), and if an opinion which contradicts another is contrary to it, obviously it is impossible for the same man at the same time to believe the same thing to be and not to be; for if a man were mistaken in this point he would have contrary opinions at the same time. It is for this reason that all who are carrying out a demonstration reduce it to this as an ultimate belief; for this is naturally the starting-point even for all the other axioms.

Chapter 4

There are some who, as we have said, both themselves assert that it is possible for the same thing to be and not to be, and say that people can judge this to be the case. And among others many writers about nature use this language. But we have now posited that it is impossible for anything at the same time to be and not to be, and by this means have shown that this is the most indisputable of all principles.—Some indeed demand that even this shall be demonstrated, but this they do through want of education, for not to know of what things one may demand demonstration, and of what one may not, argues simply want of education. For it is impossible that there should be demonstration of absolutely everything; there would be an infinite regress, so that there would still be no demonstration. But if there are things of which one should not demand demonstration, these persons cannot say what principle they regard as more indemonstrable than the present one.

We can, however, demonstrate negatively even that this view is impossible, if our opponent will only say something; and if he says nothing, it is absurd to attempt to reason with one who will not reason about anything, in so far as he refuses to reason. For such a man, as such, is seen already to be no better than a mere plant. Now negative demonstration I distinguish from demonstration proper, because in a demonstration one might be thought to be begging the question, but if another person is responsible for the assumption we shall have negative proof, not demonstration. The starting-point for all such arguments is not the demand that our opponent shall say that something either is or is not (for this one might perhaps take to be a begging of the question), but that he shall say something which is significant both for himself and for another; for this is necessary, if he

really is to say anything. For, if he means nothing, such a man will not be capable of reasoning, either with himself or with another. But if any one grants this, demonstration will be possible; for we shall already have something definite. The person responsible for the proof, however, is not he who demonstrates but he who listens; for while disowning reason he listens to reason. [And again he who admits this has admitted that something is true apart from demonstration, so that not everything will be 'so and not so'.]

First then this at least is obviously true, that the word 'be' or 'not be' has a definite meaning, so that not everything will be 'so and not so'.—Again, if 'man' has one meaning, let this be 'two-footed animal'; by having one meaning I understand this:—if 'man' means 'X', then if *A* is a man 'X' will be what 'being a man' means for him. And it makes no difference even if one were to say a word has several meanings, if only they are limited in number; for to each formula there might be assigned a different word. For instance, we might say that 'man' has not one meaning but several, one of which would be defined as 'two-footed animal', while there might be also several other formulae if only they were limited in number; for a peculiar name might be assigned to each of the formulae. If, however, they were not limited but one were to say that the word has an infinite number of meanings, obviously reasoning would be impossible; for not to have one meaning is to have no meaning, and if words have no meaning reasoning with other people, and indeed with oneself, has been annihilated; for it is impossible to think of anything if we do not think of one thing; but if this is possible, one name might be assigned to this thing. Let it be assumed then, as was said at the beginning, that the name has a meaning and has one meaning; it is impossible, then, that 'being a man' should mean precisely 'not being a man', if 'man' is not only predicable of one subject but also has one meaning (for we do not identify 'having one meaning' with 'being predicable of one subject', since on that assumption even 'musical' and 'white' and 'man' would have had one meaning, so that all things would have been one; for they would all have had the same meaning).

And it will not be possible for the same thing to be and not to be, except in virtue of an ambiguity, just as one whom we call 'man', others might call 'not-man'; but the point in question is not this, whether the same thing can at the same time be and not be a man in name, but whether it can in fact. Now if 'man' and 'not-man' mean nothing different, obviously 'not being a man' will mean nothing different from 'being a man'; so that 'being a man' will be 'not being a man'; for they will be one. For being one means this—what we find in the case of 'raiment' and 'dress'—viz. that the definitory formula is one. And if 'being a man' and 'not being a man' are to be one, they must mean one thing. But it was shown earlier that they mean different things. Therefore, if it is true to say of anything that it is a man, it must be a two-footed animal; for this was what 'man' meant; and if this is necessary, it is impossible that the same thing should not be a two-footed animal; for this is what 'being necessary' means—that it is impossible for the thing not to be. It is, then, impossible that it should be at the same time true to say the same thing is a man and is not a man.

The same account holds good with regard to 'not being man', for 'being man' and 'being not-man' mean different things, since even 'being white' and 'being man' are different; for the former terms are much more opposed, so that they must *a fortiori* mean different things. And if any one says that 'white' means one and the same thing as 'man', again we shall say the same as what was said before, that it would follow that *all* things are one, and not only opposites. But if this is impossible, then what has been said will follow, if our opponent answers our question.

And if, when one asks the question simply, he adds the contradictories, he is not answering the question. For there is nothing to prevent the same thing from being both man and white and

countless other things: but still if one asks whether it is true to call this a man or not our opponent must give an answer which means one thing, and not add that it is also white and large. For, besides other reasons, it is impossible to enumerate the accidents, which are infinite in number; let him, then, enumerate either all or none. Similarly, therefore, even if the same thing is a thousand times man and not-man, we must not add, in answering the question whether this is a man, that it is also at the same time not a man, unless we are bound to add also all the other accidents, all that the subject is or is not; and if we do this, we are not observing the rules of argument.

And in general those who use this argument do away with substance and essence. For they must say that all attributes are accidents, and that there is no such thing as 'being essentially man' or 'animal'. For if there is to be any such thing as 'being essentially man' this will not be 'being not-man' or 'not being man' (yet these are negations of it); for there was some one thing which it meant, and this was the substance of something. And denoting the substance of a thing means that the essence of the thing is nothing else. But if its being essentially man is to be the same as either being essentially not-man or essentially not being man, then its essence *will* be something else. Therefore our opponents must say that there cannot be such a definition of anything, but that all attributes are accidental; for this is the distinction between substance and accident—'white' is accidental to man, because though he is white, whiteness is not his essence. But if *all* statements are accidental, there will be nothing primary about which they are made, if the accidental always implies predication about a subject. The predication, then, must go on *ad infinitum*. But this is impossible; for not even more than two terms can be combined in accidental predication. For (1) an accident is not an accident of an accident, unless it be because both are accidents of the same subject. I mean, for instance, the white is musical and the latter is white, only because both are accidental to man. But (2) Socrates is musical, not in this sense, that both terms are accidental to something else. Since then some predicates are accidental in this and some in that sense, (a) those which are accidental in the latter sense, in which white is accidental to Socrates, cannot form an infinite series in the upward direction,—e. g. Socrates the white has not yet another accident; for no unity can be got out of such a sum. Nor again (b) will 'white' have another term accidental to it, e. g. 'musical'. For this is no more accidental to that than that is to this; and at the same time we have drawn the distinction, that while some predicates are accidental in this sense, others, are so in the sense in which 'musical' is accidental to Socrates; and the accident is an accident of an accident not in cases of the latter kind, but only in cases of the other kind, so that not *all* terms will be accidental. There must, then, be something which denotes substance. And it has been shown that, if this is so, contradictory statements cannot be predicated at the same time.

Again, if all contradictory statements are true of the same subject at the same time, evidently all things will be one. For the same thing will be a trireme, a wall, and a man, if it is equally possible to affirm and to deny anything of anything,—and this premise must be accepted by those who share the views of Protagoras. For if any one thinks that the man is not a trireme, evidently he is not a trireme; so that he also is a trireme, if, as they say, contradictory statements are both true. And we thus get the doctrine of Anaxagoras, that all things are mixed together; so that nothing really exists. They seem, then, to be speaking of the indeterminate, and, while fancying themselves to be speaking of being, they are speaking about non-being; for that which exists potentially and not actually is the indeterminate. But they must predicate of every subject every attribute and the negation of it indifferently. For it is absurd if of every subject its own negation is to be predicable, while the negation of something else which cannot be predicated of it is not predicable of it; for instance, if it is true to say of a man that he is not a man, evidently it is also true to say that he

is not a trireme. If, then, the affirmative can be predicated, the negative must be predicable too; and if the affirmative is not predicable, the negative, at least, will be more predicable than {{Left sidenote | 1008 a the negative of the subject itself. If, then, even the latter negative is predicable, the negative of 'trireme' will be also predicable; and, if this is predicable, the affirmative will be so too.—Those, then, who maintain this view are driven to this conclusion, and to the further conclusion that it is not necessary either to assert or to deny. For if it is true that a thing is man and not-man, evidently also it will be neither man nor not-man. For to the two assertions there answer two negations. And if the former is treated as a single proposition compounded out of two, the latter also is a single proposition opposite to the former.

Again, either the theory is true in all cases, and a thing is both white and not-white, and being and non-being, and all other contradictories are similarly compatible, or the theory is true of some statements and not of others. And if not of all, the exceptions will be contradictories of which admittedly only one is true; but if of all, again either the negation will be true wherever the assertion is, and the assertion true wherever the negation is, or the negation will be true where the assertion is, but the assertion not always true where the negation is. And (1) in the latter case there will be something which fixedly *is not*, and this will be an indisputable belief; and if non-being is indisputable and knowable, the opposite assertion will be more knowable. But (2) if what it is necessary to deny it is equally necessary to assert, it is either true or not true to separate the predicates and say, for instance, that a thing is white, and again that it is not-white. And if (a) it is not true to apply the predicates separately, our opponent is not really applying them, and nothing at all exists; but how could non-existent things speak or walk, as he does? Also all things will on this view be one, as has been already said, and man and God and trireme and their contradictories will be the same. For if contradictories can be predicated alike of each subject, one thing will in no wise differ from another; for if it differ, this difference will be something true and peculiar to it. And (b) if one may with truth apply the predicates separately, the above-mentioned result follows none the less.

Further, it follows that all would then be right and all would be in error, and our opponent himself confesses himself to be in error.—And at the same time our discussion with him is evidently about nothing at all; for he says nothing. For he says neither 'yes' nor 'no', but both 'yes' and 'no'; and again he denies both of these and says 'neither yes nor no'; for otherwise there would already be something definite.—Again, if when the assertion is true, the negation is false, and when this is true, the affirmation is false, it will not be possible to assert and deny the same thing truly at the same time. But perhaps they might say we had assumed 1008 b the very question at issue.

Again, is he in error who judges either that the thing is so or that it is not so, and is he right who judges both? If he is not right, what can they mean by saying that the nature of existing things is of this kind? And if he is not right, but more right than he who judges in the other way, being will already be of a definite nature, and this will be true, and not at the same time also not true. But if all are alike both right and wrong, one who believes this can neither speak nor say anything intelligible; for he says at the same time both 'yes' and 'no'. And if he makes no judgement but 'thinks' and 'does not think', indifferently, what difference will there be between him and the plants?—Thus, then, it is in the highest degree evident that neither any one of those who maintain this view nor any one else is really in this position. For why does a man walk to Megara and not stay at home thinking he is walking? Why does he not walk early some morning into a well or over a precipice, if one happens to be in his way? Why do we observe him guarding against this, evidently not thinking that falling in is alike good and not good? Evidently he judges one thing to

be better and another worse. And if this is so, he must judge one thing to be man and another to be not-man, one thing to be sweet and another to be not-sweet. For he does not aim at and judge all things alike, when, thinking it desirable to drink water or to see a man, he proceeds to aim at these things; yet he ought, if the same thing were alike man and not-man. But, as was said, there is no one who does not obviously avoid some things and not others. Therefore, as it seems, all men make unqualified judgements, if not about all things, still about what is better and worse. And if this is not knowledge but opinion, they should be all the more anxious about the truth, as a sick man should be more anxious about his health than one who is healthy; for he who has opinions is, in comparison with the man who knows, not in a healthy state as far as the truth is concerned.

Again, however much all things may be 'so and not so', still there is a more and a less in the nature of things; for we should not say that two and three are equally even, nor is he who thinks four things are five equally wrong with him who thinks they are a thousand. If then they are not equally wrong, obviously one is less wrong and therefore more right. If then that which has more of any quality is nearer the 1009 a norm, there must be some truth to which the more true is nearer. And even if there is not, still there is already something more certain and true, and we shall have got rid of the unqualified doctrine which would prevent us from determining anything in our thought.

CHAPTER V

Again, from the same opinion proceeds the doctrine of Protagoras, and both doctrines must be alike true or alike untrue. For on the one hand, if all opinions and appearances are true, all statements must be at the same time true and false. For many men hold beliefs in which they conflict with one another, and all think those mistaken who have not the same opinions as themselves; so that the same thing must be and not be. And on the other hand, if this is so, all opinions must be true; for those who are mistaken and those who are right are opposed to one another in their opinions; if, then, reality is such as the view in question supposes, all will be right in their beliefs. Evidently, then, both doctrines proceed from the same way of thinking.

But the same method of discussion must not be used with all opponents; for some need persuasion, and others compulsion. Those who have been driven to this position by difficulties in their thinking can easily be cured of their ignorance; for it is not their expressed argument but their thought that one has to meet. But those who argue for the sake of argument can be convinced only by emending the argument as expressed in words.

Those who really feel the difficulties have been led to this opinion by observation of the sensible world. (1) They think that contradictions or contraries are true at the same time, because they see contraries coming into existence out of the same thing. If, then, that which is not cannot come to be, the thing must have existed before as both contraries alike, as Anaxagoras says all is mixed in all, and Democritus too; for *he* says the void and the full exist alike in every part, and yet one of these is being, and the other non-being. To those, then, whose belief rests on these grounds, we shall say that in a sense they speak rightly and in a sense they err. For 'that which is' has two meanings, so that in some sense a thing can come to be out of that which is not, while in some sense it cannot, and the same thing can at the same time be and not be — but not in the same respect. For the same thing can be potentially at the same time two contraries, but it cannot actually. And again we shall ask them to believe that among existing things there is another kind of substance to which neither movement nor destruction nor generation at all belongs.

And (2) similarly some have inferred from the sensible world 'the truth of appearances'. For they think that the truth should not be determined by the large or small number of those who hold

a belief, and that the same thing is thought sweet by some who taste it, and bitter by others, so that if all were ill or all were mad, and only two or three were well or sane, these would be thought ill and mad, and not the others. And again, many of the other animals receive about the same things impressions contrary to ours; and even to the senses of each individual, things do not always seem the same. Which, then, of these impressions are true and which are false is not obvious; for the one set is no more true than the other, but both are alike. And this is why Democritus, at any rate, says that either there is no truth or to us at least it is not evident. And in general it is because these thinkers suppose knowledge to be sensation, and this to be a physical alteration, that they say that what appears to our senses must be true; for it is for these reasons that Empedocles and Democritus and, one may almost say, all the others have fallen victims to opinions of this sort. For Empedocles says that when men change their condition they change their knowledge;

For wisdom increases in men according to their present state.

And elsewhere he says:

So far as their nature changes, so far to them always
Come changed thoughts into mind.

And Parmenides also expresses himself in the same way:

For as in each case the much-bent limbs are composed,
So is the mind of men; for in each and all men
'Tis one thing thinks — the substance of their limbs:
For that of which there is more is thought.

A saying of Anaxagoras to some of his friends is also related, — that things would be for them such as they supposed them to be. And they say that Homer also evidently had this opinion, because he made Hector, when he was unconscious from the blow, lie 'thinking other thoughts', — which implies that even those who are bereft of thought have thoughts, though not the same. Evidently, then, if both are forms of thought, the real things also are at the same time 'so and not so'. And it is in this direction that the consequences are most difficult. For if those who have seen most of what truth is possible for us (and these are those who seek and love it most) — if these have such opinions and express these views about the truth, is it not natural that beginners in philosophy should lose heart? For to seek the truth would be to follow flying game.

But the reason for this opinion is that while these thinkers were inquiring into the truth of that which is, they thought 'that which is' was identical with the sensible world; in this, however, there is largely present the nature of the indeterminate — of that which exists in the peculiar sense which we have explained; and, therefore, while they speak plausibly, they do not say what is true. For it befits us to put the matter so rather than as Epicharmus put it against Xenophanes. And again, they held these views because they saw that all this world of nature is in movement, and that about that which changes no true statement can be made; at least, regarding that which everywhere in every respect is changing nothing could truly be affirmed. It was this belief that blossomed into the most extreme of the views above mentioned, that of the professed Heracliteans, such as was held by Cratylus, who finally did not think it right to say anything but only moved his finger, and criticized Heraclitus for saying that it is impossible to step twice into the same river; for *he* thought one could not do it even once.

But we shall say in answer to this argument also, that there is some real sense in their thinking that the changing, when it is changing, does not exist. Yet it is after all disputable; for that which is losing a quality has something of that which is being lost, and of that which is coming to be, something must already be. And in general if a thing is perishing, there will be present something that exists; and if a thing is coming to be, there must be something from which it comes to be and something by which it is generated, and this process cannot be *ad infinitum*. But leaving these arguments, let us insist on this, that it is not the same thing to change in quantity and in quality. Grant that in quantity a thing is not constant; still it is in respect of its form that we know each thing. — And again, it would be fair to criticize those who hold this view for asserting about the whole material universe what they saw only in a minority even of sensible things. For only that region of the sensible world which immediately surrounds us is always in process of destruction and generation; but this is — so to speak — not even a fraction of the whole, so that it would have been juster to acquit this part of the world because of the other part, than to condemn the other because of this. And again, obviously we shall make to them also the same reply that we made before; we must show them and persuade them that there is something whose nature is changeless. Indeed, from the assertion that things at the same time are and are not, there follows the assertion that all things are at rest rather than that they are in movement; for there is nothing into which they can change, since all attributes belong already to all subjects.

Regarding the nature of truth, we must maintain that not everything which appears is true. Firstly, they contend that sensation — at least of the object peculiar to the sense in question — is not false; we answer that appearance is not the same as sensation. — Again, it is fair to express surprise at our opponents for raising the question whether magnitudes are as great, and colours are of such a nature, as they appear to people at a distance, or as they appear to those close at hand, and whether they are such as they appear to the sick or to the healthy, and whether those things are heavy which appear so to the weak or those which appear so to the strong, and whether truth is what appears to the sleeping or to the waking. For obviously they do not think these to be open questions; no one, at least, if when he is in Libya he fancies one night that he is in Athens, straightway starts for the Odeum. And again with regard to the future, as Plato says, surely the opinion of the physician and that of the ignorant man are not equally weighty, for instance, on the question whether a man will get well or not. — And again, among sensations themselves the sensation of a foreign object and that of the appropriate object, or that of a kindred object and that of the object of the sense in question, are not equally authoritative, but in the case of colour, sight, not taste, has the authority, and in the case of flavour, taste, not sight; each of which senses never says at the same moment of the same object that it at the same time is 'so and not so'. — But not even at different moments does one sense disagree about the quality, but only about that to which the quality belongs. I mean, for instance, the same wine might seem, if either it or one's body changed, at one time sweet and at another time not sweet; but at least the sweet, such as it is when it exists, has never yet changed, but one is always right about it, and that which is to be sweet must of necessity be of such and such a nature. Yet all these views destroy this distinction, so that as there is no essence of anything, so nothing is of necessity; for the necessary cannot be in this way and also in that, so that if anything is of necessity, it will not be both 'so and not so'.

And, in general, if only the sensible exists, there would be nothing if animate things were not; for there would be no faculty of sense. The view that the sensations also would not exist is doubtless true (for they are affections of the perceiver), but that the substrata which cause the sensation should not exist even apart from sensation is impossible. For sensation is surely not

the sensation of itself, but there is something beyond the sensation, which must be prior to the sensation; for that which moves is prior in nature to that which is moved, and if they are correlative terms, this is no less the case.

Chapter 6

There are, both among those who have these convictions and among those who merely profess these views, some who raise a difficulty by asking, who is the judge of the healthy man, and in general who is likely to judge rightly on each class of questions. But such inquiries are like puzzling over the question whether we are now asleep or awake. And all such questions have the same meaning. These people demand that a reason shall be given for everything; for they seek a starting-point, and they wish to get this by demonstration, while it is obvious from their actions that they have no conviction. But their mistake is what we have stated it to be; they seek a reason for that for which no reason can be given; for the starting-point of demonstration is not demonstration.

These, then, might be easily persuaded of this truth, for it is not difficult to grasp; but those who seek merely compulsion in ailment seek what is impossible; for they demand to be made to contradict themselves, while they are contradicting themselves from the very first. — But if not all things are relative, but some are self-existent, not everything that appears will be true; for that which appears appears to some one; so that he who says all things that appear are true, makes all things relative. And, therefore, those who ask for an irresistible argument, and at the same time demand to be called to account for their views, must guard themselves by saying that the truth is not that what appears exists, but that what appears exists *for him to whom it appears, and when, and in the sense in which, and in the way in which it appears*. And if they give an account of their view, but do not give it in this way, they will soon find themselves contradicting themselves. For it is possible that a thing may for the same man appear as honey to the sight, but not to the taste, and that, as we have two eyes, things may not appear the same to each, if the eyes are unlike. For to those who for the reasons named above say that what appears is true, and therefore that all things are alike false and true, for things do not appear either the same to all men or always the same to the same man, but (often have contrary appearances at the same time (for touch says there are two objects when we cross our fingers, while sight says there is one), — to these we shall say ‘yes, but not to the same sense and in the same part of it and in the same way and at the same time’, so that what appears is under these qualifications true. But perhaps for this reason those who argue thus not because they feel a difficulty but for the sake of argument, should say that this is not true, but true for this man. And as has been already said, they must make everything relative — relative to thought and perception, so that nothing either has come to be or will be without some one’s first thinking so. But if things *have* come to be or will be, evidently not all things will be relative to opinion. — Again, if a thing is one, it is in relation to one thing or to a definite number of things; and if the same thing is both half and equal, still the equal is not correlative to the double. If, then, the same thing is a man, in relation to that which thinks, and is that which is thought, in relation to that which thinks, that which *thinks* will not be a man, but only that which *is thought*. Again, if each thing is to be relative to that which thinks, that which thinks will be relative to an infinity of specifically different things.

Let this, then, suffice to show (1) that the most indisputable of all beliefs is that contradictory statements are not at the same time true, and (2) what consequences follow from the denial of this belief, and (3) why people do deny it. Now since it is impossible that contradictories should be at the same time true of the same thing, obviously contraries also cannot belong at the same time

to the same thing. For of the contraries, no less than of the contradictories, one is a privation — and a privation of substance ; and privation is the denial of a predicate to a determinate genus. If, then, it is impossible to affirm and deny truly at the same time, it is also impossible that contraries should belong to a subject at the same time, unless both belong to it in particular relations, or one in a particular relation and one without qualification.

Chapter 7

But on the other hand there cannot be an intermediate between contradictories, but of one subject we must either affirm or deny any one predicate. This is clear, in the first place, if we define what the true and the false are. To say of what is that it is not, or of what is not that it is, is false, while to say of what is that it is, and of what is not that it is not, is true; so that he who says of anything that it is, or that it is not, will say either what is true or what is false; but neither what is nor what is not is said to be or not to be. — Again, either the intermediate between the contradictories will be so in the way in which grey is between black and white, or as that which is neither man nor horse is between man and horse, (1) If it were of the latter kind, it could not change into the extremes, for change is from not-good to good, or from good to not-good (but as a matter of fact when there is an intermediate it is always observed to change into the extremes). For there is no change except to opposites and to their intermediate. (2) But if it is really intermediate, in this way too there is a difficulty — there would have to be a change to white, which was not from not-white; but as it is, this is never seen. — Again, the understanding either affirms or denies every object of understanding or reason; this becomes obvious if we define when thought is true and when false. When it connects in one way by assertion or negation, it is true, and when it does so in the other way, it is false. — Again, there must be an intermediate between *all* contradictories, if one is not arguing merely for the sake of argument; so that it will be possible for a man to say what is neither true nor untrue. And there will be a middle between that which is and that which is not, so that there will also be a kind of change intermediate between generation and destruction. — Again, in all classes in which the negation of an attribute means the assertion of its contrary, even in these there will be an intermediate ; for instance, in the sphere of numbers there will be number which is neither odd nor not-odd. But this is impossible, as is obvious from the definition. - Again, the process will go on *ad infinitum*, and the number of realities will be not only made half as great again, but even greater. For again it will be possible to deny this intermediate with reference both to its assertion and to its negation, and this new term will be some definite thing; for its essence is something different. — Again, when a man, on being asked whether a thing is white, says 'no' he has denied nothing except that it is; and its not being is a negation.

Some people have acquired this opinion as other paradoxical opinions have been acquired; when men cannot refute eristical arguments, they give in to the argument and agree that the conclusion is true. This, then, is why some argue in such fashion; others do so because they demand a reason for everything. And the starting-point in dealing with all such people is definition. Now the definition rests on the necessity of their meaning something; for the formula, of which the word is a sign, becomes its definition. — The doctrine of Heraclitus, that all things are and are not, seems to make everything true, while that of Anaxagoras, that there is an intermediate between the terms of a contradiction, seems to make everything false; for when things are mixed, the mixture is neither good nor not- good, so that one cannot say anything that is true.

Chapter 8

In view of these distinctions it is obvious that the one-sided theories which some people express about all things cannot be valid — on the one hand the theory that nothing is true (for, say they,

there is nothing to prevent every statement from being like the statement 'the diagonal of a square is commensurate with the side'), — on the other hand the theory that everything is true. — These views are practically the same as that of Heraclitus; for that which says that 'all things are true and all are false' also makes each of these statements separately, so that since they are impossible, the double statement must be impossible too. — Again, there are obviously contradictories which cannot be at the same time true. Nor on the other hand can all statements be false; yet this would *seem* more possible in view of what has been said. — But against all such arguments we must postulate, as we said above, not that something is or is not, but that people mean something, so that we must argue from a definition, having got what falsity or truth means. If that which it is true to affirm is nothing other than that which it is false to deny, it is impossible that all statements should be false; for one side of the contradiction must be true. — Again, if it is necessary with regard to everything either to assert or to deny it, it is impossible that both should be false; for it is *one* side of the contradiction that is false. — Further, all such arguments are exposed to the often-expressed objection, that they destroy themselves. For he who says that everything is true makes the statement contrary to his own also true, so that his own is not true (for the contrary statement denies that it is true), while he who says everything is false makes himself also false. — And if the former person excepts the contrary statement, saying it alone is not true, while the latter excepts his own as being alone not false, none the less they are driven to postulate the truth or falsehood of an infinite number of statements; for that which says the true statement is true, is true, and this process will go on to infinity.

Evidently again those who say all things are at rest are not right, nor are those who say all things are in movement. For if all things are at rest, the same statements will always be true and the same always false,— but they obviously are not; for he who makes a statement himself at one time was not and again will not be. And if all things are in motion, nothing will be true; everything therefore will be false. But it has been shown that this is impossible. Again, it must be that which is that changes; for change is from something to something. But again it is not the case that all things are at rest or in motion *sometimes*, and nothing *for ever*; for there is something which always moves the things that are in motion, and the first mover must itself be unmoved.

Book V

Layout 2

Chapter 1

'Beginning' means (1) that part of a thing from which one would start first, e.g. a line or a road has a beginning in either of the contrary directions. (2) That from which each thing would best be originated, e.g. we must sometimes begin to learn not from the first point and the beginning of the thing, but from the point from which we should learn most easily. (3) That from which (as an immanent part) a thing first arises, e.g. as the keel of a ship and the foundation of a house, while in animals some suppose the heart, others the brain, others some other part, to be of this nature. (4) That from which (*not* as an immanent part) a thing first arises, and from which the movement or the change naturally first proceeds, as a child comes from the father and the mother, and a fight from abusive language. (5) That at whose will that which is moved is moved and that which changes changes, e.g. the magistracies in cities, and oligarchies and monarchies and tyrannies, are called ἀρχαί, and so are the arts, and of these especially the architectonic arts. (6) That from which a thing can first be known; for this also is called the beginning of the thing, e.g. the hypotheses are the beginnings of demonstrations. (Causes are spoken of in an equal number of senses; for all causes are beginnings.) It is common, then, to all beginnings to be the first point from which a thing either is or comes to be or is known; but of these some are immanent in the thing and others are outside. Therefore the nature of a thing is a beginning, and so are the elements of a thing, and thought and will, and essence, and the final cause — for the good and the beautiful are the beginning both of the knowledge and of the movement of many things.

Chapter 2

'Cause' means (1) that from which (as immanent material) a thing comes into being, e.g. the bronze of the statue and the silver of the saucer, and the classes which include these. (2) The form or pattern, i.e. the formula of the essence, and the classes which include this (e.g. the ratio 2:1 and number in general are causes of the octave) and the parts of the formula. (3) That from which the change or the freedom from change first begins, e.g. the adviser is a cause of the action, and the father a cause of the child, and in general the maker a cause of the thing made and the change-producing of the changing. (4) The end, i.e. that for the sake of which a thing is, e.g. health is the cause of walking. For why does one walk? We say 'that one may be healthy', and in speaking thus we think we have given the cause. The same is true of all the means that intervene before the end, when something else has put the process in motion (as e.g. thinning or purging or drugs or instruments intervene before health is reached); for all these are for the sake of the end, though they differ from one another in that some are instruments and others are actions.

These, then, are practically all the senses in which causes are spoken of, and as they are spoken of in several senses it follows that there are several causes of the same thing, and in no accidental sense, e.g. both the art of sculpture and the bronze are causes of the statue not in virtue of anything else but *qua* statue; not, however, in the same way, but the one as matter and the other as source of the movement. And things can be causes of one another, e.g. exercise of good condition, and the latter of exercise; not, however, in the same way, but the one as end and the other as source of movement.—Again, the same thing is sometimes cause of contraries; for that which when present causes a particular thing, we sometimes charge, when absent, with the contrary, e.g. we impute the shipwreck to the absence of the steers man, whose presence was the cause of safety; and both — the presence and the privation — are causes as sources of movement.

All the causes now mentioned fall under four senses which are the most obvious. For the

letters are the causes of syllables, and the material is the cause of manufactured things, and fire and earth and all such things are the causes of bodies, and the parts are causes of the whole, and the hypotheses are causes of the conclusion, in the sense that so they are that out of which these respectively are made; but of these some are cause as *substratum* (e.g. the parts), others as *essence* (the whole, the synthesis, and the form). The semen, the physician, the adviser, and in general the agent, are all *sources of change* or of rest. The remainder are causes as the *end* and the good of the other things; for that, for the sake of which other things are, is naturally the best and the end of the other things; let us take it as making no difference whether we call it good or apparent good.

These, then, are the causes, and this is the number of their kinds, but the *varieties* of causes are many in number, though when summarized these also are comparatively few. Causes are spoken of in many senses, and even of those which are of the same kind some are causes in a prior and others in a posterior sense, e.g. both 'the physician' and 'the professional man' are causes of health, and 'the ratio 2:1' and 'number' are causes of the octave, and the classes that include any particular cause are always causes of the particular effect. Again, there are accidental causes and the classes which include these, e.g. while in one sense 'the sculptor' causes the statue, in another sense 'Polyclitus' causes it, because the sculptor happens to be Polyclitus; and the classes that include the accidental cause are also causes, e.g. 'man' — or in general 'animal' — is the cause of the statue, because Polyclitus is a man, and a man is an animal. Of accidental causes also some are more remote or nearer than others, as, for instance, if 'the white' and 'the musical' were called causes of the statue, and not only 'Polyclitus' or 'man'. But besides all these varieties of causes, whether proper or accidental, some are called causes as being able to act, others as acting, e.g. the cause of the house's being built is the builder, or the builder when building. — The same variety of language will be found with regard to the effects of causes, e.g. a thing may be called the cause of this statue or of a statue or in general of an image, and of this bronze or of bronze or of matter in general; and similarly in the case of accidental effects. Again, both accidental and proper causes may be spoken of in combination, e.g. we may say not 'Polyclitus' nor 'the sculptor', but 'Polyclitus the sculptor'.

Yet all these are but six in number, while each is spoken of in two ways; for (1) they are causes either as the individual, or as the class that includes the individual, or as the accidental, or as the class that includes the accidental, and these either as combined, or as taken simply; and (2) all may be taken as acting or as having a capacity. But they differ inasmuch as the acting causes and the individuals exist, or do not exist, simultaneously with the things of which they are causes, e.g. this particular man who is curing, with this particular man who is recovering health, and this particular builder with this particular thing that is being built; but the potential causes are not always in this case; for the house does not perish at the same time as the builder.

Chapter 3

'Element' means the primary component immanent in a thing, and indivisible in kind into other kinds, e.g. the elements of speech are the parts of which speech consists and into which it is ultimately divided, while *they* are no longer divided into other forms of speech different in kind from them. If they *are* divided, their parts are of the same kind, as a part of water is water (while a part of the syllable is not a syllable). Similarly those who speak of the elements of bodies mean the things into which bodies are ultimately divided, while *they* are no longer divided into other things differing in kind; and whether the things of this sort are one or more, they call these elements. The elements of geometrical proofs, and in general the elements of demonstrations, have a similar character; for the primary demonstrations, each of which is implied in many demonstrations,

are called elements of demonstrations; and the primary syllogisms, which have three terms and proceed by means of one middle, are of this nature.

People also transfer the word 'element' from this meaning and apply it to that which, being one and small, is useful for many purposes; for which reason the small and simple and indivisible is called an element. Hence come the facts that the most universal things are elements (because each of them being one and simple is present in a plurality of things, either in all or in as many as possible), and that unity and the point are thought by some to be first principles. Now, since the so-called genera are universal and indivisible (for there is no analytic formula of them), some say the genera are elements, and more so than the differentia, because the genus is more universal; for where the differentia is present, the genus accompanies it, but where the genus is, the differentia is not always. It is common to all the meanings that the element of each thing is the first component immanent in each.

Chapter 4

'Nature' means (1) the genesis of growing things — the meaning which would be suggested if one were to pronounce the υ in φύσις long. (2) The primary immanent element in a thing, from which its growth proceeds. (3) The source from which the primary movement in each natural object is present in it in virtue of its own essence. Those things are said to grow which derive increase from something else by contact and organic unity, or organic adhesion as in the case of embryos. Organic unity differs from contact; for in the latter case there need not be anything besides the contact, but in organic unities there is something identical in both parts, which makes them grow together instead of merely touching, and be one in respect of continuity and quantity, though not of quality.— (4) 'Nature' means the primary matter of which any non-natural object consists or out of which it is made, which cannot be modified or changed from its own potency, as e.g. utensils, and wood the nature of wooden things; and so in all other cases; for when a product is made out of these materials, the first matter is preserved throughout. In this way people call the elements of *natural* objects also their nature, some naming fire, others earth, others air, others water, others something else of the sort, and some naming more than one of these, and others all of them. — (5) 'Nature' means the *essence* of natural objects, as with those who say the nature is the primary mode of composition, or as Empedocles says:—

Nothing that is has a nature,
But only mixing and parting of the mixed,
And nature is but a name applied to them by men.

Hence as regards the things that are or come to be by nature, though that *from which* they naturally come to be or are is already present, we say they have not their nature yet, unless they have their form or shape. That which comprises both of these — matter and form — exists *by* nature, e.g. the animals and their parts; and nature is both the first matter (and this in two senses, either first, counting from the thing, or first in general, e.g. in the case of works in bronze, bronze is first with reference to them, but in general perhaps water is first, if all things that can be melted are water), and the form or essence, which is the end of the process of becoming. And from this sense of 'nature' every essence in general is in fact, by an extension of meaning, called a 'nature', because the nature of a thing is one kind of essence.

From what has been said, then, it is plain that nature in the primary and strict sense is the essence of things which have in themselves, as such, a source of movement; for the matter is called

the nature because it is qualified to receive this, and processes of becoming and growing are called nature because they are movements proceeding from this. And nature in this sense is the source of the movement of natural objects, being present in them somehow, either potentially or actually.

Chapter 5

'The necessary' means (1) that without which, as a condition, a thing cannot live, e.g. breathing and food are necessary for an animal; for it is incapable of existing without these. — (2) The conditions without which good cannot be or come to be, or without which we cannot get rid or be freed of evil, e.g. drinking the medicine is necessary in order that we may be cured of disease, and sailing to Aegina is necessary in order that we may get our money. — (3) The compulsory and compulsion, i.e. that which impedes and hinders contrary to impulse and purpose. For the compulsory is called necessary; whence the necessary is painful, as Evenus says: 'For every necessary thing is ever irksome.' And compulsion is a form of necessity, as Sophocles says: 'Force makes this action a necessity.' And necessity is held to be something that cannot be persuaded — and rightly, for it is contrary to the movement which accords with purpose and with reasoning. — (4) We say that that which cannot be otherwise is necessarily so. And from this sense of 'necessary' all the others are somehow derived; for as regards the compulsory we say that it is necessary to act or to be acted on, only when we cannot act according to impulse because of the compelling force, — which implies that necessity is that because of which the thing cannot be otherwise; and similarly as regards the conditions of life and of good, when in the one case good, in the other life and being, are not possible without certain conditions, these are necessary, and this cause is a kind of necessity. — Again, (5) demonstration is a necessary thing, because the conclusion cannot be otherwise, if there has been demonstration in the full sense; and the causes of this necessity are the first premises, i.e. the fact that the propositions from which the syllogism proceeds cannot be otherwise.

Now some things owe their necessity to something other than themselves; others do not, while they are the source of necessity in other things. Therefore the necessary in the primary and strict sense is the simple; for this does not admit of more states than one, so that it does not admit even of one state and another; for already it would admit of more than one. If, then, there are certain eternal and unmovable things, nothing compulsory or against their nature attaches to them.

Chapter 6

'One' means (1) that which is one by accident, (2) that which is one by its own nature, (1) Instances of the accidentally one are 'Coriscus' and 'musical', and 'musical Coriscus' (for it is the same thing to say 'Coriscus' and 'musical', and 'musical Coriscus'), and 'musical' and 'just', and 'musical Coriscus' and 'just Coriscus'. For all these are called one by accident, 'just' and 'musical' because they are accidents of one substance, 'musical' and 'Coriscus' because the one is an accident of the other; and similarly in a sense 'musical Coriscus' is one with 'Coriscus', because one of the parts of the concept is an accident of the other, i.e. 'musical' is an accident of Coriscus; and 'musical Coriscus' is one with 'just Coriscus', because both have parts which are accidents of one and the same subject. The case is similar if the accident is predicated of a class or of any universal term, e.g. if one says that man is the same as 'musical man'; for this is either because 'musical' is an accident of man, which is one substance, or because both are accidents of some individual, e.g. Coriscus. Both, however, do not belong to him in the same way, but one doubtless as genus and in the substance, the other as a state or affection of the substance.

The things, then, that are called one by accident, are called so in this way. (2) Of things that are called one in virtue of their own nature some (*a*) are so called because they are continuous, e.g. a bundle is made one by a band, and pieces of wood are made one by glue; and a line, even

if it is bent, is called one if it is continuous, as each part of the body is, e.g. the leg or the arm. Of these themselves, the continuous by nature are more one than the continuous by art. A thing is called continuous which has by its own nature one movement and cannot have any other; and the movement is one when it is indivisible, and indivisible in time. Those things are continuous by their own nature which are one not merely by contact; for if you put pieces of wood touching one another, you will not say these are one piece of wood or one body or one *continuum* of any other sort. Things, then, that are continuous in any way are called one, even if they admit of being bent, and still more those which cannot be bent, e.g. the shin or the thigh is more one than the leg, because the movement of the leg need not be one. And the straight line is more one than the bent; but that which is bent and has an angle we call both one and not one, because its movement may be either simultaneous or not simultaneous; but that of the straight line is always simultaneous, and no part of it which has magnitude rests while another moves, as in the bent line.

(b) Things are called one in another sense because the substratum does not differ in kind; it does not differ in the case of things whose kind is indivisible to the sense. The substratum meant is either the nearest to, or the furthest from, the final state. For, on the one hand, wine is said to be one and water is said to be one, *qua* indivisible in kind; and, on the other hand, *all* juices, e.g. oil and wine, are said to be one, and so are all things that can be melted, because the ultimate substratum of all is the same; for all of these are water or air.

(c) Those things are called one whose genus is one though distinguished by opposite differentiae; and these are all called one because the genus which underlies the differentiae is one (e.g. horse, man, and dog are one, because all are animals), and in a way similar to that in which the matter is one.

These are sometimes called one in this way, but sometimes it is the higher genus that is said to be the same (if they are *infimae species* of their genus) — the genus above the proximate genera, e.g. the isosceles and the equilateral are one and the same *figure* because both are triangles, but they are not the same triangles.

(d) Two things are called one, when the formula which states the essence of one is indivisible from another formula which shows the essence of the other (though *in itself* every formula is divisible). Thus even that which has increased or is diminishing is one, because its formula is one, as, in the case of planes, is the formula of their form. In general those things, the thought of whose essence is indivisible, and cannot separate them either in time or in place or in formula, are most of all one, and of these especially those which are substances. For in general those things that do not admit of division are one in so far as they do not admit of it, e.g. if something *qua* man does not admit of division, it is one man; if *qua* animal, it is one animal; if *qua* magnitude, it is one magnitude. — Now most things are called one because they do or have or suffer or are related to something else that is one, but the things that are primarily called one are those whose substance is one, — and one either in continuity or in form or in formula; for we count as more than one either things that are not continuous, or those whose form is not one, or those whose formula is not one.

(e) While in a sense we call anything one if it is a quantity and continuous, in a sense we do not unless it is a whole, i.e. unless it has one form; e.g. if we saw the parts of a shoe put together anyhow we should not call them one all the same (unless because of their continuity); we do this only if they are put together so as to be a shoe and have already some one form. This is why the circle is of all lines most truly one, because it is whole and complete.

The *essence* of 'one' is to be a beginning of number; for the first measure is the beginning, for that by which we first know each class is the first measure of the class; the one, then, is the beginning

of the knowable regarding each class. But the one is not the same in all classes. For here it is a quartertone, and there it is the vowel or the consonant; and there is another unit of weight and another of movement. But everywhere the one is indivisible either in quantity or in kind. That which is indivisible in quantity and *qua* quantity is called a unit if it is not divisible in any dimension and is without position, a point if it is not divisible in any dimension and has position, a line if it is divisible in one dimension, a plane if in two, a body if divisible in quantity in all— i.e. in three — dimensions. And, reversing the order, that which is divisible in two dimensions is a plane, that which is divisible in one a line, that which is in no way divisible in quantity is a point or a unit, — that which has not position a unit, that which has position a point.

Again, some things are one in number, others in species, others in genus, others by analogy; in number those whose matter is one, in species those whose definition is one, in genus those to which the same figure of predication applies, by analogy those which are related as a third thing is to a fourth. The latter kinds of unity are always found when the former are, e.g. things that are one in number are one in species, while things that are one in species are not all one in number; but things that are one in species are all one in genus, while things that are so in genus are not all one in species but are all one by analogy; while things that are one by analogy are not all one in genus.

Evidently 'many' will have meanings opposite to those of 'one'; some things are many because they are not continuous, others because their matter — either the proximate matter or the ultimate — is divisible in kind, others because the formulae which state their essence are more than one.

Chapter 7

Things are said to 'be' (1) in an accidental sense, (2) by their own nature.

(1) In an accidental sense, e.g., we say 'the just is musical', and 'the man is musical' and 'the musical is a man', just as we say 'the musical builds', because the builder happens to be musical or the musical happens to be a builder; for here 'one thing is another' means 'one is an accident of another'. So in the cases we have mentioned; for when we say 'the man is musical' and 'the musical is a man', or 'the white is musical' or 'the musical is white', the last two mean that both attributes are accidents of the same thing, which *is*; the first that the attribute is an accident of that which *is*; while 'the musical is a man' means that 'musical' is an accident of man. In this sense, too, the not-white is said to *be*, because that of which it is an accident *is*. Thus when one thing is said in an accidental sense to be another, this is either because both belong to the same thing, and this *is*, or because that to which the attribute belongs *is*, or because the subject which has as an attribute that of which it is itself predicated, itself *is*.

(2) The kinds of essential being are those that are indicated by the figures of predication; for the senses of 'being' are just as many as these figures. Since some predicates indicate what the subject is, others its quality, others quantity, others relation, others activity or passivity, others its 'where', others its 'when', 'being' has a meaning answering to each of these. For there is no difference between * 'the man is recovering' and 'the man recovers', nor between 'the man is walking' or 'cutting' and 'the man walks' or 'cuts'; and similarly in all other cases.

(3) 'Being' and 'is' mean that a statement is true, 'not being' that it is not true but false, — and this alike in affirmation and negation; e.g. 'Socrates *is* musical' means that this is true, or 'Socrates *is* not-white' means that this is true; but 'the diagonal of the square *is* not commensurate with the side' means that it is false to say it is.

(4) Again, 'being' and 'that which is', in these cases we have mentioned, sometimes mean being potentially, and sometimes being actually. For we say both of that which sees potentially and of that which sees actually, that it is 'seeing', and both of that which can use knowledge and of that

which is using it, that it knows, and both of that to which rest is already present and of that which can rest, that it rests. And similarly in the case of substances we say the Hermes is in the stone, and the half of the line is in the line, and we say of that which is not yet ripe that it is corn. *When* a thing is potential and when it is not yet potential must be explained, elsewhere.

Chapter 8

'Substance' means (1) the simple bodies, i.e. earth and fire and water and everything of the sort, and in general bodies and the things composed of them, both animals and divine beings, and the parts of these. All these are called substance because they are not predicated of a subject but everything else is predicated of them. — (2) That which, being present in such things as are not predicated of a subject, is the cause of their being, as the soul is of the being of animals. — (3) The parts which are present in such things, limiting them and marking them as individuals, and by whose destruction the whole is destroyed, as the body is by the destruction of the plane, as some say, and the plane by the destruction of the line; and in general number is thought by some to be of this nature; for if it is destroyed, they say, nothing exists, and it limits all things. — (4) The essence, the formula of which is a definition, is also called the substance of each thing.

It follows, then, that substance has two senses, (a) the ultimate substratum, which is no longer predicated of anything else, and (b) that which is a 'this' and separable — and of this nature is the shape or form of each thing.

Chapter 9

'The same' means (1) that which is the same in an accidental sense, e.g. 'white' and 'musical' are the same because they are accidents of the same thing, and 'man' and 'musical' because the one is an accident of the other; and 'the musical' is 'man' because it is an accident of man. And the complex notion is the same as either of the simple ones and each of these is the same as it; for 'man' and 'musical' are said to be the same as 'musical man', and this is the same as they. This is why all of these statements are made not universally; for it is not true to say that *every* man is the same as 'musical'; for universal attributes belong to things in virtue of their own nature, but accidents do not belong to them in virtue of their own nature, but are predicated without qualification only of the individuals. For 'Socrates' and 'musical Socrates' are thought to be the same; but 'Socrates' is not predicable of more than one subject, and therefore we do not say 'every Socrates' as we say 'every man'.

Some things are said to be the same in this sense, others (2) are the same by their own nature, as that which is one is in some cases one by its own nature; for both the things whose matter is one either in kind or in number, and those whose essence is one, are said to be the same. Clearly, therefore, sameness is a unity of the being either of more than one thing or of one thing when it is treated as more than one, i.e. when we say a thing is the same as itself; for we treat it as two.

Things are called 'other' if either their kinds or their matters or the formulae of their essence are more than one; and in general 'other' has meanings opposite to those of 'the same'.

'Different' is applied to (1) those things which though other are the same in some respect, only not in number but either in species or in genus or by analogy; (2) those whose genus is other, and contraries, and all things that have their otherness in their essence.

Those things are called 'like' which have the same attributes in every respect, and those which have more attributes the same than different, and those whose quality is one; and that which shares with another thing the greater number or the more important of the attributes (each of them one of two contraries) in respect of which things are capable of altering, is like that other thing. The senses of 'unlike' are opposite to those of 'like'.

Chapter 10

The term 'opposite' is applied to contradictories, and contraries, and relative terms, and privation and possession, and the extremes from which and into which generation and dissolution take place; and the attributes that cannot be present at the same time in that which is receptive of both, are said to be opposed, — either themselves or their constituents. Grey and white colour do not belong at the same time to the same thing; therefore their constituents are opposed.

The term 'contrary' is applied (1) to those attributes that differ in genus, which cannot belong at the same time to the same subject, (2) to the most different of the things in the same genus, (3) to the most different of the attributes in the same receptive material, (4) to the most different of the things that fall under the same faculty, (5) to the things whose difference is greatest either absolutely or in genus or in species. The other things that are called contrary are so called, some because they possess contraries of the above kind, some because they are receptive of such, some because they are productive of or susceptible to such, or are producing or suffering them, or are losses or acquisitions, or possessions or privations, of such. Since 'one' and 'being' have many senses, the other concepts which are derived from these, and therefore 'same', 'other', and 'contrary', must correspond, so that they must be different for each category.

The term 'other in species' is applied to things which being of the same genus are not subordinate the one to the other, or which being in the same genus have a difference, or which have a contrariety in their substance; and contraries are other than one another in species (either all contraries or those which are so called in the primary sense), and so are those things whose formulae differ in the *infima species* of the genus (e.g. man and horse are indivisible in genus, but their formulae are different), or which being in the same substance have a difference. 'The same in species' has the various meanings opposite to these.

Chapter 11

The words 'prior' and 'posterior' are applied (1) to some things (on the assumption that there is a first, i.e. a beginning, in each class) because they are nearer some beginning determined either absolutely and by nature, or by reference to something or in some place or by certain people, e.g. things are prior in place because they are nearer either to some place determined by nature, e.g. the middle or the last place, or to some chance object; and that which is further is posterior. — Other things are prior in time; some by being further from the present, i.e. in the case of past events (for the Trojan war is prior to the Persian, because it is further from the present), others by being nearer the present, i.e. in the case of future events (for the Nemean games are prior to the Pythian, if we treat the present as beginning and first point, because they are nearer the present). — Other things are prior in movement; for the things that are nearer the first mover are prior (e.g. the boy is prior to the man); and the prime mover also is a beginning absolutely. — Others are prior in power; for that which exceeds in power, i.e. the more powerful, is prior; and such is that according to whose will the other — i.e. the posterior — must follow, so that if the prior does not set it in motion the other does not move, and if it sets it in motion it does move; and here will is a beginning. — Others are prior in arrangement; these are the things that are placed at certain intervals in reference to some one definite thing according to rule, e.g. in the chorus the second man is prior to the third, and in the lyre the second-lowest string is prior to the lowest; for in the one case the leader and in the other the middle string is the beginning.

These, then, are called prior in this sense, but (2) in another sense that which is prior for knowledge is treated as absolutely prior; of these, the things that are prior in formula are different from those that are prior in perception. For in formula universals are prior, in perception individuals.

And in formula also the accident is prior to the whole, e.g. 'musical' to 'musical man', for the formula cannot exist as a whole without the part; yet musicalness cannot exist unless there is *some one* who is musical.

(3) The attributes of prior things are called prior, e.g. straightness is prior to smoothness; for one is an attribute of a line as such, and the other of a surface.

Some things then are called prior and posterior in this sense, others (4) in respect of nature and substance, i.e. those which can be without other things, while the others cannot be without *them*, — a distinction which Plato used. If we consider the various senses of 'being', firstly the subject is prior (so that substance is prior); secondly, according as potency or actuality is taken into account, different things are prior, for some things are prior in respect of potency, others in respect of actuality, e.g. in potency the half line is prior to the whole line and the part to the whole and the matter to the concrete substance, but in actuality these are posterior; for it is only when the whole is dissolved that they will exist in actuality. In a sense, therefore, all things that are called prior and posterior are so called according to this fourth sense; for some things can exist without others in respect of generation, e.g. the whole without the parts, and others in respect of dissolution, e.g. the part without the whole. And the same is true in all other cases.

Chapter 12

'Potency' means (1) a source of movement or change, which is in another thing than the thing moved or in the same thing *qua* other, e.g. the art of building is a potency which is not in the thing built, while the art of healing, which is a potency, might be in the man healed, but not in him *qua* healed. 'Potency' then means the source, in general, of change or movement in another thing or in the same thing *qua* other, and also the source of a thing's being moved by another thing or by itself *qua* other. For in virtue of that principle, in virtue of which the patient suffers anything, we call it 'capable' of suffering; and this we do sometimes if it suffers anything at all, sometimes not in respect of everything it suffers, but only if it suffers a change for the better. — (2) The capacity of performing this well or according to intention; for sometimes we say of those who merely can walk or speak but not well or not as they intend, that they *cannot* speak or walk. The case of passivity is similar. — (3) The states in virtue of which things are absolutely impassive or unchangeable, or not easily changed for the worse, are called potencies; for things are broken and crushed and bent and in general destroyed not by having a potency but by not having one and by lacking something, and things are impassive with respect to such processes if they are scarcely and slightly affected by them, because of a 'potency' and because they 'can' do something and are in some positive state.

As 'potency' has so many meanings, the 'potent' or 'capable' in one sense will mean that which can begin a movement (or a change in general, for even that which can bring things to rest is a 'potent' thing) in another thing or in itself *qua* other; and in one sense that over which something else has such a potency; and in one sense that which has a potency of changing into something, whether for the worse or for the better (for even that which perishes is thought to be 'capable' of perishing, for it would not have perished if it had not been capable of it; but, as a matter of fact, it has a certain disposition and cause and principle which fits it to suffer this; — sometimes it is thought to be of this sort because it has something, sometimes because it is deprived of something; but if privation is in a sense 'having' or 'habit', everything will be capable by having something, so that things are capable both by having a positive habit and principle, and by having the privation of the positive principle, if it is possible to *have* a privation; and if privation is *not* in a sense 'habit', 'capable' is used in two distinct senses); and a thing is capable in another sense because neither any other thing, nor itself *qua* other, has a potency or principle which can destroy it. Again, all these are

capable either merely because the thing might chance to happen or not to happen, or because it might do so *well*. This sort of potency is found also in lifeless things, e.g. in instruments; for we say one lyre can be made to 'speak', and another cannot be made to 'speak' at all, if it has not a good tone.

Incapacity is privation of capacity — i.e. of such a principle as has been described — either in general or in the case of something that would naturally have the capacity, or even at the time when it would naturally already have it; for the senses in which we should call a boy and a man and a eunuch incapable of begetting are distinct. — Again, to either kind of capacity there is an opposite incapacity — both to that which only *can* produce movement and to that which can produce it well.

Some things, then, are called ἀδύνατα in virtue of this kind of incapacity, while others are so in another sense, i.e. in that in which we couple δυνατόν and ἀδύνατον. The impossible is that of which the contrary is of necessity true, e.g. that the diagonal of a square is commensurate with the side is impossible, because such a statement is a falsity such that not only is the contrary true but it is *necessary* that the diagonal should be incommensurate; that it is commensurate, then, is not only false but of necessity false. The contrary of this, the possible, is found when it is not necessary that the contrary is false, e.g. that a man should be seated is possible; for that he is not seated is not of necessity false. — The possible, then, in one sense, as has been said, means that which is not of necessity false; in one, that which is true; in one, that which is capable of being true. — A 'potency' or 'power' in geometry is so called by a change of meaning. — These senses of 'capable' or 'possible' involve no reference to potency. But the senses which involve a reference to potency all refer to the one primary kind of potency; and this is a source of change in another thing or in the same thing *qua* other. For other things are called 'capable', some because something else has such a potency over them, some because it has not, some because it has it in a particular way. The same is true of the things that are incapable. Therefore the proper definition of the primary kind of potency will be 'a source of change in another thing or in the same thing *qua* other'.

Chapter 13

'Quantity' means that which is divisible into two or more constituent parts of which each is by nature a 'one' and a 'this'. A quantity is a plurality if it is numerable, a magnitude if it is measurable. 'Plurality' means that which is divisible potentially into non-continuous parts, 'magnitude' that which is divisible into continuous parts; in magnitude, that which is continuous in one dimension is length, in two breadth, in three depth. Of these, limited plurality is number, limited length is a line, breadth a surface, depth a solid.

Again, some things are called quantities in virtue of their own nature, others incidentally, e.g. the line is a quantity by its own nature, the musical is one incidentally. Of the things that are quantities by their own nature some are so as substances, e.g. the line is a quantity (for 'a certain kind of quantity' is present in the formula which states what it is), and others are modifications and states of this kind of substance, e.g. much and little, long and short, broad and narrow, deep and shallow, heavy and light, and the other terms of the sort. And also great and small, and greater and smaller, both in themselves and when taken relatively to each other, are by their own nature attributes of quantity; but these names are transferred to other things also. Of things that are quantities incidentally, some are so called in the sense in which it was said that 'musical' and 'white' were quantities, viz. because that to which they belong is a quantity, and some are quantities in the way in which movement and time are so; for these are called quantities and continuous because the things of which these are attributes are divisible. I mean not that which is moved, but the space through which it is moved; for because that is a quantity movement also is a quantity, and because

this is a quantity time is so.

Chapter 14

'Quality' means (1) the differentia of the essence, e.g. man is an animal of a certain quality because he is two-footed, and the horse is so because it is four-footed; and a circle is a figure of particular quality because it is without angles, — which shows that the essential differentia is a quality. — This, then, is one meaning of quality — essential differentia, but (2) there is another sense in which it applies to the unmovable objects of mathematics; i.e. the numbers have a certain quality, e.g. the composite numbers which are not in one dimension only, but of which the plane and the solid are copies (these are those which have two or three factors); and in general that which exists in the essence of numbers besides quantity is quality; for the essence of each is what it is once, e.g. that of 6 is not what it is twice or thrice, but what it is once; for 6 is once 6.

(3) All the attributes of substances in motion (e.g. heat and cold, whiteness and blackness, heaviness and lightness, and the others of the sort), in virtue of which, when they change, bodies are said to alter. (4) Quality in respect of virtue and vice and, in general, of evil and good.

Quality, then, seems to have practically two meanings, and one of these is the more proper. The primary quality is the essential differentia, and of this the quality in numbers is a part; for it is a differentia of essences, but either not of things in motion or not of them *qua* in motion. Secondly, there are the modifications of things in motion *qua* in motion, and the differentiae of movements. Virtue and vice fall among these modifications; for they indicate differentiae of the movement or activity, according to which the things in motion act or are acted on well or badly; for that which can be moved or act in one way is good, and that which can do so in another — the contrary — way is vicious. Good and evil indicate quality especially in living things, and among these especially in those which have purpose.

Chapter 15

Things are relative (1) as double to half and treble to a third, and in general that which contains something else many times to that which is contained many times in something else, and that which exceeds to that which is exceeded; (2) as that which can heat to that which can be heated, and that which can cut to that which can be cut, and in general the active to the passive; (3) as the measured to the measure and the known to knowledge and the perceived to perception.

(1) Relative terms of the first kind are numerically related either indefinitely or definitely, either to various numbers or to 1, e.g. the double is in a definite numerical relation to 1, and that which is 'many times as great' is in a numerical, but not in a definite, relation to 1, i.e. not in this or in that relation to it; the relation of that which is of something else to that something is a definite numerical relation to a number; that which is $n + 1 / n$ times something else is in an indefinite relation to that something, as that which is 'many times as great' is in an indefinite relation to 1; the relation of that which exceeds to that which is exceeded is numerically quite indefinite; for number is always commensurate, but this relation may involve a 'non-commensurate number'; for that which exceeds is, in relation to that which is exceeded, so much and something more; and this something is indefinite; for it can, indifferently, be either equal or not equal to that which is exceeded. — All these relations are numerically expressed and are determinations of number, and so in another way are the equal and the like and the same, for all refer to unity. Those things are the same whose substance is one; those are like whose quality is one; those are equal whose quantity is one; and 1 is the beginning and measure of number, so that all these relations imply number, though not in the same way.

(2) The active and the passive imply an active and a passive potency and the actualization of

the potencies, e.g. that which is capable of heating is related to that which is capable of being heated, because it *can* heat it, and, again, that which heats is related to that which is heated and that which cuts to that which is cut, because they actually do these things. But *numerical* relations are not actualized except in the sense which has been elsewhere stated; actualizations in the sense of movement they have not. Of relations which imply potency some further imply particular periods of time, e.g. that which has made is relative to that which has been made and that which will make to that which will be made. For it is in this way that a father is called *father of his son*; for the one has acted, and the other has been acted on in a certain way. Further, some relative terms imply *privation* of potency, i.e. 'incapable' and terms of this sort, e.g. 'invisible'.

Relative terms which imply number or potency, therefore, are all relative because their very essence includes in its nature a reference to something else, not because something else is referred to *it*; but (3) that which is measured or known or thought is called relative because something else is referred to it. For 'that which is thought' implies that the thought of it exists, but the thought is not relative to 'that of which it is the thought'; for we should then have said the same thing twice. Similarly sight is the sight of something, not 'of that of which it is the sight' (though of course it is true to say this); in fact it is relative to colour or to something else of the sort. But according to the other way of speaking the same thing would be said twice, — 'it is the sight of that which is the object of sight.'

Things that are by their own nature called relative are called so sometimes in these senses, sometimes because the classes that include them are of this sort, e.g. medicine is a relative term because its genus, science, is thought to be a relative term. Further, there are the properties in virtue of which the things that have them are called relative, e.g. equality is relative because the equal is, and likeness because the like is. Other things are relative by accident, e.g. a man is relative because he happens to be double of something and double is a relative term; or the white is relative, if the same thing happens to be double and white.

Chapter 16

'The complete' means (1) that outside which it is not possible to find even one of the parts proper to it, e.g. the complete time of each thing is that outside which it is not possible to find any time which is a part proper to it. — (2) That which in respect of excellence and goodness cannot be excelled in its kind, e.g. a doctor is complete and a flute-player is complete, when they lack nothing in respect of their proper kind of excellence. And thus we transfer the word to bad things, and speak of a complete scandal-monger and a complete thief; indeed we even call them *good*, i.e. a good thief and a good scandalmonger. And excellence is a completion; for each thing is complete and every substance is complete, when in respect of its proper kind of excellence it lacks no part of its natural magnitude. — (3) The things which have attained a good end are called complete; for things are complete in virtue of having attained their end. Therefore, since the end is something ultimate, we transfer the word to bad things and say a thing has been completely spoilt, and completely destroyed; when it in no wise falls short of destruction and badness, but is at its last point. This is why death is by a figure of speech called the end, because both are last things. The ultimate purpose is also an end. — Things, then, that are called complete in virtue of their *own* nature are so called in all these senses, some because they lack nothing in respect of goodness and cannot be excelled and no part proper to them can be found outside, others in general because they cannot be exceeded in their several classes and no part proper to them is outside; the *others* are so called in virtue of these first two kinds, because they either make or have something of the sort or are adapted to it or in some way or other are referred to the things that are called complete in the

primary sense.

Chapter 17

'Limit' means the last point of each thing, i.e. the first point beyond which it is not possible to find any part, and the first point within which every part is; it is applied to the form, whatever it may be, of a spatial magnitude or of a thing that has magnitude, and to the end of each thing (and of this nature is that towards which the movement and the action are — not that from which they are, though sometimes it is both, that from which and that to which the movement is — and the final cause), and to the substance of each thing, and the essence of each; for this is the limit of knowledge; and if of knowledge, of the thing also. Evidently, therefore, 'limit' has as many senses as 'beginning', and yet more; for the beginning is a limit, but not every limit is a beginning.

Chapter 18

'That in virtue of which' has several meanings, (1) the form or substance of each thing, e.g. that in virtue of which a man is good is the good itself, (2) the proximate subject in which an attribute is naturally found, e.g. colour in a surface. 'That in virtue of which,' then, in the primary sense is the form, and in a secondary sense the matter of each thing and the proximate substratum of each. — In general 'that in virtue of which' will be found in the same number of senses as 'cause'; for we say 'in virtue of what has he come?' or 'for what end has he come?'; and 'in virtue of what has he inferred wrongly, or inferred at all?' or 'what is the cause of the inference, or of the wrong inference?' — Further (3) *καθό* is used in reference to position, e.g. 'in which he stands' or 'in which he walks'; for all such phrases indicate place and position.

Therefore 'in virtue of itself' must have several meanings. It applies to (1) the essence of each thing, e.g. Callias is in virtue of himself Callias and the essence of Callias; (2) whatever is present in the 'what', e.g. Callias is in virtue of himself an animal. For 'animal' is present in the formula that defines him; Callias is a particular animal. — (3) Whatever attribute a thing receives in itself directly or in one of its parts, e.g. a surface is white in virtue of itself, and a man is alive in virtue of himself; for the soul, in which life directly resides, is a part of the man. — (4) That which has no cause other than itself; man has more than one cause - animal, two-footed — but man is man in virtue of himself. — (5) Whatever attributes belong to a thing alone and *qua* alone; hence also that which exists separately is 'in virtue of itself'.

Chapter 19

'Disposition' means the arrangement of that which has parts, in respect either of place or of potency or of kind; for there must be a certain position, as the word 'disposition' shows.

Chapter 20

'Having' means (1) a kind of activity of the haver and the had — something like an action or movement. When one thing makes and one is made, between them there is a making; so too between him who has a garment and the garment which he has there is a having. This sort of having, then, evidently we cannot *have*; for the process will go on to infinity, if we can have the having of what we have. — (2) 'Having' or 'habit' means a disposition according to which that which is disposed is either well or ill disposed, either in itself or with reference to something else, e.g. health is a 'habit'; for it is such a disposition. — (3) We speak of a 'habit' if there is a portion of such a disposition; therefore the excellence of the parts is a 'habit'.

Chapter 21

'Affection' means (1) a quality in respect of which a thing can be altered, e.g. white and black, sweet and bitter, heaviness and lightness, and all others of the kind. — (2) The already actualized alterations. — (3) Especially, injurious alterations and movements, and, above all, painful injuries.

— (4) Experiences pleasant or painful when on a large scale are called affections.

Chapter 22

We speak of 'privation' (1) if something has not one of the attributes which a thing might naturally have, even if this thing itself would not naturally have it, e.g. a plant is said to be 'deprived' of eyes. — (2) If, though either the thing itself or its genus would naturally have an attribute, it has it not, e.g. a blind man and a mole are in different senses 'deprived' of sight; the latter in contrast with its genus, the former in contrast with his own normal nature. — (3) If, though it would naturally have the attribute, and when it would naturally have it, it has it not; for blindness is a privation, but one is not 'blind' at any and every age, but only if one has not sight at the age at which one would naturally have it. Similarly a thing suffers privation when it has not an attribute in those circumstances, or in that respect and in that relation and in that sense, in which it would naturally have it. — (4) The violent taking away of anything is called privation.

There are just as many kinds of privations as there are of words with negative prefixes or affixes; for a thing is called unequal because it has not equality though it would naturally have it, and invisible either because it has no colour at all or because it has a poor colour, and footless either because it has no feet at all or because it has imperfect feet. Again, a privative term may be used because the thing has little of the attribute (and this means having it in a sense imperfectly), e.g. 'kernelless'; or because it has it not easily or not well (e.g. we call a thing indivisible not only if it cannot be divided but also if it cannot be easily or well divided); or because it has not the attribute at all; for it is not the one-eyed man but he who is sightless in both eyes that is called blind. This is why not every man is good or bad, just or unjust, but there is also an intermediate state.

Chapter 23

To 'have' or 'hold' means many things, (1) To treat a thing according to one's own nature or according to one's own impulse, so that fever is said to have a man, and tyrants to have their cities, and people to have the clothes they wear. — (2) That in which a thing is present as in something receptive is said to have the thing, e.g. the bronze has the form of the statue, and the body has the disease. — (3) As that which contains holds that which is contained; for a thing is said to be held by that in which it is contained, e.g. we say that the vessel holds the liquid and the city holds men and the ship sailors; and so too that the whole holds the parts. — (4) That which hinders a thing from moving or acting according to its own impulse is said to hold it, as pillars hold the incumbent weights, and as the poets make Atlas hold the heavens, implying that otherwise they would collapse on the earth, as some of the natural philosophers also say. In this way that which holds things together is said to hold the things it holds together, since they would otherwise separate, each according to its own impulse.

'Being in something' has similar and corresponding meanings to 'holding' or 'having'.

Chapter 24

'To come *from* something' means (1) to come from something as from matter, and this in two senses, either in respect of the highest genus or in respect of the lowest species, e.g. in a sense all things that can be melted come from water, but in a sense the statue comes from bronze. — (2) As from the first moving principle, e.g. the fight comes from abusive language, because this is the source of the fight. — (3) From the compound of matter and shape, as the parts come from the whole and the verse from the *Iliad* and the stones from the house; <the stones are to the house as part to whole,> for arrangement as a house is their end, and only that which attains an end is complete. — (4) As the form from its part, e.g. man from 'two-footed' and syllable from 'letter'; for this is a different sense from that in which the statue comes from bronze; for the composite

substance comes from the sensible matter, but the form also comes from the matter of the form. — These, then, are some of the meanings of 'coming from something', but sometimes (5) one of these senses is applicable only to part of a whole, e.g. the child comes from its father and mother and plants come from the earth, because they come from a part of those things. — (6) It means coming after a thing in time, e.g. night comes from day and storm from fine weather, because the one comes after the other. Of these things some are so described because they admit of change into one another, as in the cases now mentioned; some merely because they are successive in time, e.g. the voyage took place 'from' the equinox, because it took place after the equinox, and the Thargelia come 'from' the Dionysia, because after the Dionysia.

Chapter 25

'Part' means (1) that into which a quantity can in any way be divided; for that which is taken from a quantity *qua* quantity is always called a part of it, e.g. two is called in a sense a part of three. — (2) It means, of the parts in the first sense, only those which measure the whole; this is why two, though in one sense it is, in another is not, a part of three. — (3) The elements into which the kind might be divided apart from the quantity, are also called parts of it; for which reason we say the species are parts of the genus. — (4) The elements into which the whole is divided, or of which it consists — 'the whole' meaning either the form or that which has the form; e.g. of the bronze sphere or of the bronze cube both the bronze — i.e. the matter in which the form is — and the characteristic angle are parts. — (5) The elements in the formula which explains a thing are parts of the whole; this is why the genus is called a part of the species, though in another sense the species is part of the genus.

Chapter 26

'A whole' means (1) that from which is absent none of the parts of which it is said to be naturally a whole, and (2) that which so contains the things it contains that they form a unity; and this in two senses — either as each and all one, or as making up the unity between them. For (a) that which is true of a whole class and is said to hold good as a whole (which implies that it is a kind of whole) is true of a whole in the sense that it contains many things by being predicated of each, and that each and all of them, e.g. man, horse, god, are one, because all are living things. But (b) the continuous and limited is a whole, when there is a unity consisting of several parts present in it, especially if they are present only potentially, but, failing this, even if they are present actually. Of these things themselves, those which are so by nature are wholes in a higher degree than those which are so by art, as we said in the case of unity also, wholeness being in fact a sort of oneness.

Again, as quantities have a beginning and a middle and an end, those to which the position does not make a difference are called totals, and those to which it does, wholes, and those which admit of both descriptions are both wholes and totals. These are the things whose nature remains the same after transposition, but whose form does not, e.g. wax or a coat; they are called both wholes and totals; for they have both characteristics. Water and all liquids and number are called totals, but 'the whole number' or 'the whole water' one does not speak of, except by an extension of meaning. To things, to which *qua* one the term 'total' is applied, the term 'all' is applied when they are treated as separate; 'this total number,' 'all these units.'

Chapter 27

It is not any chance quantitative thing that can be said to be 'mutilated'; it must be both divisible and a whole. For two is not 'mutilated' if one of the two ones is taken away (for the part removed by mutilation is never equal to the remainder), nor in general is any number thus mutilated; for it is also necessary that the essence remain; if a cup is mutilated, it must still be a cup; but the number

is no longer the same. Further, even if things consist of unlike parts, not even these things can all be said to be mutilated, for in a sense a number has unlike parts, e.g. two and three. But in general of the things to which their position makes no difference, e.g. water or fire, none can be mutilated; to be mutilated, things must be such as in virtue of their essence have a certain position. Again, they must be continuous; for a musical scale consists of unlike parts and has position, but cannot become mutilated. Besides, not even the things that are wholes are mutilated by the privation of *any* part. For the parts removed must be neither those which determine the essence nor any chance parts, irrespective of their position; e.g. a cup is not mutilated if it is bored through; but only if the handle or a projecting part is removed. And a man is mutilated not if the flesh or the spleen is removed, but if an extremity is, and that not every extremity but one which when completely removed cannot grow again. Therefore baldness is not a mutilation.

Chapter 28

The term 'race' or 'genus' is used (1) if there is continuous generation of things which have the same form, e.g. 'while the race of men lasts' means 'while the generation of them goes on continuously'. — (2) It means that which first brought things into existence; for so some are called Hellenes by race and others Ionians, because the former proceed from Hellen and the latter from Ion as their first begetter. And the word is used in reference to the begetter more than to the matter, though people also get a race-name from the female, e.g. 'the descendants of Pyrrha.' — (3) There is genus in the sense in which 'plane' is the genus of plane figures and 'solid' of solids; for each of the figures is in the one case a plane of such and such a kind, and in the other a solid of such and such a kind; and this is what underlies the differentiae. Again, in formulae their first constituent element, which is included in the essence, is the genus, whose differentiae the qualities are said to be. — 'Genus' then is used in all these ways, (1) in reference to continuous generation of the same kind, (2) in reference to the first mover which is of the same kind as the things it moves, (3) as matter; for that to which the differentia or quality belongs is the substratum, which we call matter.

Those things are said to be 'other in genus' whose ultimate substratum is different, and which are not analysed the one into the other nor both into the same thing (e.g. form and matter are different in genus); and things which belong to different categories of being; for some of the things that are said to 'be' signify essence, others a quality, others the other categories we have before distinguished; these also are not analysed either into one another or into some one thing.

Chapter 29

'The false' means (1) that which is false as a *thing*, and that (*a*) because it is not put together or cannot be put together, e.g. 'that the diagonal of a square is commensurate with the side' or 'that you are sitting'; for one of these is false always, and the other sometimes; it is in these two senses that they are non-existent. (*b*) There are things which exist, but whose nature it is to appear either not to be such as they are or to be things that do not exist, e.g. a sketch or a dream; for these are something, but are not the things the appearance of which they produce in us. We call things false in this way, then, — either because they themselves do not exist, or because the appearance which results from them is that of something that does not exist.

(2) A false *conception* is the conception of non-existent objects, in so far as it is false. Hence every conception is false when applied to something other than that of which it is true, e.g. the conception of a circle is false when applied to a triangle. In a sense there is one conception of each thing, i.e. the conception of its essence, but in a sense there are many, since the thing itself and the thing itself modified in a certain way are somehow the same, e.g. Socrates and musical Socrates. The false conception is not the conception of anything, except in a qualified sense. Hence Antisthenes

foolishly claimed that nothing could be described except by its own conception, — one predicate to one subject; from which it followed that there could be no contradiction, and almost that there could be no error. But it is possible to describe each thing not only by its own conception, but also by that of something else. This may be done altogether falsely indeed, but in some ways it may be done truly, e.g. eight may be described as a double number by the use of the conception of two.

These things, then, are called false in these senses, but (3) a false *man* is one who is ready at and fond of such conceptions, not for any other reason but for their own sake, and one who is good at impressing such conceptions on other people, just as we say *things* are false, which produce a false appearance. This is why the proof in the *Hippias* that the same man is false and true is misleading. For it assumes that he is false who can deceive (i.e. the man who knows and is wise); and further that he who is *willingly* bad is better. This is a false result of *induction*; for a man who limps willingly is better than one who does so unwillingly; by 'limping' Plato means 'mimicking a limp', for if the man were actually lame willingly, he would perhaps be worse in this case as in the corresponding case of moral character.

Chapter 30

'Accident' means that which attaches to something and can be truly asserted, but neither of necessity nor usually, e.g. if one in digging a hole for a plant found treasure. This — the finding of treasure — happens by accident to the man who digs the hole; for neither does the one come of necessity from the other or after the other, nor, if a man plants, does he usually find treasure. And a musical man might be white; but since this does not happen of necessity nor usually, we call it an accident. Therefore since there are attributes and they attach to a subject, and some of them attach in a particular place and at a particular time, whatever attaches to a subject, but not because it is this subject, at this time or in this place, will be an accident. Therefore there is no definite cause for an accident, but a chance cause, i.e. an indefinite one. Going to Aegina was an accident, if the man went not in order to get there, but because he was carried out of his way by a storm or captured by pirates. The accident has happened or exists, — not in virtue of itself, however, but of something else; for the *storm* was the cause of his coming to a place for which he was not sailing, and this was Aegina.

'Accident' has also another meaning, i.e. what attaches to each thing in virtue of itself but is not in its essence, as having its angles equal to two right angles attaches to the triangle. And accidents of this sort may be eternal, but no accident of the other sort is. This is explained elsewhere.

Book VI

Layout 2

Chapter 1

We are seeking the principles and the causes of the things that are, and obviously of things *qua* being. For there is a cause of health and of good condition, and the objects of mathematics have principles and elements and causes, and in general every science which is ratiocinative or at all involves reasoning deals with causes and principles, exact or indeterminate; but all these sciences mark off some particular being — some genus and inquire into this, but not into being simply nor *qua* being, nor do they offer any discussion of the essence of the things of which they treat; but starting from the essence — some making it plain to the senses, others assuming it as a hypothesis — they then demonstrate, more or less cogently, the essential attributes of the genus with which they deal. It is obvious, therefore, from such a review of the sciences, that there is no demonstration of substance or of the essence, but some other way of exhibiting it. And similarly the sciences omit the question whether the genus with which they deal exists or does not exist, because it belongs to the same line of thought to show what it is and that it is.

And since natural science, like other sciences, confines itself to one class of beings, i.e. to that sort of substance which has the principle of its movement and rest present in itself, evidently it is neither practical nor productive. For the principle of production is in the producer — it is either reason or art or some potency, while the principle of action is in the doer — viz. will, for that which is done and that which is willed are the same. Therefore, if all thought is either practical or productive or theoretical, physics must be a theoretical science, but it will theorize about such being as admits of being moved, and only about that kind of substance which in respect of its definition is for the most part not separable from matter. Now, we must not fail to notice the nature of the essence and of its definition, for, without this, inquiry is but idle. Of things defined, i.e. of essences, some are like 'snub', and some like 'concave'. And these differ because 'snub' is bound up with matter (for what is snub is a concave *nose*), while concavity is independent of perceptible matter. If then all natural things are analogous to the snub in their nature — e.g. nose, eye, face, flesh, bone, and, in general, animal; leaf, root, bark, and, in general, plant (for none of these can be defined without reference to movement — they always have matter), it is clear how we must seek and define the 'what' in the case of natural objects, and also why it belongs to the student of nature to study soul to some extent, i.e. so much of it as is not independent of matter. — That physics, then, is a theoretical science, is plain from these considerations. Mathematics also is theoretical; but whether its objects are immovable and separable from matter, is not at present clear; it is clear, however, that it *considers some* mathematical objects *qua* immovable and *qua* separable from matter. But if there is something which is eternal and immovable and separable, clearly the knowledge of it belongs to a theoretical science, — not, however, to physics (for physics deals with certain movable things) nor to mathematics, but to a science prior to both. For physics deals with things which are inseparable from matter but not immovable, and some parts of mathematics deal with things which are immovable, but probably not separable, but embodied in matter; while the first science deals with things which are both separable and immovable. Now all causes must be eternal, but especially these; for they are the causes of so much of the divine as appears to us. There must, then, be three theoretical philosophies, mathematics, physics, and what we may call theology, since it is obvious that if the divine is present anywhere, it is present in things of this sort. And the highest science must deal with the highest genus, so that the theoretical sciences are superior to the other sciences, and this to the other theoretical sciences. One might indeed raise the question

whether first philosophy is universal, or deals with one genus, i.e. some one kind of being; for not even the mathematical sciences are all alike in this respect, — geometry and astronomy deal with a certain particular kind of thing, while universal mathematics applies alike to all. We answer that if there is no substance other than those which are formed by nature, natural science will be the first science; but if there is an immovable substance, the science of this must be prior and must be first philosophy, and universal in this way, because it is first. And it will belong to this to consider being *qua* being — both what it is and the attributes which, belong to it *qua* being.

Chapter 2

But since the unqualified term 'being' has several meanings, of which one was seen to be the accidental, and another the true ('non-being' being the false), while besides these there are the figures of predication, e.g. the 'what', quality, quantity, place, time, and any similar meanings which 'being' may have; and again besides all these there is that which 'is' potentially or actually: — since 'being' has many meanings, we must first say regarding the *accidental*, that there can be no scientific treatment of it. This is confirmed by the fact that no science — practical, productive, or theoretical — troubles itself about it. For on the one hand he who produces a house does not produce all the attributes that come into being along with the house; for these are innumerable; the house that is made may be pleasant for some people, hurtful to some, and useful to others, and different — to put it shortly — from all things that are; and the science of building does not aim at producing any of these attributes. And in the same way the geometer does not consider the attributes which attach thus to figures, nor whether 'triangle' is different from 'triangle whose angles are equal to two right angles'. — And this happens naturally enough; for the accidental is practically a mere name. And therefore Plato was in a sense not wrong in saying that sophistic deals with that which is not. For the arguments of the sophists deal, we may say, above all with the accidental; e.g. the question whether 'musical' and 'lettered' are different or the same, and whether 'musical Coriscus' and 'Coriscus' are the same, and whether 'everything which is, but is not eternal, has come to be', with the paradoxical conclusion that if one who was musical has come to be lettered, he must also have been lettered and have come to be musical, — and all the other arguments of this sort; the accidental is obviously akin to non-being. And this is clear also from arguments such as the following: of things which are in another sense there is generation and decay, but of things which are accidentally there is not. But still we must, as far as we can, say, regarding the accidental, what is its nature and from what cause it proceeds; for it will perhaps at the same time become clear why there is no science of it.

Since, among things which are, some are always in the same state and are of necessity (not necessity in the sense of compulsion but that which means the impossibility of being otherwise), and some are not of necessity nor always, but for the most part, this is the principle and this the cause of the existence of the accidental; for that which is neither always nor for the most part, we call accidental. For instance, if in the dog-days there is wintry and cold weather, we say this is an accident, but not if there is sultry heat, because the latter is always or for the most part so, but not the former. And it is an accident that a man is white (for this is neither always nor for the most part so), but it is not by accident that he is an animal. And that the builder produces health is an accident, because it is the nature not of the builder but of the doctor to do this, — but the builder happened to be a doctor. Again, a confectioner, aiming at giving pleasure, may make something wholesome, but not in virtue of the confectioner's art; and therefore we say it was an accident, and while there is a sense in which he makes it, in the full sense he does not make it. — For some accidental results sometimes tend to be produced by alien potencies but to others there corresponds

no determinate art nor potency; for of things which are or come to be by accident, the cause also is accidental. Therefore, since not all things are or come to be of necessity and always, but the majority of things are *for the most part*, the accidental must exist; for instance a white man is not always nor for the most part musical, but since this sometimes happens, it must be accidental. If not, everything will be of necessity. The matter, therefore, which is capable of being otherwise than as it usually is, is the cause of the accidental. And we must take as our starting-point the question whether everything is either always or for the most part. Surely this is impossible. There is, then, besides these something which is fortuitous and accidental. But while the usual exists, can nothing be said to be always, or are there eternal things? This must be considered later, but that there is no science of the accidental is obvious; for all science is either of that which is always or of that which is for the most part. For how else is one to learn or to teach another? The thing must be determined as occurring either always or for the most part, e.g. that honey-water is useful for a patient in a fever is true for the most part. But one will not be able to state when that which is contrary to the usual law happens, e.g. 'on the day of new moon'; for if one can say this, 'on the day of new moon' is itself the statement of a universal or a usual law; but the accidental is contrary to such laws. We have stated, then, what the accidental is and from what cause it arises, and that there is no science which deals with it.

Chapter 3

That there are principles and causes which are generable and destructible without ever being in course of being generated or destroyed, is obvious. For otherwise all things will be of necessity, since that which is being generated or destroyed must have a cause which is not accidentally its cause. Whether is *A* to be or not? It will be if *B* happens; and if not, not. And *B* will be if *C* happens. And thus if time is constantly subtracted from a limited extent of time, one will obviously come to the present. This man, then, will die by disease or violence, *if* he goes out; and he will do this if he is thirsty; and he will be thirsty if something else happens; and thus we shall come to that which is now present, or to some past event. For instance, he will go out if he is thirsty; and he will be thirsty if he is eating something pungent; and this is either the case or not; so that he will of necessity die, or of necessity not die. And similarly if one jumps over to the past, the same account will hold good; for this — I mean the past condition — is already present in something. Everything, therefore, that is to be, will be of necessity, e.g. it is necessary that he who lives shall one day die; for already some link in the series has been forged — e.g. the presence of contraries in the same body. But whether he dies by disease or by violence, is not yet determined, but depends on the happening of something else. Clearly then the process goes back to a certain starting-point, but this no longer points to something further. This then will be the starting-point for the fortuitous, and will have nothing else as cause of its coming to be. But to what sort of starting-point and what sort of cause we thus refer the fortuitous — whether to matter or to the purpose or to the motive power, must be carefully considered.

Chapter 4

Let us dismiss the accidental; for we have sufficiently determined its nature. But since that which *is* in the sense of being true, or *is not* in the sense of being false, depends on combination and separation, and truth and falsehood together depend on the decision between the two sides of a contradiction (for the true judgement affirms where the subject and predicate really are combined, and denies where they are separated, while the false judgement predicates the contradictory of this — it is another question, how it happens that we think things together or apart; by 'together' and 'apart' I mean thinking them so that there is no succession in the thoughts but they become a unity

—; for falsity and truth are not in things — it is not as if the good were true, and the bad were in itself false — but in thought; while with regard to simple concepts and essences falsity and truth do not exist even in thought): — we must consider later what has to be discussed with regard to that which is or is not in this sense; but since the combination and the separation are in thought and not in the things, and that which is in this sense is a different sort of 'being' from the things that are in the full sense (for the thought attaches or removes either the 'what' or quality or quantity or one of the other categories), that which *is* accidentally and that which *is* in the sense of being true must be dismissed. For the cause of the former is indeterminate, and that of the latter is some affection of the thought, and both are related to the remaining genus of being, and do not indicate any separate class of being. Therefore let these be dismissed, and let us consider the causes and the principles of being itself, *qua* being. It was clear in our discussion of the various meanings of terms, that 'being' has several meanings.

Book VII

Layout 2

Chapter 1

There are several senses in which a thing may be said to 'be', as we pointed out previously in our book on the various senses of words; for in one sense the 'being' meant is 'what a thing is' or the individual thing, and in another sense it means that a thing is of a certain quality or quantity or has some such predicate asserted of it. While 'being' has all these senses, obviously that which 'is' primarily is the 'what', which indicates the substance of the thing. For when we say of what quality a thing is, we say that it is good or beautiful, but not that it is three cubits long or that it is a man; but when we say *what* it is, we do not say 'white' or 'hot' or 'three cubits long', but 'man' or 'God'. And all other things are said to be because they are, some of them, quantities of that which *is* in this primary sense, others qualities of it, others affections of it, and others some other determination of it. And so one might raise the question whether walking and being healthy and sitting are, each of them, existent or non-existent, and similarly in any other case of this sort; for none of them is either self-subsistent or capable of being separated from substance, but rather, if anything, it is that which walks or is seated or is healthy that is an existent thing. Now these are seen to be more real because there is something definite which underlies them; and this is the substance or individual, which is implied in such a predicate; for 'good' or 'sitting' apart from that which sits or is good has no meaning. Clearly then it is in virtue of this category that each of the others *is*. Therefore that which is primarily and *is* simply (not 'is something') must be substance.

Now there are several senses in which a thing is said to be first; but substance is first in every sense — (1) in formula, (2) in order of knowledge, (3) in time. For (3) of the other categories none can exist independently, but only substance. And (1) in formula also this is first; for in the formula of each term the formula of its substance must be present. And (2) we think we know each thing most fully, when we know what it is, e.g. what man is or what fire is, rather than when we know its quality, its quantity, or where it is; since we know each of these predicates also, only when we know *what* the quantity or the quality *is*.

And indeed the question which was raised of old and is raised now and always, and is always the subject of doubt, viz. what being is, is just the question, what is substance? For it is this that some assert to be one, others more than one, and that some assert to be limited in number, others unlimited. And so we also must consider chiefly and primarily and almost exclusively what that is which is in this sense.

Chapter 2

Substance is thought to belong most obviously to bodies; and so we say that both animals and plants and their parts are substances, and so are natural bodies such as fire and water and earth and everything of the sort, and all things that are parts of these or composed of these (either of parts or of the whole bodies), e.g. the heaven and its parts, stars and moon and sun. But whether these alone are substances, or there are also others, or only some of these, or some of these and some other things are substances, or none of these but only some other things, must be considered. Some think the limits of body, i.e. surface, line, point, and unit, are substances, and more so than body or the solid.

Further, some do not think there is anything substantial besides sensible things, but others think there are eternal substances which are more in number and more real, e.g. Plato posited two kinds of substance — the Forms and the objects of mathematics — as well as a third kind, viz. the substance of

sensible bodies. And Speusippus made still more kinds of substance, beginning with the One, and making principles for each kind of substance, one for numbers, another for spatial magnitudes, and then another for the soul; and in this way he multiplies the kinds of substance. And some say Forms and numbers have the same nature, and other things come after them, e.g. lines and planes, until we come to the substance of the material universe and to sensible bodies.

Regarding these matters, then, we must inquire which of the common statements are right and which are not right, and what things are substances, and whether there are or are not any besides sensible substances, and how sensible substances exist, and whether there is a separable substance (and if so why and how) or there is no substance separable from sensible substances; and we must first sketch the nature of substance.

Chapter 3

The word 'substance' is applied, if not in more senses, still at least to four main objects; for both the essence and the universal and the genus are thought to be the substance of each thing, and fourthly the substratum. Now the substratum is that of which the others are predicated, while it is itself not predicated of anything else. And so we must first determine the nature of this; for that which underlies a thing primarily is thought to be in the truest sense its substance. And in one sense matter is said to be of the nature of substratum, in another, shape, and in a third sense, the compound of these. By the matter I mean, for instance, the bronze, by the shape the plan of its form, and by the compound of these (the concrete thing) the statue. Therefore if the form is prior to the matter and more real, the compound of both will be prior also for the same reason.

We have now outlined the nature of substance, showing that it is that which is not predicated of a subject, but of which all else is predicated. But we must not merely state the matter thus; for this is not enough. The statement itself is obscure, and further, on this view, *matter* becomes substance. For if this is not substance, it is beyond our power to say what else is. When all else is taken away evidently nothing but matter remains. For of the other elements some are affections, products, and potencies of bodies, while length, breadth, and depth are quantities and not substances. For a quantity is not a substance; but the substance is rather that to which these belong primarily. But when length and breadth and depth are taken away we see nothing left except that which is bounded by these, whatever it be; so that to those who consider the question thus matter alone must seem to be substance. By matter I mean that which in itself is neither a particular thing nor of a certain quantity nor assigned to any other of the categories by which being is determined. For there is something of which each of these is predicated, so that its being is different from that of each of the predicates; for the predicates other than substance are predicated of substance, while substance is predicated of matter. Therefore the ultimate substratum is of itself neither a particular thing nor of a particular quantity nor otherwise positively characterized; nor yet negatively, for negations also will belong to it only by accident.

If we adopt this point of view, then, it follows that matter is substance. But this is impossible; for both separability and individuality are thought to belong chiefly to substance. And so form and the compound of form and matter would be thought to be substance, rather than matter. The substance compounded of both, i.e. of matter and shape, may be dismissed; for it is posterior and its nature is obvious. And matter also is in a sense manifest. But we must inquire into the third kind of substance; for this is the most difficult.

It is agreed that there are some substances among sensible things, so that we must look first among these. For it is an advantage to advance to that which is more intelligible. For learning proceeds for all in this way — through that which is less intelligible by nature to that which is more

intelligible; and just as in conduct our work is to start from what is good for each and make what is good in itself good for each, so it is our work to start from what is more intelligible to oneself and make what is intelligible by nature intelligible to oneself. Now what is intelligible and primary for particular sets of people is often intelligible to a very small extent, and has little or nothing of reality. But yet one must start from that which is barely intelligible but intelligible to oneself, and try to understand what is intelligible in itself, passing, as has been said, by way of those very things which one understands.

Chapter 4

Since at the start we distinguished the various marks by which we determine substance, and one of these was thought to be the essence, we must investigate this. And first let us make some abstract linguistic remarks about it. The essence of each thing is what it is said to be *propter se*. For being you is not being musical; for you are not by your very nature musical. What, then, you are by your very nature is your essence.

But not the whole of this is the essence of a thing; not that which it is *propter se* as a surface is *propter se* white, because being a surface is not *identical* with being white. But again the combination of both — 'being a white surface' — is not the essence of surface. Why? Because 'surface' itself is repeated. The formula, therefore, in which the term itself is not present but its meaning is expressed, this is the formula of the essence of each thing. Therefore if to be a white surface is to be a smooth surface, to be white and to be smooth are one and the same.

But since there are compounds of substance with the other categories (for there is a substrate for each category, e.g. for quality, quantity, time, place, and motion), we must inquire whether there is a formula of the essence of each of them, i.e. whether to these compounds also there belongs an essence, e.g. to white man the essence of white man. Let the compound be denoted by 'X'. What is the essence of X? But, it may be said, this also is not a *propter se* expression. We reply that there are just two ways in which a predicate may fail to be true of a subject *propter se*, and one of these results from the addition, and the other from the omission, of a determinant. *One* kind of predicate is not *propter se* because the term that is being defined is combined with another determinant, e.g. if in defining the essence of white one were to state the formula of white *man*; *another* because in the subject another determinant is combined with that which is expressed in the formula, e.g. if X meant white man, and one were to define X as white; white man is white indeed, but its essence is not to be white. But is being-X an essence at all? Probably not.' For the essence is an individual type; but when an attribute is asserted of an alien subject, the complex is not an individual type, e.g. white man is not an individual type, since individuality belongs only to substances. Therefore there is an essence only of those things whose formula is a definition. But we have a definition not where we have a word and a formula identical in meaning (for in that case all formulae or sets of words would be definitions; for there will be some name for any set of words whatever, so that even the *Iliad* would be a definition), but where there is a formula of something primary; and primary things are those which do not imply the predication of one element in them of another, alien element. Nothing, then, which is not a species of a genus will have an *essence* — only species will have it, for in these the subject is not thought to participate in the attribute and to have it as an affection, nor to have it by accident; but for everything else as well, if it has a name, there will be a *formula of its meaning* — viz. that this attribute belongs to this subject; Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/148 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/149 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/150 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/151 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/152

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Book VIII

Layout 2

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Book IX

Layout 2

Chapter 1

We have treated of that which *is* primarily and to which all the other categories of being are referred — i.e. of substance. For it is in virtue of the concept of substance that the others also are said to be — quantity and quality and the like; for all will be found to involve the concept of substance, as we said in the first part of our work. And since 'being' is in one way divided into 'what', quality, and quantity, and is in another way distinguished in respect of potency and complete reality, and of function, let us discuss potency and complete reality. First let us explain potency in the strictest sense, which is, however, not the most useful for our present purpose. For potency and actuality extend further than the mere sphere of motion. But when we have spoken of this first kind, we shall in our discussions of actuality explain the other kinds of potency.

We have pointed out elsewhere that 'potency' and the word 'can' have several senses. Of these we may neglect all the potencies that are so called by an equivocation. For some are called so by analogy, as in geometry; and we say things can be or cannot be because in some definite way they are or are not.

But all potencies that conform to the same type are starting-points, and are called potencies in reference to one primary kind of potency, which is a starting-point of change in another thing or in the thing itself *qua* other. For one kind is a potency of being acted on, i.e. the principle in the very thing acted on, which makes it capable of being changed and acted on by another thing or by itself regarded as other; and another kind is a state of insusceptibility to change for the worse and to destruction by another thing or by the thing itself *qua* other, i.e. by a principle of change. In all these definitions is implied the formula of potency in the primary sense. — And again these so-called potencies are potencies either of acting merely or of being acted on, or of acting or being acted on *well*, so that even in the formulae of the latter the formulae of the prior kinds of potency are somehow implied.

Obviously, then, in a sense the potency of acting and of being acted on is one (for a thing may be 'capable' either because it can be acted on or because something else can be acted on by it), but in a sense the potencies are different. For the one is in the thing acted on; it is because it contains a certain motive principle, and because even the matter is a motive principle, that the thing acted on is acted on, one thing by one, another by another; for that which is oily is *inflammable*, and that which *yields* in a particular way can be *crushed*; and similarly in all other cases. But the other potency is in the agent, e.g. heat and the art of building are present, one in that which can produce heat and the other in the man who can build. And so in so far as a thing is an organic unity, it cannot be acted on by itself; for it is one and not two different things. And want of potency, or powerlessness, is the privation which is contrary to potency of this sort, so that every potency belongs to the same subject and refers to the same process as a corresponding want of potency. Privation has several senses; for it means (1) that which has not a certain quality and (2) that which might naturally have it but has not it, either (a) in general or (b) when it might naturally have it, and either (i) in some particular way, e.g. when it *completely* fails to have it, or (ii) when it in any degree fails to have it. And in certain cases if things which naturally have a quality lose it by violence, we say they suffer privation.

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Book X

Part 1

WE have said previously, in our distinction of the various meanings of words, that 'one' has several meanings; the things that are directly and of their own nature and not accidentally called one may be summarized under four heads, though the word is used in more senses. (1) There is the continuous, either in general, or especially that which is continuous by nature and not by contact nor by being together; and of these, that has more unity and is prior, whose movement is more indivisible and simpler. (2) That which is a whole and has a certain shape and form is one in a still higher degree; and especially if a thing is of this sort by nature, and not by force like the things which are unified by glue or nails or by being tied together, i.e. if it has in itself the cause of its continuity. A thing is of this sort because its movement is one and indivisible in place and time; so that evidently if a thing has by nature a principle of movement that is of the first kind (i.e. local movement) and the first in that kind (i.e. circular movement), this is in the primary sense one extended thing. Some things, then, are one in this way, qua continuous or whole, and the other things that are one are those whose definition is one. Of this sort are the things the thought of which is one, i.e. those the thought of which is indivisible; and it is indivisible if the thing is indivisible in kind or in number. (3) In number, then, the individual is indivisible, and (4) in kind, that which in intelligibility and in knowledge is indivisible, so that that which causes substances to be one must be one in the primary sense. 'One', then, has all these meanings-the naturally continuous and the whole, and the individual and the universal. And all these are one because in some cases the movement, in others the thought or the definition is indivisible.

But it must be observed that the questions, what sort of things are said to be one, and what it is to be one and what is the definition of it, should not be assumed to be the same. 'One' has all these meanings, and each of the things to which one of these kinds of unity belongs will be one; but 'to be one' will sometimes mean being one of these things, and sometimes being something else which is even nearer to the meaning of the word 'one' while these other things approximate to its application. This is also true of 'element' or 'cause', if one had both to specify the things of which it is predicable and to render the definition of the word. For in a sense fire is an element (and doubtless also 'the indefinite' or something else of the sort is by its own nature the element), but in a sense it is not; for it is not the same thing to be fire and to be an element, but while as a particular thing with a nature of its own fire is an element, the name 'element' means that it has this attribute, that there is something which is made of it as a primary constituent. And so with 'cause' and 'one' and all such terms. For this reason, too, 'to be one' means 'to be indivisible, being essentially one means a "this" and capable of being isolated either in place, or in form or thought'; or perhaps 'to be whole and indivisible'; but it means especially 'to be the first measure of a kind', and most strictly of quantity; for it is from this that it has been extended to the other categories. For measure is that by which quantity is known; and quantity qua quantity is known either by a 'one' or by a number, and all number is known by a 'one'. Therefore all quantity qua quantity is known by the one, and that by which quantities are primarily known is the one itself; and so the one is the starting-point of number qua number. And hence in the other classes too 'measure' means that by which each is first known, and the measure of each is a unit-in length, in breadth, in depth, in weight, in speed. (The words 'weight' and 'speed' are common to both contraries; for each of them has two meanings-'weight' means both that which has any amount of gravity and that which has an excess of gravity, and 'speed' both that which has any amount of movement and that which has an excess of movement; for even the slow has a certain speed and the comparatively light a certain

weight.)

In all these, then, the measure and starting-point is something one and indivisible, since even in lines we treat as indivisible the line a foot long. For everywhere we seek as the measure something one and indivisible; and this is that which is simple either in quality or in quantity. Now where it is thought impossible to take away or to add, there the measure is exact (hence that of number is most exact; for we posit the unit as indivisible in every respect); but in all other cases we imitate this sort of measure. For in the case of a furlong or a talent or of anything comparatively large any addition or subtraction might more easily escape our notice than in the case of something smaller; so that the first thing from which, as far as our perception goes, nothing can be subtracted, all men make the measure, whether of liquids or of solids, whether of weight or of size; and they think they know the quantity when they know it by means of this measure. And indeed they know movement too by the simple movement and the quickest; for this occupies least time. And so in astronomy a 'one' of this sort is the starting-point and measure (for they assume the movement of the heavens to be uniform and the quickest, and judge the others by reference to it), and in music the quarter-tone (because it is the least interval), and in speech the letter. And all these are ones in this sense—not that 'one' is something predicable in the same sense of all of these, but in the sense we have mentioned.

But the measure is not always one in number—sometimes there are several; e.g. the quarter-tones (not to the ear, but as determined by the ratios) are two, and the articulate sounds by which we measure are more than one, and the diagonal of the square and its side are measured by two quantities, and all spatial magnitudes reveal similar varieties of unit. Thus, then, the one is the measure of all things, because we come to know the elements in the substance by dividing the things either in respect of quantity or in respect of kind. And the one is indivisible just because the first of each class of things is indivisible. But it is not in the same way that every 'one' is indivisible e.g. a foot and a unit; the latter is indivisible in every respect, while the former must be placed among things which are undivided to perception, as has been said already—only to perception, for doubtless every continuous thing is divisible.

The measure is always homogeneous with the thing measured; the measure of spatial magnitudes is a spatial magnitude, and in particular that of length is a length, that of breadth a breadth, that of articulate sound an articulate sound, that of weight a weight, that of units a unit. (For we must state the matter so, and not say that the measure of numbers is a number; we ought indeed to say this if we were to use the corresponding form of words, but the claim does not really correspond—it is as if one claimed that the measure of units is units and not a unit; number is a plurality of units.)

Knowledge, also, and perception, we call the measure of things for the same reason, because we come to know something by them—while as a matter of fact they are measured rather than measure other things. But it is with us as if some one else measured us and we came to know how big we are by seeing that he applied the cubit-measure to such and such a fraction of us. But Protagoras says 'man is the measure of all things', as if he had said 'the man who knows' or 'the man who perceives'; and these because they have respectively knowledge and perception, which we say are the measures of objects. Such thinkers are saying nothing, then, while they appear to be saying something remarkable.

Evidently, then, unity in the strictest sense, if we define it according to the meaning of the word, is a measure, and most properly of quantity, and secondly of quality. And some things will be one if they are indivisible in quantity, and others if they are indivisible in quality; and so that which is one is indivisible, either absolutely or qua one.

Part 2

With regard to the substance and nature of the one we must ask in which of two ways it exists. This is the very question that we reviewed in our discussion of problems, viz. what the one is and how we must conceive of it, whether we must take the one itself as being a substance (as both the Pythagoreans say in earlier and Plato in later times), or there is, rather, an underlying nature and the one should be described more intelligibly and more in the manner of the physical philosophers, of whom one says the one is love, another says it is air, and another the indefinite.

If, then, no universal can be a substance, as has been said our discussion of substance and being, and if being itself cannot be a substance in the sense of a one apart from the many (for it is common to the many), but is only a predicate, clearly unity also cannot be a substance; for being and unity are the most universal of all predicates. Therefore, on the one hand, genera are not certain entities and substances separable from other things; and on the other hand the one cannot be a genus, for the same reasons for which being and substance cannot be genera.

Further, the position must be similar in all the kinds of unity. Now 'unity' has just as many meanings as 'being'; so that since in the sphere of qualities the one is something definite-some particular kind of thing-and similarly in the sphere of quantities, clearly we must in every category ask what the one is, as we must ask what the existent is, since it is not enough to say that its nature is just to be one or existent. But in colours the one is a colour, e.g. white, and then the other colours are observed to be produced out of this and black, and black is the privation of white, as darkness of light. Therefore if all existent things were colours, existent things would have been a number, indeed, but of what? Clearly of colours; and the 'one' would have been a particular 'one', i.e. white. And similarly if all existing things were tunes, they would have been a number, but a number of quarter-tones, and their essence would not have been number; and the one would have been something whose substance was not to be one but to be the quarter-tone. And similarly if all existent things had been articulate sounds, they would have been a number of letters, and the one would have been a vowel. And if all existent things were rectilinear figures, they would have been a number of figures, and the one would have been the triangle. And the same argument applies to all other classes. Since, therefore, while there are numbers and a one both in affections and in qualities and in quantities and in movement, in all cases the number is a number of particular things and the one is one something, and its substance is not just to be one, the same must be true of substances also; for it is true of all cases alike.

That the one, then, in every class is a definite thing, and in no case is its nature just this, unity, is evident; but as in colours the one-itself which we must seek is one colour, so too in substance the one-itself is one substance. That in a sense unity means the same as being is clear from the facts that its meanings correspond to the categories one to one, and it is not comprised within any category (e.g. it is comprised neither in 'what a thing is' nor in quality, but is related to them just as being is); that in 'one man' nothing more is predicated than in 'man' (just as being is nothing apart from substance or quality or quantity); and that to be one is just to be a particular thing.

Part 3

The one and the many are opposed in several ways, of which one is the opposition of the one and plurality as indivisible and divisible; for that which is either divided or divisible is called a plurality, and that which is indivisible or not divided is called one. Now since opposition is of four kinds, and one of these two terms is privative in meaning, they must be contraries, and neither contradictory nor correlative in meaning. And the one derives its name and its explanation from its contrary, the indivisible from the divisible, because plurality and the divisible is more

perceptible than the indivisible, so that in definition plurality is prior to the indivisible, because of the conditions of perception.

To the one belong, as we indicated graphically in our distinction of the contraries, the same and the like and the equal, and to plurality belong the other and the unlike and the unequal. 'The same' has several meanings; (1) we sometimes mean 'the same numerically'; again, (2) we call a thing the same if it is one both in definition and in number, e.g. you are one with yourself both in form and in matter; and again, (3) if the definition of its primary essence is one; e.g. equal straight lines are the same, and so are equal and equal-angled quadrilaterals; there are many such, but in these equality constitutes unity.

Things are like if, not being absolutely the same, nor without difference in respect of their concrete substance, they are the same in form; e.g. the larger square is like the smaller, and unequal straight lines are like; they are like, but not absolutely the same. Other things are like, if, having the same form, and being things in which difference of degree is possible, they have no difference of degree. Other things, if they have a quality that is in form one and same-e.g. whiteness-in a greater or less degree, are called like because their form is one. Other things are called like if the qualities they have in common are more numerous than those in which they differ-either the qualities in general or the prominent qualities; e.g. tin is like silver, qua white, and gold is like fire, qua yellow and red.

Evidently, then, 'other' and 'unlike' also have several meanings. And the other in one sense is the opposite of the same (so that everything is either the same as or other than everything else). In another sense things are other unless both their matter and their definition are one (so that you are other than your neighbour). The other in the third sense is exemplified in the objects of mathematics. 'Other or the same' can therefore be predicated of everything with regard to everything else-but only if the things are one and existent, for 'other' is not the contradictory of 'the same'; which is why it is not predicated of non-existent things (while 'not the same' is so predicated). It is predicated of all existing things; for everything that is existent and one is by its very nature either one or not one with anything else.

The other, then, and the same are thus opposed. But difference is not the same as otherness. For the other and that which it is other than need not be other in some definite respect (for everything that is existent is either other or the same), but that which is different is different from some particular thing in some particular respect, so that there must be something identical whereby they differ. And this identical thing is genus or species; for everything that differs differs either in genus or in species, in genus if the things have not their matter in common and are not generated out of each other (i.e. if they belong to different figures of predication), and in species if they have the same genus ('genus' meaning that identical thing which is essentially predicated of both the different things).

Contraries are different, and contrariety is a kind of difference. That we are right in this supposition is shown by induction. For all of these too are seen to be different; they are not merely other, but some are other in genus, and others are in the same line of predication, and therefore in the same genus, and the same in genus. We have distinguished elsewhere what sort of things are the same or other in genus.

Part 4

Since things which differ may differ from one another more or less, there is also a greatest difference, and this I call contrariety. That contrariety is the greatest difference is made clear by induction. For things which differ in genus have no way to one another, but are too far distant

and are not comparable; and for things that differ in species the extremes from which generation takes place are the contraries, and the distance between extremes-and therefore that between the contraries-is the greatest.

But surely that which is greatest in each class is complete. For that is greatest which cannot be exceeded, and that is complete beyond which nothing can be found. For the complete difference marks the end of a series (just as the other things which are called complete are so called because they have attained an end), and beyond the end there is nothing; for in everything it is the extreme and includes all else, and therefore there is nothing beyond the end, and the complete needs nothing further. From this, then, it is clear that contrariety is complete difference; and as contraries are so called in several senses, their modes of completeness will answer to the various modes of contrariety which attach to the contraries.

This being so, it is clear that one thing have more than one contrary (for neither can there be anything more extreme than the extreme, nor can there be more than two extremes for the one interval), and, to put the matter generally, this is clear if contrariety is a difference, and if difference, and therefore also the complete difference, must be between two things.

And the other commonly accepted definitions of contraries are also necessarily true. For not only is (1) the complete difference the greatest difference (for we can get no difference beyond it of things differing either in genus or in species; for it has been shown that there is no 'difference' between anything and the things outside its genus, and among the things which differ in species the complete difference is the greatest); but also (2) the things in the same genus which differ most are contrary (for the complete difference is the greatest difference between species of the same genus); and (3) the things in the same receptive material which differ most are contrary (for the matter is the same for contraries); and (4) of the things which fall under the same faculty the most different are contrary (for one science deals with one class of things, and in these the complete difference is the greatest).

The primary contrariety is that between positive state and privation-not every privation, however (for 'privation' has several meanings), but that which is complete. And the other contraries must be called so with reference to these, some because they possess these, others because they produce or tend to produce them, others because they are acquisitions or losses of these or of other contraries. Now if the kinds of opposition are contradiction and privation and contrariety and relation, and of these the first is contradiction, and contradiction admits of no intermediate, while contraries admit of one, clearly contradiction and contrariety are not the same. But privation is a kind of contradiction; for what suffers privation, either in general or in some determinate way, either that which is quite incapable of having some attribute or that which, being of such a nature as to have it, has it not; here we have already a variety of meanings, which have been distinguished elsewhere. Privation, therefore, is a contradiction or incapacity which is determinate or taken along with the receptive material. This is the reason why, while contradiction does not admit of an intermediate, privation sometimes does; for everything is equal or not equal, but not everything is equal or unequal, or if it is, it is only within the sphere of that which is receptive of equality. If, then, the comings-to-be which happen to the matter start from the contraries, and proceed either from the form and the possession of the form or from a privation of the form or shape, clearly all contrariety must be privation, but presumably not all privation is contrariety (the reason being that that has suffered privation may have suffered it in several ways); for it is only the extremes from which changes proceed that are contraries.

And this is obvious also by induction. For every contrariety involves, as one of its terms,

a privation, but not all cases are alike; inequality is the privation of equality and unlikeness of likeness, and on the other hand vice is the privation of virtue. But the cases differ in a way already described; in one case we mean simply that the thing has suffered privation, in another case that it has done so either at a certain time or in a certain part (e.g. at a certain age or in the dominant part), or throughout. This is why in some cases there is a mean (there are men who are neither good nor bad), and in others there is not (a number must be either odd or even). Further, some contraries have their subject defined, others have not. Therefore it is evident that one of the contraries is always privative; but it is enough if this is true of the first-i.e. the generic-contraries, e.g. the one and the many; for the others can be reduced to these.

Part 5

Since one thing has one contrary, we might raise the question how the one is opposed to the many, and the equal to the great and the small. For if we used the word 'whether' only in an antithesis such as 'whether it is white or black', or 'whether it is white or not white' (we do not ask 'whether it is a man or white'), unless we are proceeding on a prior assumption and asking something such as 'whether it was Cleon or Socrates that came' as this is not a necessary disjunction in any class of things; yet even this is an extension from the case of opposites; for opposites alone cannot be present together; and we assume this incompatibility here too in asking which of the two came; for if they might both have come, the question would have been absurd; but if they might, even so this falls just as much into an antithesis, that of the 'one or many', i.e. 'whether both came or one of the two':-if, then, the question 'whether' is always concerned with opposites, and we can ask 'whether it is greater or less or equal', what is the opposition of the equal to the other two? It is not contrary either to one alone or to both; for why should it be contrary to the greater rather than to the less? Further, the equal is contrary to the unequal. Therefore if it is contrary to the greater and the less, it will be contrary to more things than one. But if the unequal means the same as both the greater and the less together, the equal will be opposite to both (and the difficulty supports those who say the unequal is a 'two'), but it follows that one thing is contrary to two others, which is impossible. Again, the equal is evidently intermediate between the great and the small, but no contrariety is either observed to be intermediate, or, from its definition, can be so; for it would not be complete if it were intermediate between any two things, but rather it always has something intermediate between its own terms.

It remains, then, that it is opposed either as negation or as privation. It cannot be the negation or privation of one of the two; for why of the great rather than of the small? It is, then, the privative negation of both. This is why 'whether' is said with reference to both, not to one of the two (e.g. 'whether it is greater or equal' or 'whether it is equal or less'); there are always three cases. But it is not a necessary privation; for not everything which is not greater or less is equal, but only the things which are of such a nature as to have these attributes.

The equal, then, is that which is neither great nor small but is naturally fitted to be either great or small; and it is opposed to both as a privative negation (and therefore is also intermediate). And that which is neither good nor bad is opposed to both, but has no name; for each of these has several meanings and the recipient subject is not one; but that which is neither white nor black has more claim to unity. Yet even this has not one name, though the colours of which this negation is privatively predicated are in a way limited; for they must be either grey or yellow or something else of the kind. Therefore it is an incorrect criticism that is passed by those who think that all such phrases are used in the same way, so that that which is neither a shoe nor a hand would be intermediate between a shoe and a hand, since that which is neither good nor bad is intermediate

between the good and the bad-as if there must be an intermediate in all cases. But this does not necessarily follow. For the one phrase is a joint denial of opposites between which there is an intermediate and a certain natural interval; but between the other two there is no 'difference'; for the things, the denials of which are combined, belong to different classes, so that the substratum is not one.

Part 6

We might raise similar questions about the one and the many. For if the many are absolutely opposed to the one, certain impossible results follow. One will then be few, whether few be treated here as singular or plural; for the many are opposed also to the few. Further, two will be many, since the double is multiple and 'double' derives its meaning from 'two'; therefore one will be few; for what is that in comparison with which two are many, except one, which must therefore be few? For there is nothing fewer. Further, if the much and the little are in plurality what the long and the short are in length, and whatever is much is also many, and the many are much (unless, indeed, there is a difference in the case of an easily-bounded continuum), the little (or few) will be a plurality. Therefore one is a plurality if it is few; and this it must be, if two are many. But perhaps, while the 'many' are in a sense said to be also 'much', it is with a difference; e.g. water is much but not many. But 'many' is applied to the things that are divisible; in the one sense it means a plurality which is excessive either absolutely or relatively (while 'few' is similarly a plurality which is deficient), and in another sense it means number, in which sense alone it is opposed to the one. For we say 'one or many', just as if one were to say 'one and ones' or 'white thing and white things', or to compare the things that have been measured with the measure. It is in this sense also that multiples are so called. For each number is said to be many because it consists of ones and because each number is measurable by one; and it is 'many' as that which is opposed to one, not to the few. In this sense, then, even two is many-not, however, in the sense of a plurality which is excessive either relatively or absolutely; it is the first plurality. But without qualification two is few; for it is first plurality which is deficient (for this reason Anaxagoras was not right in leaving the subject with the statement that 'all things were together, boundless both in plurality and in smallness'-where for 'and in smallness' he should have said 'and in fewness'; for they could not have been boundless in fewness), since it is not one, as some say, but two, that make a few.

The one is opposed then to the many in numbers as measure to thing measurable; and these are opposed as are the relatives which are not from their very nature relatives. We have distinguished elsewhere the two senses in which relatives are so called:-(1) as contraries; (2) as knowledge to thing known, a term being called relative because another is relative to it. There is nothing to prevent one from being fewer than something, e.g. than two; for if one is fewer, it is not therefore few. Plurality is as it were the class to which number belongs; for number is plurality measurable by one, and one and number are in a sense opposed, not as contrary, but as we have said some relative terms are opposed; for inasmuch as one is measure and the other measurable, they are opposed. This is why not everything that is one is a number; i.e. if the thing is indivisible it is not a number. But though knowledge is similarly spoken of as relative to the knowable, the relation does not work out similarly; for while knowledge might be thought to be the measure, and the knowable the thing measured, the fact that all knowledge is knowable, but not all that is knowable is knowledge, because in a sense knowledge is measured by the knowable.-Plurality is contrary neither to the few (the many being contrary to this as excessive plurality to plurality exceeded), nor to the one in every sense; but in the one sense these are contrary, as has been said, because the former is divisible and the latter indivisible, while in another sense they are relative as knowledge

is to knowable, if plurality is number and the one is a measure.

Part 7

Since contraries admit of an intermediate and in some cases have it, intermediates must be composed of the contraries. For (1) all intermediates are in the same genus as the things between which they stand. For we call those things intermediates, into which that which changes must change first; e.g. if we were to pass from the highest string to the lowest by the smallest intervals, we should come sooner to the intermediate notes, and in colours if we were to pass from white to black, we should come sooner to crimson and grey than to black; and similarly in all other cases. But to change from one genus to another genus is not possible except in an incidental way, as from colour to figure. Intermediates, then, must be in the same genus both as one another and as the things they stand between.

But (2) all intermediates stand between opposites of some kind; for only between these can change take place in virtue of their own nature (so that an intermediate is impossible between things which are not opposite; for then there would be change which was not from one opposite towards the other). Of opposites, contradictories admit of no middle term; for this is what contradiction is—an opposition, one or other side of which must attach to anything whatever, i.e. which has no intermediate. Of other opposites, some are relative, others privative, others contrary. Of relative terms, those which are not contrary have no intermediate; the reason is that they are not in the same genus. For what intermediate could there be between knowledge and knowable? But between great and small there is one.

(3) If intermediates are in the same genus, as has been shown, and stand between contraries, they must be composed of these contraries. For either there will be a genus including the contraries or there will be none. And if (a) there is to be a genus in such a way that it is something prior to the contraries, the differentiae which constituted the contrary species-of-a-genus will be contraries prior to the species; for species are composed of the genus and the differentiae. (E.g. if white and black are contraries, and one is a piercing colour and the other a compressing colour, these differentiae—'piercing' and 'compressing'—are prior; so that these are prior contraries of one another.) But, again, the species which differ contrariwise are the more truly contrary species. And the other species, i.e. the intermediates, must be composed of their genus and their differentiae. (E.g. all colours which are between white and black must be said to be composed of the genus, i.e. colour, and certain differentiae. But these differentiae will not be the primary contraries; otherwise every colour would be either white or black. They are different, then, from the primary contraries; and therefore they will be between the primary contraries; the primary differentiae are 'piercing' and 'compressing'.)

Therefore it is (b) with regard to these contraries which do not fall within a genus that we must first ask of what their intermediates are composed. (For things which are in the same genus must be composed of terms in which the genus is not an element, or else be themselves incomposite.) Now contraries do not involve one another in their composition, and are therefore first principles; but the intermediates are either all incomposite, or none of them. But there is something compounded out of the contraries, so that there can be a change from a contrary to it sooner than to the other contrary; for it will have less of the quality in question than the one contrary and more than the other. This also, then, will come between the contraries. All the other intermediates also, therefore, are composite; for that which has more of a quality than one thing and less than another is compounded somehow out of the things than which it is said to have more and less respectively of the quality. And since there are no other things prior to the contraries and homogeneous with

the intermediates, all intermediates must be compounded out of the contraries. Therefore also all the inferior classes, both the contraries and their intermediates, will be compounded out of the primary contraries. Clearly, then, intermediates are (1) all in the same genus and (2) intermediate between contraries, and (3) all compounded out of the contraries.

Part 8

That which is other in species is other than something in something, and this must belong to both; e.g. if it is an animal other in species, both are animals. The things, then, which are other in species must be in the same genus. For by genus I mean that one identical thing which is predicated of both and is differentiated in no merely accidental way, whether conceived as matter or otherwise. For not only must the common nature attach to the different things, e.g. not only must both be animals, but this very animality must also be different for each (e.g. in the one case equinity, in the other humanity), and so this common nature is specifically different for each from what it is for the other. One, then, will be in virtue of its own nature one sort of animal, and the other another, e.g. one a horse and the other a man. This difference, then, must be an otherness of the genus. For I give the name of 'difference in the genus' an otherness which makes the genus itself other.

This, then, will be a contrariety (as can be shown also by induction). For all things are divided by opposites, and it has been proved that contraries are in the same genus. For contrariety was seen to be complete difference; and all difference in species is a difference from something in something; so that this is the same for both and is their genus. (Hence also all contraries which are different in species and not in genus are in the same line of predication, and other than one another in the highest degree-for the difference is complete-, and cannot be present along with one another.) The difference, then, is a contrariety.

This, then, is what it is to be 'other in species'-to have a contrariety, being in the same genus and being indivisible (and those things are the same in species which have no contrariety, being indivisible); we say 'being indivisible', for in the process of division contrarieties arise in the intermediate stages before we come to the indivisibles. Evidently, therefore, with reference to that which is called the genus, none of the species-of-a-genus is either the same as it or other than it in species (and this is fitting; for the matter is indicated by negation, and the genus is the matter of that of which it is called the genus, not in the sense in which we speak of the genus or family of the Heraclidae, but in that in which the genus is an element in a thing's nature), nor is it so with reference to things which are not in the same genus, but it will differ in genus from them, and in species from things in the same genus. For a thing's difference from that from which it differs in species must be a contrariety; and this belongs only to things in the same genus.

Part 9

"One might raise the question, why woman does not differ from man in species, when female and male are contrary and their difference is a contrariety; and why a female and a male animal are not different in species, though this difference belongs to animal in virtue of its own nature, and not as paleness or darkness does; both 'female' and 'male' belong to it qua animal. This question is almost the same as the other, why one contrariety makes things different in species and another does not, e.g. 'with feet' and 'with wings' do, but paleness and darkness do not. Perhaps it is because the former are modifications peculiar to the genus, and the latter are less so. And since one element is definition and one is matter, contrarieties which are in the definition make a difference in species, but those which are in the thing taken as including its matter do not make one. And so paleness in a man, or darkness, does not make one, nor is there a difference in species between the pale man and the dark man, not even if each of them be denoted by one word. For man is here

being considered on his material side, and matter does not create a difference; for it does not make individual men species of man, though the flesh and the bones of which this man and that man consist are other. The concrete thing is other, but not other in species, because in the definition there is no contrariety. This is the ultimate indivisible kind. Callias is definition + matter, the pale man, then, is so also, because it is the individual Callias that is pale; man, then, is pale only incidentally. Neither do a brazen and a wooden circle, then, differ in species; and if a brazen triangle and a wooden circle differ in species, it is not because of the matter, but because there is a contrariety in the definition. But does the matter not make things other in species, when it is other in a certain way, or is there a sense in which it does? For why is this horse other than this man in species, although their matter is included with their definitions? Doubtless because there is a contrariety in the definition. For while there is a contrariety also between pale man and dark horse, and it is a contrariety in species, it does not depend on the paleness of the one and the darkness of the other, since even if both had been pale, yet they would have been other in species. But male and female, while they are modifications peculiar to 'animal', are so not in virtue of its essence but in the matter, ie. the body. This is why the same seed becomes female or male by being acted on in a certain way. We have stated, then, what it is to be other in species, and why some things differ in species and others do not.

Part 10

Since contraries are other in form, and the perishable and the imperishable are contraries (for privation is a determinate incapacity), the perishable and the imperishable must be different in kind.

Now so far we have spoken of the general terms themselves, so that it might be thought not to be necessary that every imperishable thing should be different from every perishable thing in form, just as not every pale thing is different in form from every dark thing. For the same thing can be both, and even at the same time if it is a universal (e.g. man can be both pale and dark), and if it is an individual it can still be both; for the same man can be, though not at the same time, pale and dark. Yet pale is contrary to dark.

But while some contraries belong to certain things by accident (e.g. both those now mentioned and many others), others cannot, and among these are 'perishable' and 'imperishable'. For nothing is by accident perishable. For what is accidental is capable of not being present, but perishableness is one of the attributes that belong of necessity to the things to which they belong; or else one and the same thing may be perishable and imperishable, if perishableness is capable of not belonging to it. Perishableness then must either be the essence or be present in the essence of each perishable thing. The same account holds good for imperishableness also; for both are attributes which are present of necessity. The characteristics, then, in respect of which and in direct consequence of which one thing is perishable and another imperishable, are opposite, so that the things must be different in kind.

Evidently, then, there cannot be Forms such as some maintain, for then one man would be perishable and another imperishable. Yet the Forms are said to be the same in form with the individuals and not merely to have the same name; but things which differ in kind are farther apart than those which differ in form.

Book XI

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Book XII

Layout 2

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matter ; and of eternal things those which are not generable 25 but are movable in space have matter — not matter for generation, however, but for motion from one place to another.

(One might raise the question from what sort of non-being generation proceeds ; for * non-being ' has three senses.^)

If, then, a thing exists potentially, still it is not potentially any and every thing, but different things come from different things; nor is it satisfactory to say that *all things were together ' ; for they differ in their matter, since otherwise 30 why did an infinity of things come to be, and not one thing ? For * reason ^ ' is one, so that if matter also is one, that must have come to be in actuality which the matter was in potency." The causes and the principles, then, are three, two being the pair of contraries of which one is definition and form and the other is privation, and the third being the matter."

CHAPTER III

Next we must observe that neither the matter nor the form 35 comes to be — i. e. the proximate matter and form. For every- thing that changes is something and is changed by something and into something. That by which it is changed is the im- 1070* mediate mover ; that which is changed, the matter; that into which it is changed^the form. The process, then, will go on to infinity, if not only the bronze comes to be round but also the round or the bronze comes to be; therefore there must be a stop at some point.

Next we must observe that each substance comes into being 5 out of something that shares its name. (Natural objects and other things both rank as substances.) For things come into being either by art or by nature or by luck or by spontaneity. Now art is a principle of movement in something other

Alexander points out that a7rop^<rete ... ^1/ refers to 1. 20. The three senses are probably the absolutely non-existent, the false, and the potential.

^ Sc. the vovs of Anaxagoras* doctrine, summarized by D. L. in the words, navra xphv^oro. Jy 6ixov^ wa vovs iKOoiV avta dieK6(rfirja'ev*

^ Sc. an undifferentiated unity.

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Book XIII

Layout 2

BOOK XIII (M)

CHAPTER I

We have stated what is the substance of sensible things, dealing in the treatise on physics with the material sub- stratum, and later with the substance which has actual existence. Now since our inquiry is whether there is or is not besides the sensible substances any which is immovable and eternal, and, if there is, what it is, we must first consider what is said by others, so that, if there is anything which they say wrongly, we may not be liable to the same objections, while, if there is any opinion common to them and us, we shall not quarrel with ourselves on that account; for one must be content to state some points better than one's predecessors, and others no worse.

Two opinions are held on this subject; it is said that the objects of mathematics—i. e. numbers and lines and the like—are substances, and again that the Ideas are substances. And since (1) some recognize these as two different classes—the Ideas and the mathematical numbers, and (2) some 20 recognize both as having one nature, while (3) some others say that the mathematical substances are the only substances, we must consider the objects of mathematics, not qualifying them by any other characteristic—not asking, for instance, whether they are Ideas or not, or whether they are the principles and substances of existing things or not, but only whether as the objects of mathematics they exist or not, and if they do, how they exist ; then after this we must separately consider the Ideas themselves in a general way, and only as far as systematic treatment demands; for most of what we have to say has been repeatedly stated in popular works. And the greater part of our account must attack the Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/280 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/281 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/282 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/283 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/284 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/285 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/286 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/287 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/288 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/289 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/290 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/291 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/292 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/293 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/294 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/295 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/296 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/297 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/298 M. BOOK XIII 1083"

ideal 3? There is one over. But perhaps it is for this reason that they give the ideal 1 the middle place in odd numbers.1 (2) But 30 if each of the two units consists of both the great and the small, equalized, how will the 2, which is one thing, consist of the great and the small? Or how will it differ from the unit? Again, the unit is prior to the 2; for when it is destroyed the 2 is destroyed. It must, then, be the Idea of an Idea since it is prior to an Idea, and it must have come into being before it. From what, then? Not from the indefinite dyad, 35 for *its* function was to double.

Again, number must be either infinite or finite; for these thinkers think of number as capable of existing separately, so that it is not possible that neither of those alternatives should be true. Clearly it cannot be *infinite*; for infinite number is 1084a neither odd nor even, but the generation of numbers is always the generation either of an odd or of an even number,—when I operates in one

way² on an even number, an odd number is produced, and when 2 (or an odd number) operates in the 5 other way,³ the numbers got from 1 by doubling (or the other even numbers) are produced. Again, if every Idea is an Idea of something, and the numbers are Ideas, infinite number will be an Idea of something, either of some sensible thing or of something else. Yet this is not possible in view of their hypothesis any more than it is reasonable in itself, if they conceive⁴ of the Ideas as they do.

But if number is *finite*, how far does it go? With regard to 10 this not only the fact but the reason should be stated. But if number goes only up to 10, as some say, firstly the Forms will soon run short; e.g. if 3 is man-in-himself, what number will be the horse-in-itself? The numbers which are Ideas of the several things go up to 10. It must, then, be one of the 15 numbers within these limits; for it is these that are substances and Ideas. Yet they will run short; for the various forms of animal will exceed them. At the same time it is clear that if in this way *the* 3 is the Idea of man, the other 3's are so also (for those in the same number are similar), so that there will be

1 Cf. Diels, Vorsokratiker, ed. 2, p. 270, § 2. 2 Sc. addition. 3 Sc, multiplication. 4 10848 Io read *rarrovai y'*. Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/300 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/301 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/302 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/303 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/304 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/305 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/306 Page:Metaphysics by Aristotle Ross 1908 (deannotated).djvu/307

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